

Predict Bike Sharing Demand with AutoGluon Template

Project: Predict Bike Sharing Demand with AutoGluon

This notebook is a template with each step that you need to complete for the project.

Please fill in your code where there are explicit `?` markers in the notebook. You are welcome to add more cells and code as you see fit.

Once you have completed all the code implementations, please export your notebook as a HTML file so the reviews can view your code. Make sure you have all outputs correctly outputted.

File-> Export Notebook As... -> Export Notebook as HTML

There is a writeup to complete as well after all code implementation is done. Please answer all questions and attach the necessary tables and charts. You can complete the writeup in either markdown or PDF.

Completing the code template and writeup template will cover all of the rubric points for this project.

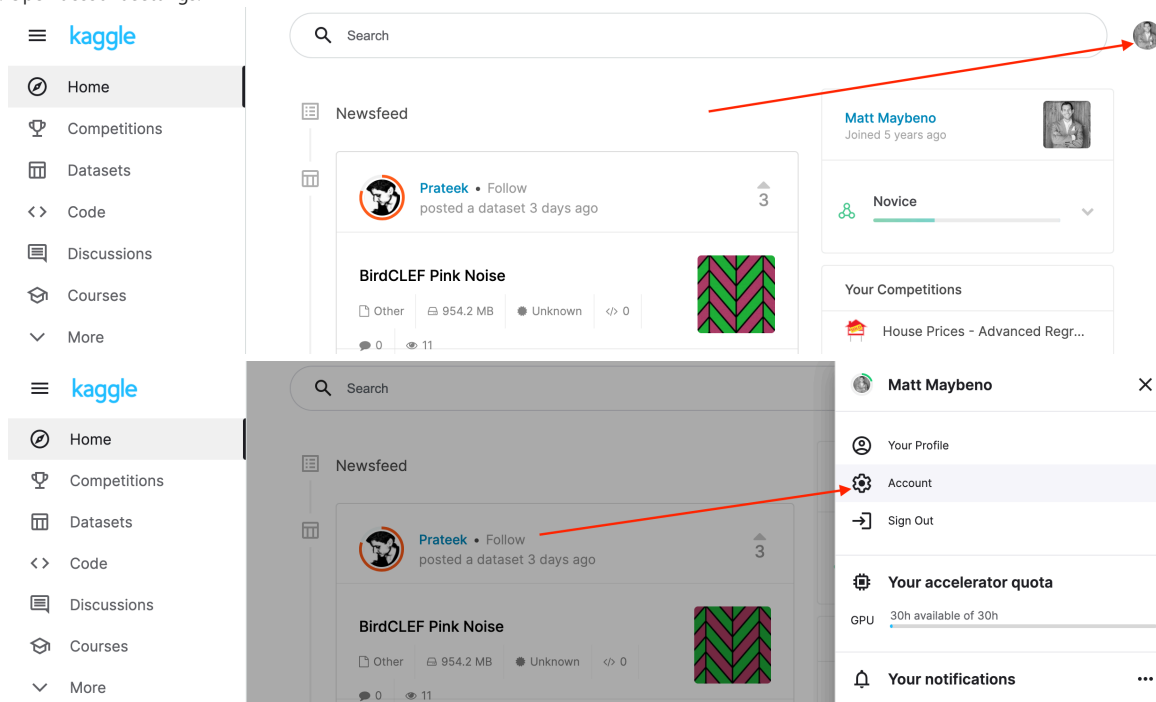
The rubric contains "Stand Out Suggestions" for enhancing the project beyond the minimum requirements. The stand out suggestions are optional. If you decide to pursue the "stand out suggestions", you can include the code in this notebook and also discuss the results in the writeup file.

Step 1: Create an account with Kaggle

Create Kaggle Account and download API key

Below is example of steps to get the API username and key. Each student will have their own username and key.

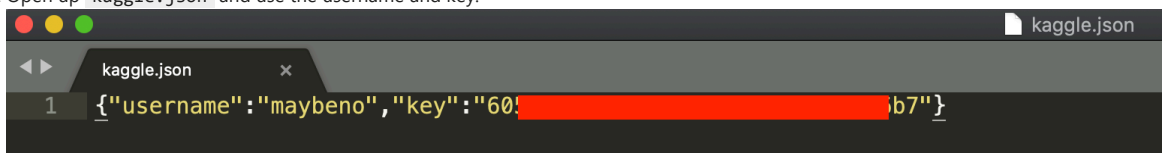
1. Open account settings.



2. Scroll down to API and click Create New API Token.

The screenshot shows the Kaggle Account page for user 'maybeno' (User ID 489628). The page has a navigation bar with links: Home, Competitions (1), Datasets, Code, Discussion, Followers, Notifications, Account (selected), and Edit Public Profile. The account details section includes: User Name (maybeno), Display Name (Matt Maybeno), Email Address (mmaybeno@gmail.com), Phone Verification (Verified), and Email Preferences. The API section is at the bottom, with a red arrow pointing to the 'Create New API Token' button. Below the API section, there is a green banner with a checkmark and the text: 'Ensure kaggle.json is in the location ~/.kaggle/kaggle.json to use the API.'

3. Open up `kaggle.json` and use the username and key.



Step 2: Download the Kaggle dataset using the kaggle python library

Open up Sagemaker Studio and use starter template

1. Notebook should be using a `ml.t3.medium` instance (2 vCPU + 4 GiB)
2. Notebook should be using kernel: `Python 3 (MXNet 1.8 Python 3.7 CPU Optimized)`

Install packages

```
In [1]: !pip install -U pip
!pip install -U setuptools wheel
```

```
!pip install -U "mxnet<2.0.0" bokeh==2.0.1
!pip install autogluon --no-cache-dir
# Without --no-cache-dir, smaller aws instances may have trouble installing
```

Requirement already satisfied: pip in /usr/local/lib/python3.11/dist-packages (26.2.1)
Requirement already satisfied: setuptools in /usr/local/lib/python3.11/dist-packages (81.0.0)
Collecting setuptools
Using cached setuptools-84.0.0-py3-none-any.whl.metadata (6.6 kB)
Requirement already satisfied: wheel in /usr/local/lib/python3.11/dist-packages (0.48.0)
Requirement already satisfied: packaging>=24.0 in /usr/local/lib/python3.11/dist-packages (from wheel) (24.2)
Using cached setuptools-84.0.0-py3-none-any.whl (818 kB)
Installing collected packages: setuptools
Attempting uninstall: setuptools
Found existing installation: setuptools 81.0.0
Uninstalling setuptools-81.0.0:
Successfully uninstalled setuptools-81.0.0
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
ipython 7.34.0 requires jedi>=0.16, which is not installed.
Successfully installed setuptools-84.0.0
Requirement already satisfied: mxnet<2.0.0 in /usr/local/lib/python3.11/dist-packages (1.9.1)
Requirement already satisfied: bokeh==2.0.1 in /usr/local/lib/python3.11/dist-packages (2.0.1)
Requirement already satisfied: PyYAML>=3.10 in /usr/local/lib/python3.11/dist-packages (from bokeh==2.0.1) (6.0.2)
Requirement already satisfied: python-dateutil>=2.1 in /usr/local/lib/python3.11/dist-packages (from bokeh==2.0.1) (2.9.0.post0)
Requirement already satisfied: Jinja2>=2.7 in /usr/local/lib/python3.11/dist-packages (from bokeh==2.0.1) (3.1.6)
Requirement already satisfied: numpy>=1.11.3 in /usr/local/lib/python3.11/dist-packages (from bokeh==2.0.1) (2.0.2)
Requirement already satisfied: pillow>=4.0 in /usr/local/lib/python3.11/dist-packages (from bokeh==2.0.1) (11.2.1)
Requirement already satisfied: packaging>=16.8 in /usr/local/lib/python3.11/dist-packages (from bokeh==2.0.1) (24.2)
Requirement already satisfied: tornado>=5 in /usr/local/lib/python3.11/dist-packages (from bokeh==2.0.1) (6.4.2)
Requirement already satisfied: typing_extensions>=3.7.4 in /usr/local/lib/python3.11/dist-packages (from bokeh==2.0.1) (4.14.1)
Collecting numpy>=1.11.3 (from bokeh==2.0.1)
Using cached numpy-1.26.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl.metadata (61 kB)
Requirement already satisfied: requests<3,>=2.20.0 in /usr/local/lib/python3.11/dist-packages (from mxnet<2.0.0) (2.32.3)
Requirement already satisfied: graphviz<0.9.0,>=0.8.1 in /usr/local/lib/python3.11/dist-packages (from mxnet<2.0.0) (0.8.4)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.20.0->mxnet<2.0.0) (3.4.2)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.20.0->mxnet<2.0.0) (3.10)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.20.0->mxnet<2.0.0) (2.4.0)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests<3,>=2.20.0->mxnet<2.0.0) (2025.7.14)
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from Jinja2>=2.7->bokeh==2.0.1) (3.0.2)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.11/dist-packages (from python-dateutil>=2.1->bokeh==2.0.1) (1.17.0)
Using cached numpy-1.26.4-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (18.3 MB)
Installing collected packages: numpy
Attempting uninstall: numpy
Found existing installation: numpy 2.0.2
Uninstalling numpy-2.0.2:
Successfully uninstalled numpy-2.0.2
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
cudf-cu12 25.2.1 requires pyarrow<20.0.0a0,>=14.0.0; platform_machine == "x86_64", but you have pyarrow 24.0.0 which is incompatible.
holoviews 1.21.0 requires bokeh>=3.1, but you have bokeh 2.0.1 which is incompatible.
opencv-python-headless 4.12.0.88 requires numpy<2.3.0,>=2; python_version >= "3.9", but you have numpy 1.26.4 which is incompatible.
thinc 8.3.6 requires numpy<3.0.0,>=2.0.0, but you have numpy 1.26.4 which is incompatible.
Successfully installed numpy-1.26.4

Requirement already satisfied: autogluon in /usr/local/lib/python3.11/dist-packages (1.6.3)
Requirement already satisfied: autogluon.core==1.6.3 in /usr/local/lib/python3.11/dist-packages (from autogluon.core[all]==1.6.3->autogluon) (1.6.3)
Requirement already satisfied: autogluon.features==1.6.3 in /usr/local/lib/python3.11/dist-packages (from autogluon) (1.6.3)
Requirement already satisfied: autogluon.tabular==1.6.3 in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (1.6.3)
Requirement already satisfied: autogluon.multimodal==1.6.3 in /usr/local/lib/python3.11/dist-packages (from autogluon) (1.6.3)
Requirement already satisfied: autogluon.timeseries==1.6.3 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries[all]==1.6.3->autogluon) (1.6.3)
Requirement already satisfied: numpy<2.6.0,>=1.25.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.26.4)
Requirement already satisfied: scipy<1.19,>=1.5.4 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.15.3)
Requirement already satisfied: scikit-learn<1.10.0,>=1.4.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.6.1)
Requirement already satisfied: networkx<4,>=3.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (3.5)
Requirement already satisfied: pandas<2.4.0,>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (2.2.2)
Requirement already satisfied: tqdm<5,>=4.38 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (4.67.1)
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (2.32.3)
Requirement already satisfied: matplotlib<3.12,>=3.7.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (3.10.0)
Requirement already satisfied: boto3<2,>=1.10 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.43.108)
Requirement already satisfied: typing-extensions<5,>=4.14.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (4.14.1)
Requirement already satisfied: autogluon.common==1.6.3 in /usr/local/lib/python3.11/dist-packages (from autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.6.3)
Requirement already satisfied: pyarrow<25.0.0,>=23.0.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.common==1.6.3->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (24.0.0)
Requirement already satisfied: psutil<7.3.0,>=5.7.3 in /usr/local/lib/python3.11/dist-packages (from autogluon.common==1.6.3->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (5.9.5)
Requirement already satisfied: joblib<1.7,>=1.2 in /usr/local/lib/python3.11/dist-packages (from autogluon.common==1.6.3->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.5.1)
Requirement already satisfied: pyyaml<5.4 in /usr/local/lib/python3.11/dist-packages (from autogluon.common==1.6.3->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (6.0.2)
Requirement already satisfied: packaging<20 in /usr/local/lib/python3.11/dist-packages (from autogluon.common==1.6.3->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (24.2)
Requirement already satisfied: ray<2.59,>=2.55.0 in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (2.58.0)
Collecting setuputils<82 (from autogluon.core[all]==1.6.3->autogluon)
 Downloading setuputils-81.0.0-py3-none-any.whl.metadata (6.6 kB)
Requirement already satisfied: hyperopt<0.2.8,>=0.2.7 in /usr/local/lib/python3.11/dist-packages (from autogluon.core[all]==1.6.3->autogluon) (0.2.7)
Requirement already satisfied: stevedore<5.5 in /usr/local/lib/python3.11/dist-packages (from autogluon.core[all]==1.6.3->autogluon) (5.4.1)
Requirement already satisfied: Pillow<13,>=10.0.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (11.2.1)
Requirement already satisfied: torch<2.14,>=2.10 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (2.13.0)
Requirement already satisfied: lightning<2.7,>=2.5.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (2.6.6)
Requirement already satisfied: transformers<5.15,>=5.3 in /usr/local/lib/python3.11/dist-packages (from transformers[sentencepiece]<5.15,>=5.3->autogluon.multimodal==1.6.3->autogluon) (5.14.1)
Requirement already satisfied: accelerate<2.0,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (1.8.1)
Requirement already satisfied: fsspec<2027.0 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<2027.0->autogluon.multimodal==1.6.3->autogluon) (2025.3.2)
Requirement already satisfied: jsonschema<4.24,>=4.18 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (4.23.0)
Requirement already satisfied: sequeval<1.3.0,>=1.2.2 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (1.2.2)
Requirement already satisfied: evaluate<0.5.0,>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (0.4.6)
Requirement already satisfied: timm<1.0.7,>=0.9.5 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (1.0.3)
Requirement already satisfied: torchvision<0.29.0,>=0.21.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (0.28.0)
Requirement already satisfied: scikit-image<0.26.0,>=0.19.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (0.25.2)
Requirement already satisfied: text-unidecode<1.4,>=1.3 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (1.3)
Requirement already satisfied: torchmetrics<1.8,>=1.2.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (1.7.4)
Requirement already satisfied: omegaconf<2.4.0,>=2.1.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (2.3.0)
Requirement already satisfied: pytorch-metric-learning<2.9,>=1.3.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (2.8.1)
Requirement already satisfied: nlpaug<1.2.0,>=1.1.10 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (1.1.11)
Requirement already satisfied: nltk<3.11,>=3.10.3 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (3.10.3)
Requirement already satisfied: openmim<0.4.0,>=0.3.7 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (0.3.9)

Requirement already satisfied: defusedxml<0.7.2,>=0.7.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (0.7.1)

Requirement already satisfied: Jinja2<3.2,>=3.0.3 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (3.1.6)

Requirement already satisfied: tensorboard<3,>=2.9 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (2.18.0)

Requirement already satisfied: pytorch<0.4,>=0.3.9 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (0.3.13)

Requirement already satisfied: nvidia-ml-py3<8.0,>=7.352.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (7.352.0)

Requirement already satisfied: pdf2image<1.19,>=1.17.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.multimodal==1.6.3->autogluon) (1.17.0)

Requirement already satisfied: xgboost-cpu<3.4,>=2.1.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (3.2.0)

Requirement already satisfied: fastai<2.8.8,>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (2.8.7)

Requirement already satisfied: catboost<1.3,>=1.2 in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (1.2.10)

Requirement already satisfied: lightgbm<4.8,>=4.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (4.5.0)

Requirement already satisfied: loguru in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (0.7.3)

Requirement already satisfied: spacy<3.9 in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (3.8.7)

Requirement already satisfied: huggingface_hub<2.0,>=1.3 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (1.33.0)

Requirement already satisfied: einops<0.9,>=0.7 in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (0.8.1)

Requirement already satisfied: fastcore<2 in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (1.14.5)

Requirement already satisfied: einx in /usr/local/lib/python3.11/dist-packages (from autogluon.tabular[all]==1.6.3->autogluon) (0.4.3)

Requirement already satisfied: safetensors<1,>=0.4 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.8.0)

Requirement already satisfied: gluonts<0.18.0,>=0.17.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.17.0)

Requirement already satisfied: statsforecast<2.1.2,>=1.7.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (2.0.3)

Requirement already satisfied: mlforecast<0.15.0,>=0.14.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.14.0)

Requirement already satisfied: utilsforecast<0.2.12,>=0.2.3 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.2.11)

Requirement already satisfied: coreforecast<0.0.17,>=0.0.12 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.0.16)

Requirement already satisfied: fugue>=0.9.0 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.9.7)

Requirement already satisfied: orjson~3.9 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (3.10.18)

Requirement already satisfied: chronos-forecasting<2.4,>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (2.3.2)

Requirement already satisfied: peft<0.20,>=0.18.1 in /usr/local/lib/python3.11/dist-packages (from autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.19.1)

Requirement already satisfied: botocore<1.44.0,>=1.43.108 in /usr/local/lib/python3.11/dist-packages (from boto3<2,>=1.10->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.43.108)

Requirement already satisfied: jmespath<2.0.0,>=0.7.1 in /usr/local/lib/python3.11/dist-packages (from boto3<2,>=1.10->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.1.0)

Requirement already satisfied: s3transfer<0.20.0,>=0.19.0 in /usr/local/lib/python3.11/dist-packages (from boto3<2,>=1.10->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (0.19.2)

Requirement already satisfied: python-dateutil<3.0.0,>=2.1 in /usr/local/lib/python3.11/dist-packages (from botocore<1.44.0,>=1.43.108->boto3<2,>=1.10->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (2.9.0.post0)

Requirement already satisfied: urllib3!=2.2.0,<3,>=1.25.4 in /usr/local/lib/python3.11/dist-packages (from botocore<1.44.0,>=1.43.108->boto3<2,>=1.10->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (2.4.0)

Requirement already satisfied: graphviz in /usr/local/lib/python3.11/dist-packages (from catboost<1.3,>=1.2->autogluon.tabular[all]==1.6.3->autogluon) (0.8.4)

Requirement already satisfied: plotly in /usr/local/lib/python3.11/dist-packages (from catboost<1.3,>=1.2->autogluon.tabular[all]==1.6.3->autogluon) (5.24.1)

Requirement already satisfied: six in /usr/local/lib/python3.11/dist-packages (from catboost<1.3,>=1.2->autogluon.tabular[all]==1.6.3->autogluon) (1.17.0)

Requirement already satisfied: datasets>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from evaluate<0.5.0,>=0.4.0->autogluon.multimodal==1.6.3->autogluon) (5.0.1)

Requirement already satisfied: dill in /usr/local/lib/python3.11/dist-packages (from evaluate<0.5.0,>=0.4.0->autogluon.multimodal==1.6.3->autogluon) (0.3.7)

Requirement already satisfied: xxhash in /usr/local/lib/python3.11/dist-packages (from evaluate<0.5.0,>=0.4.0->autogluon.multimodal==1.6.3->autogluon) (3.5.0)

Requirement already satisfied: multiprocessing in /usr/local/lib/python3.11/dist-packages (from evaluate<0.5.0,>=0.4.0->autogluon.multimodal==1.6.3->autogluon) (0.70.15)

Requirement already satisfied: fastdownload<2,>=0.0.5 in /usr/local/lib/python3.11/dist-packages (from fastai<2.8.8,>=2.3.1->autogluon.tabular[all]==1.6.3->autogluon) (0.0.7)

Requirement already satisfied: fasttransform>=0.0.2 in /usr/local/lib/python3.11/dist-packages (from fastai<2.8.8,>=2.3.1->autogluon.tabular[all]==1.6.3->autogluon) (0.0.2)

Requirement already satisfied: fastprogress>=0.2.4 in /usr/local/lib/python3.11/dist-packages (from fastai<2.8.8,>=2.3.1->autogluon.tabular[all]==1.6.3->autogluon) (1.0.3)

Requirement already satisfied: plum-dispatch in /usr/local/lib/python3.11/dist-packages (from fastai<2.8.8,>=2.3.1->autogluon.tabular[all]==1.6.3->autogluon) (2.10.1)

Requirement already satisfied: cloudpickle in /usr/local/lib/python3.11/dist-packages (from fastai<2.8.8,>=2.3.1->autogluon.tabular[all]==1.6.3->autogluon) (3.1.1)

Requirement already satisfied: aiohttp!=4.0.0a0,!4.0.0a1 in /usr/local/lib/python3.11/dist-packages (from fsspec[http]<2027.0->autogluon.multimodal==1.6.3->autogluon) (3.14.3)

Requirement already satisfied: pydantic<3,>=1.7 in /usr/local/lib/python3.11/dist-packages (from gluonts<0.18.0,>=0.17.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (2.11.7)

Requirement already satisfied: toolz~0.10 in /usr/local/lib/python3.11/dist-packages (from gluonts<0.18.0,>=0.17.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.12.1)

Requirement already satisfied: click<9.0.0,>=8.4.2 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (8.5.0)

Requirement already satisfied: filelock>=3.10.0 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (4.0.10)

Requirement already satisfied: hf-xet<2.0.0,>=1.6.0 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (1.6.0)

Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.11/dist-packages (from huggingface_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (0.28.1)

Requirement already satisfied: anyio in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (4.9.0)

Requirement already satisfied: certifi in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (2025.7.14)

Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (1.0.9)

Requirement already satisfied: idna in /usr/local/lib/python3.11/dist-packages (from httpx<1,>=0.23.0->huggingface_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (3.10)

Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.11/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggingface_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (0.16.0)

Requirement already satisfied: future in /usr/local/lib/python3.11/dist-packages (from hyperopt<0.2.8,>=0.2.7->autogluon.core[all]==1.6.3->autogluon) (1.0.0)

Requirement already satisfied: py4j in /usr/local/lib/python3.11/dist-packages (from hyperopt<0.2.8,>=0.2.7->autogluon.core[all]==1.6.3->autogluon) (0.10.9.7)

Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.11/dist-packages (from jinja2<3.2,>=3.0.3->autogluon.mmultimodal==1.6.3->autogluon) (3.0.2)

Requirement already satisfied: attrs>=22.2.0 in /usr/local/lib/python3.11/dist-packages (from jsonschema<4.24,>=4.18->autogluon.mmultimodal==1.6.3->autogluon) (25.3.0)

Requirement already satisfied: jsonschema-specifications>=2023.03.6 in /usr/local/lib/python3.11/dist-packages (from jsonschema<4.24,>=4.18->autogluon.mmultimodal==1.6.3->autogluon) (2025.4.1)

Requirement already satisfied: referencing>=0.28.4 in /usr/local/lib/python3.11/dist-packages (from jsonschema<4.24,>=4.18->autogluon.mmultimodal==1.6.3->autogluon) (0.36.2)

Requirement already satisfied: rpds-py>=0.7.1 in /usr/local/lib/python3.11/dist-packages (from jsonschema<4.24,>=4.18->autogluon.mmultimodal==1.6.3->autogluon) (0.26.0)

Requirement already satisfied: lightning-utilities<2.0,>=0.10.0 in /usr/local/lib/python3.11/dist-packages (from lightning<2.7,>=2.5.1->autogluon.mmultimodal==1.6.3->autogluon) (0.15.3)

Requirement already satisfied: pytorch-lightning in /usr/local/lib/python3.11/dist-packages (from lightning<2.7,>=2.5.1->autogluon.mmultimodal==1.6.3->autogluon) (2.6.6)

Requirement already satisfied: contourpy>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib<3.12,>=3.7.0->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.3.2)

Requirement already satisfied: cycler>=0.10 in /usr/local/lib/python3.11/dist-packages (from matplotlib<3.12,>=3.7.0->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (0.12.1)

Requirement already satisfied: fonttools>=4.22.0 in /usr/local/lib/python3.11/dist-packages (from matplotlib<3.12,>=3.7.0->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (4.58.5)

Requirement already satisfied: kiwisolver>=1.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib<3.12,>=3.7.0->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (1.4.8)

Requirement already satisfied: pyparsing>=2.3.1 in /usr/local/lib/python3.11/dist-packages (from matplotlib<3.12,>=3.7.0->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (3.2.3)

Requirement already satisfied: numba in /usr/local/lib/python3.11/dist-packages (from mlforecast<0.15.0,>=0.14.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.60.0)

Requirement already satisfied: optuna in /usr/local/lib/python3.11/dist-packages (from mlforecast<0.15.0,>=0.14.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (5.0.0)

Requirement already satisfied: window-ops in /usr/local/lib/python3.11/dist-packages (from mlforecast<0.15.0,>=0.14.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.0.15)

Requirement already satisfied: gdown>=4.0.0 in /usr/local/lib/python3.11/dist-packages (from nlpaug<1.2.0,>=1.1.10->autogluon.mmultimodal==1.6.3->autogluon) (5.2.0)

Requirement already satisfied: regex>=2021.8.3 in /usr/local/lib/python3.11/dist-packages (from nltk<3.11,>=3.10.3->autogluon.mmultimodal==1.6.3->autogluon) (2026.9.29)

Requirement already satisfied: antlr4-python3-runtime==4.9.* in /usr/local/lib/python3.11/dist-packages (from omegaconf<2.4.0,>=2.1.1->autogluon.mmultimodal==1.6.3->autogluon) (4.9.3)

Requirement already satisfied: colorama in /usr/local/lib/python3.11/dist-packages (from openmim<0.4.0,>=0.3.7->autogluon.mmultimodal==1.6.3->autogluon) (0.4.6)

Requirement already satisfied: model-index in /usr/local/lib/python3.11/dist-packages (from openmim<0.4.0,>=0.3.7->autogluon.mmultimodal==1.6.3->autogluon) (0.1.11)

Requirement already satisfied: opendatalab in /usr/local/lib/python3.11/dist-packages (from openmim<0.4.0,>=0.3.7->autogluon.mmultimodal==1.6.3->autogluon) (0.0.10)

Requirement already satisfied: pip>=19.3 in /usr/local/lib/python3.11/dist-packages (from openmim<0.4.0,>=0.3.7->autogluon.mmultimodal==1.6.3->autogluon) (26.2.1)

Requirement already satisfied: rich in /usr/local/lib/python3.11/dist-packages (from openmim<0.4.0,>=0.3.7->autogluon.mmultimodal==1.6.3->autogluon) (13.9.4)

Requirement already satisfied: tabulate in /usr/local/lib/python3.11/dist-packages (from openmim<0.4.0,>=0.3.7->autogluon.mmultimodal==1.6.3->autogluon) (0.9.0)

Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.11/dist-packages (from pandas<2.4.0,>=2.0.0->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (2025.2)

Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.11/dist-packages (from pandas<2.4.0,>=2.0.0->autogluon.core==1.6.3->autogluon.core[all]==1.6.3->autogluon) (2025.2)

Requirement already satisfied: annotated-types>=0.6.0 in /usr/local/lib/python3.11/dist-packages (from pydantic<3,>=1.7->gluonts<0.18.0,>=0.17.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.7.0)

Requirement already satisfied: pydantic-core==2.33.2 in /usr/local/lib/python3.11/dist-packages (from pydantic<3,>=1.7->gluonts<0.18.0,>=0.17.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (2.33.2)

Requirement already satisfied: typing-inspection>=0.4.0 in /usr/local/lib/python3.11/dist-packages (from pydantic<3,>=1.7->gluonts<0.18.0,>=0.17.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.4.1)

Requirement already satisfied: msgpack<2.0.0,>=1.0.0 in /usr/local/lib/python3.11/dist-packages (from ray<2.59,>=2.55.0->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (1.1.1)

Requirement already satisfied: protobuf>=3.20.3 in /usr/local/lib/python3.11/dist-packages (from ray<2.59,>=2.55.0->ray[default

t,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (5.29.5)

Requirement already satisfied: aiohttp_cors in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (0.8.1)

Requirement already satisfied: colorful in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (0.5.8)

Requirement already satisfied: py-spy>=0.2.0 in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (0.4.2)

Requirement already satisfied: grpcio>=1.42.0 in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (1.73.1)

Requirement already satisfied: opencensus in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (0.11.4)

Requirement already satisfied: opentelemetry-sdk>=1.30.0 in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (1.45.0)

Requirement already satisfied: opentelemetry-exporter-prometheus in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (0.66b0)

Requirement already satisfied: opentelemetry-proto in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (1.45.0)

Requirement already satisfied: prometheus_client>=0.7.1 in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (0.22.1)

Requirement already satisfied: smart_open in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (7.3.0.post1)

Requirement already satisfied: virtualenv!=20.21.1,>=20.0.24 in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (21.14.5)

Requirement already satisfied: tensorboardX>=1.9 in /usr/local/lib/python3.11/dist-packages (from ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (2.6.5)

Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests->autogluon.core[all]==1.6.3->autogluon) (3.4.2)

Requirement already satisfied: imageio!=2.35.0,>=2.33 in /usr/local/lib/python3.11/dist-packages (from scikit-image<0.26.0,>=0.19.1->autogluon.multimodal==1.6.3->autogluon) (2.37.0)

Requirement already satisfied: tifffile>=2022.8.12 in /usr/local/lib/python3.11/dist-packages (from scikit-image<0.26.0,>=0.19.1->autogluon.multimodal==1.6.3->autogluon) (2025.6.11)

Requirement already satisfied: lazy-loader>=0.4 in /usr/local/lib/python3.11/dist-packages (from scikit-image<0.26.0,>=0.19.1->autogluon.multimodal==1.6.3->autogluon) (0.4)

Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.11/dist-packages (from scikit-learn<1.10.0,>=1.4.0->autogluon.core[all]==1.6.3->autogluon) (3.6.0)

Requirement already satisfied: spacy-legacy<3.1.0,>=3.0.11 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (3.0.12)

Requirement already satisfied: spacy-loggers<2.0.0,>=1.0.0 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (1.0.5)

Requirement already satisfied: murmurhash<1.1.0,>=0.28.0 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (1.0.13)

Requirement already satisfied: cymem<2.1.0,>=2.0.2 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (2.0.11)

Requirement already satisfied: preshed<3.1.0,>=3.0.2 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (3.0.10)

Requirement already satisfied: thinc<8.4.0,>=8.3.4 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (8.3.6)

Requirement already satisfied: wasabi<1.2.0,>=0.9.1 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (1.1.3)

Requirement already satisfied: srsly<3.0.0,>=2.4.3 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (2.5.1)

Requirement already satisfied: catalogue<2.1.0,>=2.0.6 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (2.0.10)

Requirement already satisfied: weasel<0.5.0,>=0.1.0 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (0.4.1)

Requirement already satisfied: typer<1.0.0,>=0.3.0 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (0.16.0)

Requirement already satisfied: langcodes<4.0.0,>=3.2.0 in /usr/local/lib/python3.11/dist-packages (from spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (3.5.0)

Requirement already satisfied: language-data>=1.2 in /usr/local/lib/python3.11/dist-packages (from langcodes<4.0.0,>=3.2.0->spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (1.3.0)

Requirement already satisfied: statsmodels>=0.13.2 in /usr/local/lib/python3.11/dist-packages (from statsforecast<2.1.2,>=1.7.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.14.5)

Requirement already satisfied: pbr>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from stevedore<5.5->autogluon.core[all]==1.6.3->autogluon) (7.0.3)

Requirement already satisfied: absl-py>=0.4 in /usr/local/lib/python3.11/dist-packages (from tensorboard<3,>=2.9->autogluon.multimodal==1.6.3->autogluon) (1.4.0)

Requirement already satisfied: markdown>=2.6.8 in /usr/local/lib/python3.11/dist-packages (from tensorboard<3,>=2.9->autogluon.multimodal==1.6.3->autogluon) (3.8.2)

Requirement already satisfied: tensorboard-data-server<0.8.0,>=0.7.0 in /usr/local/lib/python3.11/dist-packages (from tensorboard<3,>=2.9->autogluon.multimodal==1.6.3->autogluon) (0.7.2)

Requirement already satisfied: werkzeug>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from tensorboard<3,>=2.9->autogluon.multimodal==1.6.3->autogluon) (3.1.3)

Requirement already satisfied: blis<1.4.0,>=1.3.0 in /usr/local/lib/python3.11/dist-packages (from thinc<8.4.0,>=8.3.4->spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (1.3.0)

Requirement already satisfied: confection<1.0.0,>=0.0.1 in /usr/local/lib/python3.11/dist-packages (from thinc<8.4.0,>=8.3.4->spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (0.1.5)

Collecting numpy<2.6.0,>=1.25.0 (from autogluon.core[all]==1.6.3->autogluon.core[all]==1.6.3->autogluon)

Downloading numpy-2.4.6-cp311-cp311-manylinux_2_27_x86_64.manylinux_2_28_x86_64.whl.metadata (6.6 kB)

Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.11/dist-packages (from torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (1.13.3)

```

imodal==1.6.3->autogluon) (1.14.0)
Requirement already satisfied: cuda-toolkit==13.0.3 in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (13.0.3.0)
Requirement already satisfied: cuda-bindings<14,>=13.0.3 in /usr/local/lib/python3.11/dist-packages (from torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (13.4.3)
Requirement already satisfied: nvidia-cudnn-cu13==9.20.0.48 in /usr/local/lib/python3.11/dist-packages (from torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (9.20.0.48)
Requirement already satisfied: nvidia-cusparselt-cu13==0.8.1 in /usr/local/lib/python3.11/dist-packages (from torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (0.8.1)
Requirement already satisfied: nvidia-nccl-cu13==2.29.7 in /usr/local/lib/python3.11/dist-packages (from torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (2.29.7)
Requirement already satisfied: nvidia-nvshmem-cu13==3.4.5 in /usr/local/lib/python3.11/dist-packages (from torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (3.4.5)
Requirement already satisfied: triton==3.7.1 in /usr/local/lib/python3.11/dist-packages (from torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (3.7.1)
Requirement already satisfied: nvidia-cublas==13.1.1.3.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (13.1.1.3)
Requirement already satisfied: nvidia-cuda-nvrtc==13.0.88.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (13.0.88)
Requirement already satisfied: nvidia-cuda-runtime==13.0.96.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (13.0.96)
Requirement already satisfied: nvidia-cufft==12.0.0.61.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (12.0.0.61)
Requirement already satisfied: nvidia-nvjitlink<14,>=13.0.88 in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (13.4.92)
Requirement already satisfied: nvidia-cufile==1.15.1.6.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (1.15.1.6)
Requirement already satisfied: nvidia-cuda-cupti==13.0.85.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (13.0.85)
Requirement already satisfied: nvidia-curand==10.4.0.35.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (10.4.0.35)
Requirement already satisfied: nvidia-cusolver==12.0.4.66.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (12.0.4.66)
Requirement already satisfied: nvidia-cusparselt==12.6.3.3.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (12.6.3.3)
Requirement already satisfied: nvidia-nvtx==13.0.85.* in /usr/local/lib/python3.11/dist-packages (from cuda-toolkit[cublas,cudart,cufft,cufile,cupti,curand,cusolver,cusparsenvjitlink,nvrtc,nvtx]==13.0.3; platform_system == "Linux"->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (13.0.85)
Requirement already satisfied: cuda-pathfinder==1.4.2 in /usr/local/lib/python3.11/dist-packages (from cuda-bindings<14,>=13.0.3->torch<2.14,>=2.10->autogluon.multimodal==1.6.3->autogluon) (1.8.3)
Requirement already satisfied: tokenizers<0.23.0,>=0.22.0 in /usr/local/lib/python3.11/dist-packages (from transformers<5.15,>=5.3->transformers[sentencepiece]<5.15,>=5.3->autogluon.multimodal==1.6.3->autogluon) (0.22.2)
Requirement already satisfied: sentencepiece!=0.1.92,>=0.1.91 in /usr/local/lib/python3.11/dist-packages (from transformers[sentencepiece]<5.15,>=5.3->autogluon.multimodal==1.6.3->autogluon) (0.2.0)
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.11/dist-packages (from typer<1.0.0,>=0.3.0->spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (1.5.4)
Requirement already satisfied: cloudpathlib<1.0.0,>=0.7.0 in /usr/local/lib/python3.11/dist-packages (from weasel<0.5.0,>=0.1.0->spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (0.21.1)
Requirement already satisfied: wraps in /usr/local/lib/python3.11/dist-packages (from smart_open->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (1.17.2)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<2027.0->autogluon.multimodal==1.6.3->autogluon) (2.6.1)
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<2027.0->autogluon.multimodal==1.6.3->autogluon) (1.4.0)
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<2027.0->autogluon.multimodal==1.6.3->autogluon) (1.7.0)
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<2027.0->autogluon.multimodal==1.6.3->autogluon) (6.6.3)
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<2027.0->autogluon.multimodal==1.6.3->autogluon) (0.3.2)
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.11/dist-packages (from aiohttp!=4.0.0a0,!4.0.0a1->fsspec[http]<2027.0->autogluon.multimodal==1.6.3->autogluon) (1.20.1)
Requirement already satisfied: triad>=1.0.1 in /usr/local/lib/python3.11/dist-packages (from fugue>=0.9.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (1.0.2)
Requirement already satisfied: adagio>=0.2.6 in /usr/local/lib/python3.11/dist-packages (from fugue>=0.9.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.2.6)
Requirement already satisfied: beautifulsoup4 in /usr/local/lib/python3.11/dist-packages (from gdown>=4.0.0->nlpaug<1.2.0,>=1.1.10->autogluon.multimodal==1.6.3->autogluon) (4.13.4)
Requirement already satisfied: marisa-trie>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from language-data>=1.2->langcode<4.0.0,>=3.2.0->spacy<3.9->autogluon.tabular[all]==1.6.3->autogluon) (1.2.1)
Requirement already satisfied: llvmlite<0.44,>=0.43.0dev0 in /usr/local/lib/python3.11/dist-packages (from numba->mlforecast<0.15.0,>=0.14.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (0.43.0)
Downloading numpy-2.0.2-cp311-cp311-manylinux_2_17_x86_64_manylinux2014_x86_64.whl.metadata (60 kB)
Requirement already satisfied: opentelemetry-api==1.45.0 in /usr/local/lib/python3.11/dist-packages (from opentelemetry-sdk==1.30.0->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (1.45.0)

```

Requirement already satisfied: opentelemetry-semantic-conventions==0.66b0 in /usr/local/lib/python3.11/dist-packages (from open
telemetry-sdk>=1.30.0->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra ==
"all"->autogluon.core[all]==1.6.3->autogluon) (0.66b0)
Requirement already satisfied: markdown-it-py==2.2.0 in /usr/local/lib/python3.11/dist-packages (from rich->openmim<0.4.0,>=0.
3.7->autogluon.multimodal==1.6.3->autogluon) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.11/dist-packages (from rich->openmim<0.4.0,>=
0.3.7->autogluon.multimodal==1.6.3->autogluon) (2.19.2)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.11/dist-packages (from markdown-it-py==2.2.0->rich->openmim
<0.4.0,>=0.3.7->autogluon.multimodal==1.6.3->autogluon) (0.1.2)
Requirement already satisfied: patsy>=0.5.6 in /usr/local/lib/python3.11/dist-packages (from statsmodels>=0.13.2->statsforecast
<2.1.2,>=1.7.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (1.0.1)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.11/dist-packages (from sympy>=1.13.3->torch<2.14,>=
2.10->autogluon.multimodal==1.6.3->autogluon) (1.3.0)
Requirement already satisfied: distlib<1,>=0.3.7 in /usr/local/lib/python3.11/dist-packages (from virtualenv!=20.21.1,>=20.0.24
->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.cor
e[all]==1.6.3->autogluon) (0.4.3)
Requirement already satisfied: platformdirs<5,>=3.9.1 in /usr/local/lib/python3.11/dist-packages (from virtualenv!=20.21.1,>=2
0.0.24->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autoglu
on.core[all]==1.6.3->autogluon) (4.3.8)
Requirement already satisfied: python-discovery>=1.6 in /usr/local/lib/python3.11/dist-packages (from virtualenv!=20.21.1,>=20.
0.24->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluo
n.core[all]==1.6.3->autogluon) (1.6.1)
Requirement already satisfied: sniffio>=1.1 in /usr/local/lib/python3.11/dist-packages (from anyio->httpx<1,>=0.23.0->huggingfa
ce_hub<2.0,>=1.3->huggingface_hub[torch]<2.0,>=1.3; extra == "all"->autogluon.tabular[all]==1.6.3->autogluon) (1.3.1)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.11/dist-packages (from beautifulsoup4->gdown>=4.0.0->nlp
aug<1.2.0,>=1.1.10->autogluon.multimodal==1.6.3->autogluon) (2.7)
Requirement already satisfied: frozendict in /usr/local/lib/python3.11/dist-packages (from einx->autogluon.tabular[all]==1.6.3-
>autogluon) (2.4.6)
Requirement already satisfied: ordered-set in /usr/local/lib/python3.11/dist-packages (from model-index->openmim<0.4.0,>=0.3.7-
>autogluon.multimodal==1.6.3->autogluon) (4.1.0)
Requirement already satisfied: opencensus-context>=0.1.3 in /usr/local/lib/python3.11/dist-packages (from opencensus->ray[defau
lt,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]==1.
6.3->autogluon) (0.1.3)
Requirement already satisfied: google-api-core<3.0.0,>=1.0.0 in /usr/local/lib/python3.11/dist-packages (from opencensus->ray[d
efault,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra == "all"->autogluon.core[all]=
=1.6.3->autogluon) (2.25.1)
Requirement already satisfied: googleapis-common-protos<2.0.0,>=1.56.2 in /usr/local/lib/python3.11/dist-packages (from google-
api-core<3.0.0,>=1.0.0->opencensus->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13")
and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (1.70.0)
Requirement already satisfied: proto-plus<2.0.0,>=1.22.3 in /usr/local/lib/python3.11/dist-packages (from google-api-core<3.0.
0,>=1.0.0->opencensus->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra ==
"all"->autogluon.core[all]==1.6.3->autogluon) (1.26.1)
Requirement already satisfied: google-auth<3.0.0,>=2.14.1 in /usr/local/lib/python3.11/dist-packages (from google-api-core<3.0.
0,>=1.0.0->opencensus->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.13") and extra ==
"all"->autogluon.core[all]==1.6.3->autogluon) (2.38.0)
Requirement already satisfied: cachetools<6.0,>=2.0.0 in /usr/local/lib/python3.11/dist-packages (from google-auth<3.0.0,>=2.1
4.1->google-api-core<3.0.0,>=1.0.0->opencensus->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_versio
n != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (5.5.2)
Requirement already satisfied: pyasn1-modules>=0.2.1 in /usr/local/lib/python3.11/dist-packages (from google-auth<3.0.0,>=2.14.
1->google-api-core<3.0.0,>=1.0.0->opencensus->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version
!= "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (0.4.2)
Requirement already satisfied: rsa<5,>=3.1.4 in /usr/local/lib/python3.11/dist-packages (from google-auth<3.0.0,>=2.14.1->googl
e-api-core<3.0.0,>=1.0.0->opencensus->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_version != "3.1
3") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (4.9.1)
Requirement already satisfied: pyasn1>=0.1.3 in /usr/local/lib/python3.11/dist-packages (from rsa<5,>=3.1.4->google-auth<3.0.0,
>=2.14.1->google-api-core<3.0.0,>=1.0.0->opencensus->ray[default,tune]<2.59,>=2.55.0; (platform_system != "Windows" or python_v
ersion != "3.13") and extra == "all"->autogluon.core[all]==1.6.3->autogluon) (0.6.1)
Requirement already satisfied: pycryptodome in /usr/local/lib/python3.11/dist-packages (from opendatalab->openmim<0.4.0,>=0.3.7
>autogluon.multimodal==1.6.3->autogluon) (3.24.0)
Requirement already satisfied: openxlab in /usr/local/lib/python3.11/dist-packages (from opendatalab->openmim<0.4.0,>=0.3.7->au
togluon.multimodal==1.6.3->autogluon) (0.0.11)
Requirement already satisfied: alembic>=1.5.0 in /usr/local/lib/python3.11/dist-packages (from optuna->mlforecast<0.15.0,>=0.1
4.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (1.20.0)
Requirement already satisfied: colorlog in /usr/local/lib/python3.11/dist-packages (from optuna->mlforecast<0.15.0,>=0.14.0->au
togluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (6.12.0)
Requirement already satisfied: sqlalchemy>=1.4.2 in /usr/local/lib/python3.11/dist-packages (from optuna->mlforecast<0.15.0,>=
0.14.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (2.0.41)
Requirement already satisfied: Mako in /usr/lib/python3/dist-packages (from alembic>=1.5.0->optuna->mlforecast<0.15.0,>=0.14.0-
>autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (1.1.3)
Requirement already satisfied: greenlet>=1 in /usr/local/lib/python3.11/dist-packages (from sqlalchemy>=1.4.2->optuna->mlforeca
st<0.15.0,>=0.14.0->autogluon.timeseries==1.6.3->autogluon.timeseries[all]==1.6.3->autogluon) (3.2.3)
Requirement already satisfied: tenacity>=6.2.0 in /usr/local/lib/python3.11/dist-packages (from plotly->catboost<1.3,>=1.2->aut
ogluon.tabular[all]==1.6.3->autogluon) (8.5.0)
Requirement already satisfied: beartype>=0.16.2 in /usr/local/lib/python3.11/dist-packages (from plum-dispatch->fastai<2.8.8,>=
2.3.1->autogluon.tabular[all]==1.6.3->autogluon) (0.22.9)
Requirement already satisfied: PySocks!=1.5.7,>=1.5.6 in /usr/local/lib/python3.11/dist-packages (from requests[socks]->gdown>=
4.0.0->nlpaug<1.2.0,>=1.1.10->autogluon.multimodal==1.6.3->autogluon) (1.7.1)
Downloading setuptools-81.0.0-py3-none-any.whl (1.1 MB)
1.1/1.1 MB 15.3 MB/s 0:00:00

Downloading numpy-2.0.2-cp311-cp311-manylinux_2_17_x86_64.manylinux2014_x86_64.whl (19.5 MB)

19.5/19.5 MB 66.7 MB/s 0:00:00eta 0:00:01

Installing collected packages: setuptools, numpy

Attempting uninstall: setuptools

Found existing installation: setuptools 84.0.0

Uninstalling setuptools-84.0.0:

Successfully uninstalled setuptools-84.0.0 0/2 [setuptools]

Attempting uninstall: numpy

Found existing installation: numpy 1.26.4 0/2 [setuptools]

```

Uninstalling numpy-1.26.4:0m ————— 1/2 [numpy]
Successfully uninstalled numpy-1.26.4 ————— 1/2 [numpy]
2/2 [numpy]32m1/2 [numpy]
ERROR: pip's dependency resolver does not currently take into account all the packages that are installed. This behaviour is the source of the following dependency conflicts.
ipython 7.34.0 requires jedi>=0.16, which is not installed.
mxnet 1.9.1 requires numpy<2.0.0,>1.16.0, but you have numpy 2.0.2 which is incompatible.
cudf-cu12 25.2.1 requires pyarrow<20.0.0a0,>=14.0.0; platform_machine == "x86_64", but you have pyarrow 24.0.0 which is incompatible.
holoviews 1.21.0 requires bokeh>=3.1, but you have bokeh 2.0.1 which is incompatible.
Successfully installed numpy-2.0.2 setuptools-81.0.0

```

Setup Kaggle API Key

```

In [2]: # Fill in your user name and key from creating the kaggle account and API token file
# import json
# kaggle_username = "micaellesouza"
# kaggle_key = "<your-kaggle-api-key>"

# # Save API token the kaggle.json file
# # with open("/root/.kaggle/kaggle.json", "w") as f:
# #     f.write(json.dumps({"username": kaggle_username, "key": kaggle_key}))

```

Download and explore dataset

Go to the bike sharing demand competition and agree to the terms

Data Fields

datetime - hourly date + timestamp
season - 1 = spring, 2 = summer, 3 = fall, 4 = winter
holiday - whether the day is considered a holiday

>_ kaggle competitions download -c bike-sharing-demand

Data Explorer

1.06 MB

- sampleSubmission.csv
- test.csv
- train.csv

Summary

- 3 files
- 23 columns

Download All

< sampleSubmission.csv (139.51 KB)

Competition Rules

To see this data you need to agree to the [competition rules](#).
By clicking "I understand and accept" you agree to be bound to these rules.

I understand and agree

```

In [3]: # Download the dataset, it will be in a .zip file so you'll need to unzip it as well.
!kaggle competitions download -c bike-sharing-demand
# If you already downloaded it you can use the -o command to overwrite the file
!unzip -o bike-sharing-demand.zip

```

```

Traceback (most recent call last):
  File "/usr/local/bin/kaggle", line 4, in <module>
    from kaggle.cli import main
  File "/usr/local/lib/python3.11/dist-packages/kaggle/__init__.py", line 6, in <module>
    api.authenticate()
  File "/usr/local/lib/python3.11/dist-packages/kaggle/api/kaggle_api_extended.py", line 434, in authenticate
    raise IOError('Could not find {}. Make sure it\'s located in'
OSError: Could not find kaggle.json. Make sure it's located in /root/.config/kaggle. Or use the environment method. See setup instructions at https://github.com/Kaggle/kaggle-api/
unzip: cannot find or open bike-sharing-demand.zip, bike-sharing-demand.zip.zip or bike-sharing-demand.zip.ZIP.

```

```

In [4]: !pip install kagglehub

```

Requirement already satisfied: kagglehub in /usr/local/lib/python3.11/dist-packages (0.3.12)
 Requirement already satisfied: packaging in /usr/local/lib/python3.11/dist-packages (from kagglehub) (24.2)
 Requirement already satisfied: pyyaml in /usr/local/lib/python3.11/dist-packages (from kagglehub) (6.0.2)
 Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from kagglehub) (2.32.3)
 Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from kagglehub) (4.67.1)
 Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.11/dist-packages (from requests->kagglehub) (3.4.2)
 Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.11/dist-packages (from requests->kagglehub) (3.10)
 Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.11/dist-packages (from requests->kagglehub) (2.4.0)
 Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.11/dist-packages (from requests->kagglehub) (2025.7.14)

```
In [9]: import kagglehub

kagglehub.login()
```

VBox(children=(HTML(value='<center> <img\nsrc=https://www.kaggle.com/static/images/site-logo.png\nalt=\'Kaggle...

```
In [10]: # Download Latest version
path = kagglehub.competition_download('bike-sharing-demand')

print("Path to competition files:", path)
```

Downloading from https://www.kaggle.com/api/v1/competitions/data/download-all/bike-sharing-demand...

100%|██████████| 189k/189k [00:00<00:00, 55.9MB/s]

Extracting files...

Path to competition files: /root/.cache/kagglehub/competitions/bike-sharing-demand

```
In [11]: import os

for file in os.listdir(path):
    print(file)
```

test.csv
 train.csv
 sampleSubmission.csv

```
In [12]: import pandas as pd
from autogluon.tabular import TabularPredictor
```

```
In [13]: # Create the train dataset in pandas by reading the csv
# Set the parsing of the datetime column so you can use some of the `dt` features in pandas later
train = pd.read_csv(path+"/train.csv")
train.head()
```

```
Out[13]:
```

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	casual	registered	count
0	2011-01-01 00:00:00	1	0	0	1	9.84	14.395	81	0.0	3	13	16
1	2011-01-01 01:00:00	1	0	0	1	9.02	13.635	80	0.0	8	32	40
2	2011-01-01 02:00:00	1	0	0	1	9.02	13.635	80	0.0	5	27	32
3	2011-01-01 03:00:00	1	0	0	1	9.84	14.395	75	0.0	3	10	13
4	2011-01-01 04:00:00	1	0	0	1	9.84	14.395	75	0.0	0	1	1

Features from dataset card

Data Fields

datetime - hourly date + timestamp

season - 1 = spring, 2 = summer, 3 = fall, 4 = winter

holiday - whether the day is considered a holiday

workingday - whether the day is neither a weekend nor holiday

weather - 1: Clear, Few clouds, Partly cloudy, Partly cloudy

2: Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist

3: Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds

4: Heavy Rain + Ice Pallets + Thunderstorm + Mist, Snow + Fog

temp - temperature in Celsius

atemp - "feels like" temperature in Celsius

humidity - relative humidity

windspeed - wind speed

casual - number of non-registered user rentals initiated

registered - number of registered user rentals initiated

count - number of total rentals

```
In [14]: train.info()
```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10886 entries, 0 to 10885
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   datetime    10886 non-null  object
1   season      10886 non-null  int64
2   holiday     10886 non-null  int64
3   workingday  10886 non-null  int64
4   weather     10886 non-null  int64
5   temp        10886 non-null  float64
6   atemp       10886 non-null  float64
7   humidity    10886 non-null  int64
8   windspeed   10886 non-null  float64
9   casual      10886 non-null  int64
10  registered  10886 non-null  int64
11  count       10886 non-null  int64
dtypes: float64(3), int64(8), object(1)
memory usage: 1020.7+ KB

```

```

In [15]: def cast_format(df):
         df['datetime'] = pd.to_datetime(df['datetime'])
         df['season'] = df['season'].astype('category')
         df['holiday'] = df['holiday'].astype('category')
         df['workingday'] = df['workingday'].astype('category')
         df['weather'] = df['weather'].astype('category')
         return df

```

```

In [16]: train = cast_format(train)

```

```

In [18]: train.info()

```

```

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10886 entries, 0 to 10885
Data columns (total 12 columns):
#   Column      Non-Null Count  Dtype
---  ---
0   datetime    10886 non-null  datetime64[ns]
1   season      10886 non-null  category
2   holiday     10886 non-null  category
3   workingday  10886 non-null  category
4   weather     10886 non-null  category
5   temp        10886 non-null  float64
6   atemp       10886 non-null  float64
7   humidity    10886 non-null  int64
8   windspeed   10886 non-null  float64
9   casual      10886 non-null  int64
10  registered  10886 non-null  int64
11  count       10886 non-null  int64
dtypes: category(4), datetime64[ns](1), float64(3), int64(4)
memory usage: 723.7 KB

```

```

In [17]: # Simple output of the train dataset to view some of the min/max/varition of the dataset features.
         train.describe()

```

```

Out[17]:

```

	datetime	temp	atemp	humidity	windspeed	casual	registered	count
count	10886	10886.00000	10886.000000	10886.000000	10886.000000	10886.000000	10886.000000	10886.000000
mean	2011-12-27 05:56:22.399411968	20.23086	23.655084	61.886460	12.799395	36.021955	155.552177	191.574132
min	2011-01-01 00:00:00	0.82000	0.760000	0.000000	0.000000	0.000000	0.000000	1.000000
25%	2011-07-02 07:15:00	13.94000	16.665000	47.000000	7.001500	4.000000	36.000000	42.000000
50%	2012-01-01 20:30:00	20.50000	24.240000	62.000000	12.998000	17.000000	118.000000	145.000000
75%	2012-07-01 12:45:00	26.24000	31.060000	77.000000	16.997900	49.000000	222.000000	284.000000
max	2012-12-19 23:00:00	41.00000	45.455000	100.000000	56.996900	367.000000	886.000000	977.000000
std	NaN	7.79159	8.474601	19.245033	8.164537	49.960477	151.039033	181.144454

```

In [19]: train.columns

```

```

Out[19]: Index(['datetime', 'season', 'holiday', 'workingday', 'weather', 'temp',
               'atemp', 'humidity', 'windspeed', 'casual', 'registered', 'count'],
              dtype='object')

```

```

In [20]: train.drop(columns=['casual','registered'], inplace=True)

```

```

In [21]: train.columns

```

```

Out[21]: Index(['datetime', 'season', 'holiday', 'workingday', 'weather', 'temp',
               'atemp', 'humidity', 'windspeed', 'count'],
              dtype='object')

```

```
In [22]: # Create the test pandas dataframe in pandas by reading the csv, remember to parse the datetime!
test = pd.read_csv(path+"/test.csv")
test.head()
```

```
Out[22]:
```

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed
0	2011-01-20 00:00:00	1	0	1	1	10.66	11.365	56	26.0027
1	2011-01-20 01:00:00	1	0	1	1	10.66	13.635	56	0.0000
2	2011-01-20 02:00:00	1	0	1	1	10.66	13.635	56	0.0000
3	2011-01-20 03:00:00	1	0	1	1	10.66	12.880	56	11.0014
4	2011-01-20 04:00:00	1	0	1	1	10.66	12.880	56	11.0014

```
In [23]: test = cast_format(test)
```

```
In [24]: # Same thing as train and test dataset
submission = pd.read_csv(path+'/sampleSubmission.csv')
submission.head()
```

```
Out[24]:
```

	datetime	count
0	2011-01-20 00:00:00	0
1	2011-01-20 01:00:00	0
2	2011-01-20 02:00:00	0
3	2011-01-20 03:00:00	0
4	2011-01-20 04:00:00	0

Step 3: Train a model using AutoGluon's Tabular Prediction

Requirements:

- We are predicting `count`, so it is the label we are setting.
- Ignore `casual` and `registered` columns as they are also not present in the test dataset.
- Use the `root_mean_squared_error` as the metric to use for evaluation.
- Set a time limit of 10 minutes (600 seconds).
- Use the preset `best_quality` to focus on creating the best model.

```
In [27]: # import ray
# if ray.is_initialized():
#     ray.shutdown()

predictor = TabularPredictor(
    label="count", eval_metric="root_mean_squared_error"
).fit(
    train_data=train,
    time_limit=600,
    presets="best_quality",
    fit_strategy="parallel"
    #dynamic_stacking=False,
    #num_cpus=2,
    #ag_args_ensemble={"fold_fitting_strategy": "sequential_Local"},
)
```

```

No path specified. Models will be saved in: "AutogluonModels/ag-20261004_182830"
Verbosity: 2 (Standard Logging)
===== System Info =====
AutoGluon Version: 1.6.3
Python Version: 3.11.13
Operating System: Linux
Platform Machine: x86_64
Platform Version: #1 SMP Thu Apr 30 18:17:14 UTC 2026
CPU Count: 2
Pytorch Version: 2.13.0+cu130
CUDA Version: 13.0
GPU Memory: GPU 0: 14.56/14.56 GB
Total GPU Memory: Free: 14.56 GB, Allocated: 0.00 GB, Total: 14.56 GB
GPU Count: 1
Memory Avail: 8.91 GB / 12.67 GB (70.3%)
Disk Space Avail: 188.22 GB / 241.91 GB (77.8%)
=====
Presets specified: ['best_quality']
Using hyperparameters preset: hyperparameters='zeroshot'
Setting dynamic_stacking from 'auto' to True. Reason: Enable dynamic_stacking when use_bag_holdout is disabled. (use_bag_holdout=False)
Stack configuration (auto_stack=True): num_stack_levels=1, num_bag_folds=8, num_bag_sets=1
DyStack is enabled (dynamic_stacking=True). AutoGluon will try to determine whether the input data is affected by stacked overfitting and enable or disable stacking as a consequence.
This is used to identify the optimal `num_stack_levels` value. Copies of AutoGluon will be fit on subsets of the data. Then holdout validation data is used to detect stacked overfitting.
Running DyStack for up to 150s of the 600s of remaining time (25%).
DyStack: Disabling memory safe fit mode in DyStack because GPUs were detected and num_gpus='auto' (GPUs cannot be used in memory safe fit mode). If you want to use memory safe fit mode, manually set `num_gpus=0`.
Running DyStack sub-fit ...
Beginning AutoGluon training ... Time limit = 150s
AutoGluon will save models to "/content/AutogluonModels/ag-20261004_182830/ds_sub_fit/sub_fit_ho"
Train Data Rows: 9676
Train Data Columns: 9
Label Column: count
Problem Type: regression
Preprocessing data...
Using Feature Generators to preprocess the data ...
Fitting AutoMLPipelineFeatureGenerator...
    Available Memory: 9122.47 MB
    Train Data (Original) Memory Usage: 0.41 MB (0.0% of available memory)
    Inferring data type of each feature based on column values. Set feature_metadata_in to manually specify special dtypes of the features.
    Stage 1 Generators:
        Fitting AsTypeFeatureGenerator...
        Note: Converting 2 features to boolean dtype as they only contain 2 unique values.
    Stage 2 Generators:
        Fitting FillNaFeatureGenerator...
    Stage 3 Generators:
        Fitting IdentityFeatureGenerator...
        Fitting CategoryFeatureGenerator...
        Fitting CategoryMemoryMinimizeFeatureGenerator...
        Fitting DatetimeFeatureGenerator...
    Stage 4 Generators:
        Fitting DropUniqueFeatureGenerator...
    Stage 5 Generators:
        Fitting DropDuplicatesFeatureGenerator...
    Types of features in original data (raw dtype, special dtypes):
        ('category', []) : 4 | ['season', 'holiday', 'workingday', 'weather']
        ('datetime', []) : 1 | ['datetime']
        ('float', []) : 3 | ['temp', 'atemp', 'windspeed']
        ('int', []) : 1 | ['humidity']
    Types of features in processed data (raw dtype, special dtypes):
        ('category', []) : 2 | ['season', 'weather']
        ('float', []) : 3 | ['temp', 'atemp', 'windspeed']
        ('int', []) : 1 | ['humidity']
        ('int', ['bool']) : 2 | ['holiday', 'workingday']
        ('int', ['datetime_as_int']) : 5 | ['datetime', 'datetime.year', 'datetime.month', 'datetime.day', 'datetime.dayofweek']
    0.1s = Fit runtime
    9 features in original data used to generate 13 features in processed data.
    Train Data (Processed) Memory Usage: 0.70 MB (0.0% of available memory)
Data preprocessing and feature engineering runtime = 0.11s ...
AutoGluon will gauge predictive performance using evaluation metric: 'root_mean_squared_error'
This metric's sign has been flipped to adhere to being higher_is_better. The metric score can be multiplied by -1 to get the metric value.
To change this, specify the eval_metric parameter of Predictor()
Large model count detected (110 configs) ... Only displaying the first 3 models of each family. To see all, set `verbosity=3`.
User-specified model hyperparameters to be fit:
{
    'NN_TORCH': [{}, {'activation': 'elu', 'dropout_prob': 0.10077639529843717, 'hidden_size': 108, 'learning_rate': 0.002735937344002146, 'num_layers': 4, 'use_batchnorm': True, 'weight_decay': 1.356433327634438e-12, 'ag_args': {'name_suffix': '_r79', 'priority': -2}}, {'activation': 'elu', 'dropout_prob': 0.11897478034205347, 'hidden_size': 213, 'learning_rate': 0.0010474382260641949, 'num_layers': 4, 'use_batchnorm': False, 'weight_decay': 5.594471067786272e-10, 'ag_args': {'name_suffix': '_r22', 'priority': -7}}],
    'GBM': [{'extra_trees': True, 'ag_args': {'name_suffix': 'XT'}}, {}, {'learning_rate': 0.03, 'num_leaves': 128, 'feature_fraction': 0.9, 'min_data_in_leaf': 3, 'ag_args': {'name_suffix': 'Large', 'priority': 0, 'hyperparameter_tune_kwargs': None}}],

```

```

'CAT': [{}, {'depth': 6, 'grow_policy': 'SymmetricTree', 'l2_leaf_reg': 2.1542798306067823, 'learning_rate': 0.06864209
415792857, 'max_ctr_complexity': 4, 'one_hot_max_size': 10, 'ag_args': {'name_suffix': '_r177', 'priority': -1}}, {'depth': 8,
'grow_policy': 'Depthwise', 'l2_leaf_reg': 2.7997999596449104, 'learning_rate': 0.031375015734637225, 'max_ctr_complexity': 2,
'one_hot_max_size': 3, 'ag_args': {'name_suffix': '_r9', 'priority': -5}}],
'XGB': [{}, {'colsample_bytree': 0.6917311125174739, 'enable_categorical': False, 'learning_rate': 0.01806387608752396
7, 'max_depth': 10, 'min_child_weight': 0.6028633586934382, 'ag_args': {'name_suffix': '_r33', 'priority': -8}}, {'colsample_by
tree': 0.6628423832984077, 'enable_categorical': False, 'learning_rate': 0.08775715546881824, 'max_depth': 5, 'min_child_weigh
t': 0.6294123374222513, 'ag_args': {'name_suffix': '_r89', 'priority': -16}}],
'FASTAI': [{}, {'bs': 256, 'emb_drop': 0.5411770367537934, 'epochs': 43, 'layers': [800, 400], 'lr': 0.0151984885831815
9, 'ps': 0.23782946566604385, 'ag_args': {'name_suffix': '_r191', 'priority': -4}}, {'bs': 2048, 'emb_drop': 0.0507041132260581
1, 'epochs': 29, 'layers': [200, 100], 'lr': 0.08974235041576624, 'ps': 0.10393466140748028, 'ag_args': {'name_suffix': '_r10
2', 'priority': -11}}],
'RF': [{'criterion': 'gini', 'ag_args': {'name_suffix': 'Gini', 'problem_types': ['binary', 'multiclass']}}, {'criterio
n': 'entropy', 'ag_args': {'name_suffix': 'Entr', 'problem_types': ['binary', 'multiclass']}}, {'criterion': 'squared_error',
'ag_args': {'name_suffix': 'MSE', 'problem_types': ['regression', 'quantile']}},
'XT': [{'criterion': 'gini', 'ag_args': {'name_suffix': 'Gini', 'problem_types': ['binary', 'multiclass']}}, {'criterio
n': 'entropy', 'ag_args': {'name_suffix': 'Entr', 'problem_types': ['binary', 'multiclass']}}, {'criterion': 'squared_error',
'ag_args': {'name_suffix': 'MSE', 'problem_types': ['regression', 'quantile']}}],
}
AutoGluon will fit 2 stack levels (L1 to L2) ...
Fitting 106 L1 models, fit_strategy="parallel" ...
Note: fit_strategy='parallel', but `num_cpus=2`. Running parallel mode with fewer than 12 CPUs is not recommended and has been
disabled. You can override this by specifying `os.environ["AG_FORCE_PARALLEL"] = "True"`. Falling back to fit_strategy='sequen
tial' ...
Fitting model: LightGBMXT_BAG_L1 ... Training model for up to 99.90s of the 149.87s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.77%)
-131.4309          = Validation score    (-root_mean_squared_error)
61.03s           = Training runtime
6.76s            = Validation runtime
Fitting model: LightGBM_BAG_L1 ... Training model for up to 34.66s of the 84.63s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.78%)
-131.4201          = Validation score    (-root_mean_squared_error)
35.31s           = Training runtime
1.49s            = Validation runtime
Fitting model: WeightedEnsemble_L2 ... Training model for up to 149.89s of the 43.29s of remaining time.
Fitting 1 model on all data | Fitting with cpus=2, gpus=0, mem=0.0/8.7 GB
Ensemble Weights: {'LightGBMXT_BAG_L1': 0.5, 'LightGBM_BAG_L1': 0.5}
-130.2356          = Validation score    (-root_mean_squared_error)
0.02s            = Training runtime
0.0s              = Validation runtime
Fitting 106 L2 models, fit_strategy="parallel" ...
Note: fit_strategy='parallel', but `num_cpus=2`. Running parallel mode with fewer than 12 CPUs is not recommended and has been
disabled. You can override this by specifying `os.environ["AG_FORCE_PARALLEL"] = "True"`. Falling back to fit_strategy='sequen
tial' ...
Fitting model: LightGBMXT_BAG_L2 ... Training model for up to 43.26s of the 43.24s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.79%)
-130.4773          = Validation score    (-root_mean_squared_error)
28.6s            = Training runtime
0.11s            = Validation runtime
Fitting model: LightGBM_BAG_L2 ... Training model for up to 10.40s of the 10.38s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.81%)
-130.419           = Validation score    (-root_mean_squared_error)
28.43s           = Training runtime
0.12s            = Validation runtime
Fitting model: WeightedEnsemble_L3 ... Training model for up to 149.89s of the -23.56s of remaining time.
Fitting 1 model on all data | Fitting with cpus=2, gpus=0, mem=0.0/8.5 GB
Ensemble Weights: {'LightGBM_BAG_L2': 0.389, 'LightGBMXT_BAG_L2': 0.333, 'LightGBM_BAG_L1': 0.278}
-129.8663          = Validation score    (-root_mean_squared_error)
0.02s            = Training runtime
0.0s              = Validation runtime
AutoGluon training complete, total runtime = 173.6s ... Best model: WeightedEnsemble_L3 | Estimated inference throughput: 142.6
rows/s (1210 batch size)
TabularPredictor saved. To load, use: predictor = TabularPredictor.load("/content/AutogluonModels/ag-20261004_182830/ds_sub_fi
t/sub_fit_ho")
Deleting DyStack predictor artifacts (clean_up_fits=True) ...
Leaderboard on holdout data (DyStack):

```

	model	score_holdout	score_val	eval_metric	pred_time_test	pred_time_val	fit_time	pred_tim
e_test_marginal	pred_time_val_marginal	fit_time_marginal	stack_level	can_infer	fit_order			
0	LightGBMXT_BAG_L2	-129.540371	-130.477349	root_mean_squared_error	7.875868	8.363123	124.941299	
0.142048		0.112816	28.599857	2	True	4		
1	WeightedEnsemble_L3	-129.880480	-129.866269	root_mean_squared_error	7.975284	8.485730	153.388392	
0.003301		0.000696	0.020662	3	True	6		
2	WeightedEnsemble_L2	-130.278200	-130.235559	root_mean_squared_error	7.737152	8.251051	96.358997	
0.003332		0.000744	0.017555	2	True	3		
3	LightGBM_BAG_L2	-130.755516	-130.418971	root_mean_squared_error	7.829936	8.372218	124.767873	
0.096115		0.121912	28.426431	2	True	5		
4	LightGBM_BAG_L1	-130.766214	-131.420119	root_mean_squared_error	1.188053	1.494282	35.309240	
1.188053		1.494282	35.309240	1	True	2		
5	LightGBMXT_BAG_L1	-131.158241	-131.430890	root_mean_squared_error	6.545768	6.756024	61.032202	
6.545768		6.756024	61.032202	1	True	1		

```

1          = Optimal num_stack_levels (Stacked Overfitting Occurred: False)
183s       = DyStack runtime | 417s = Remaining runtime
Starting main fit with num_stack_levels=1.
For future fit calls on this dataset, you can skip DyStack to save time: `predictor.fit(..., dynamic_stacking=False, nu

```

```

m_stack_levels=1)`
Beginning AutoGluon training ... Time limit = 417s
AutoGluon will save models to "/content/AutoGluonModels/ag-20261004_182830"
Train Data Rows:    10886
Train Data Columns: 9
Label Column:      count
Problem Type:      regression
Preprocessing data...
Using Feature Generators to preprocess the data ...
Fitting AutoMLPipelineFeatureGenerator...
    Available Memory:      8999.52 MB
    Train Data (Original) Memory Usage: 0.46 MB (0.0% of available memory)
    Inferring data type of each feature based on column values. Set feature_metadata_in to manually specify special dtypes
of the features.
    Stage 1 Generators:
        Fitting AsTypeFeatureGenerator...
            Note: Converting 2 features to boolean dtype as they only contain 2 unique values.
    Stage 2 Generators:
        Fitting FillNaFeatureGenerator...
    Stage 3 Generators:
        Fitting IdentityFeatureGenerator...
        Fitting CategoryFeatureGenerator...
        Fitting CategoryMemoryMinimizeFeatureGenerator...
        Fitting DatetimeFeatureGenerator...
    Stage 4 Generators:
        Fitting DropUniqueFeatureGenerator...
    Stage 5 Generators:
        Fitting DropDuplicatesFeatureGenerator...
    Types of features in original data (raw dtype, special dtypes):
        ('category', []) : 4 | ['season', 'holiday', 'workingday', 'weather']
        ('datetime', []) : 1 | ['datetime']
        ('float', [])    : 3 | ['temp', 'atemp', 'windspeed']
        ('int', [])      : 1 | ['humidity']
    Types of features in processed data (raw dtype, special dtypes):
        ('category', []) : 2 | ['season', 'weather']
        ('float', [])    : 3 | ['temp', 'atemp', 'windspeed']
        ('int', [])      : 1 | ['humidity']
        ('int', ['bool']) : 2 | ['holiday', 'workingday']
        ('int', ['datetime_as_int']) : 5 | ['datetime', 'datetime.year', 'datetime.month', 'datetime.day', 'datetime.da
yofweek']
    0.1s = Fit runtime
    9 features in original data used to generate 13 features in processed data.
    Train Data (Processed) Memory Usage: 0.79 MB (0.0% of available memory)
Data preprocessing and feature engineering runtime = 0.18s ...
AutoGluon will gauge predictive performance using evaluation metric: 'root_mean_squared_error'
    This metric's sign has been flipped to adhere to being higher_is_better. The metric score can be multiplied by -1 to ge
t the metric value.
    To change this, specify the eval_metric parameter of Predictor()
Large model count detected (110 configs) ... Only displaying the first 3 models of each family. To see all, set `verbosity=3`.
User-specified model hyperparameters to be fit:
{
    'NN_TORCH': [{}, {'activation': 'elu', 'dropout_prob': 0.10077639529843717, 'hidden_size': 108, 'learning_rate': 0.0027
35937344002146, 'num_layers': 4, 'use_batchnorm': True, 'weight_decay': 1.356433327634438e-12, 'ag_args': {'name_suffix': '_r7
9', 'priority': -2}}, {'activation': 'elu', 'dropout_prob': 0.11897478034205347, 'hidden_size': 213, 'learning_rate': 0.0010474
382260641949, 'num_layers': 4, 'use_batchnorm': False, 'weight_decay': 5.594471067786272e-10, 'ag_args': {'name_suffix': '_r2
2', 'priority': -7}}],
    'GBM': [{}, {'extra_trees': True, 'ag_args': {'name_suffix': 'XT'}}, {}, {'learning_rate': 0.03, 'num_leaves': 128, 'featur
e_fraction': 0.9, 'min_data_in_leaf': 3, 'ag_args': {'name_suffix': 'Large', 'priority': 0, 'hyperparameter_tune_kwargs': Non
e}}],
    'CAT': [{}, {'depth': 6, 'grow_policy': 'SymmetricTree', 'l2_leaf_reg': 2.1542798306067823, 'learning_rate': 0.06864209
415792857, 'max_ctr_complexity': 4, 'one_hot_max_size': 10, 'ag_args': {'name_suffix': '_r177', 'priority': -1}}, {'depth': 8,
'grow_policy': 'Depthwise', 'l2_leaf_reg': 2.7997999596449104, 'learning_rate': 0.031375015734637225, 'max_ctr_complexity': 2,
'one_hot_max_size': 3, 'ag_args': {'name_suffix': '_r9', 'priority': -5}}],
    'XGB': [{}, {'colsample_bytree': 0.6917311125174739, 'enable_categorical': False, 'learning_rate': 0.01806387608752396
7, 'max_depth': 10, 'min_child_weight': 0.6028633586934382, 'ag_args': {'name_suffix': '_r33', 'priority': -8}}, {'colsample_by
tree': 0.6628423832084077, 'enable_categorical': False, 'learning_rate': 0.08775715546881824, 'max_depth': 5, 'min_child_weigh
t': 0.6294123374222513, 'ag_args': {'name_suffix': '_r89', 'priority': -16}}],
    'FASTAI': [{}, {'bs': 256, 'emb_drop': 0.5411770367537934, 'epochs': 43, 'layers': [800, 400], 'lr': 0.0151984885831815
9, 'ps': 0.23782946566604385, 'ag_args': {'name_suffix': '_r191', 'priority': -4}}, {'bs': 2048, 'emb_drop': 0.0507041132260581
1, 'epochs': 29, 'layers': [200, 100], 'lr': 0.08974235041576624, 'ps': 0.10393466140748028, 'ag_args': {'name_suffix': '_r10
2', 'priority': -11}}],
    'RF': [{'criterion': 'gini', 'ag_args': {'name_suffix': 'Gini', 'problem_types': ['binary', 'multiclass']}}, {'criterio
n': 'entropy', 'ag_args': {'name_suffix': 'Entr', 'problem_types': ['binary', 'multiclass']}}, {'criterion': 'squared_error',
'ag_args': {'name_suffix': 'MSE', 'problem_types': ['regression', 'quantile']}},
    'XT': [{'criterion': 'gini', 'ag_args': {'name_suffix': 'Gini', 'problem_types': ['binary', 'multiclass']}}, {'criterio
n': 'entropy', 'ag_args': {'name_suffix': 'Entr', 'problem_types': ['binary', 'multiclass']}}, {'criterion': 'squared_error',
'ag_args': {'name_suffix': 'MSE', 'problem_types': ['regression', 'quantile']}},
    ]
AutoGluon will fit 2 stack levels (L1 to L2) ...
Fitting 106 L1 models, fit_strategy="parallel" ...
Note: fit_strategy='parallel', but `num_cpus=2`. Running parallel mode with fewer than 12 CPUs is not recommended and has been
disabled. You can override this by specifying `os.environ["AG_FORCE_PARALLEL"] = "True"`. Falling back to fit_strategy='sequent
ial' ...
Fitting model: LightGBMXT_BAG_L1 ... Training model for up to 277.49s of the 416.32s of remaining time.
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.78%)
    -131.1592      = Validation score    (-root_mean_squared_error)
    68.05s        = Training    runtime

```

```

7.09s = Validation runtime
Fitting model: LightGBM_BAG_L1 ... Training model for up to 202.62s of the 341.45s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.80%)
-130.675 = Validation score (-root_mean_squared_error)
38.47s = Training runtime
2.34s = Validation runtime
Fitting model: RandomForestMSE_BAG_L1 ... Training model for up to 159.82s of the 298.65s of remaining time.
Fitting 1 model on all data (use_child_oof=True) | Fitting with cpus=2, gpus=0, mem=0.1/8.6 GB
-116.6254 = Validation score (-root_mean_squared_error)
15.4s = Training runtime
0.55s = Validation runtime
Fitting model: CatBoost_BAG_L1 ... Training model for up to 143.02s of the 281.85s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=4.46%)
-132.3846 = Validation score (-root_mean_squared_error)
127.29s = Training runtime
0.22s = Validation runtime
Fitting model: ExtraTreesMSE_BAG_L1 ... Training model for up to 12.31s of the 151.15s of remaining time.
Fitting 1 model on all data (use_child_oof=True) | Fitting with cpus=2, gpus=0, mem=0.1/8.4 GB
-124.5641 = Validation score (-root_mean_squared_error)
6.45s = Training runtime
0.57s = Validation runtime
Fitting model: NeuralNetFastAI_BAG_L1 ... Training model for up to 4.00s of the 142.83s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.10%)
Time limit exceeded... Skipping NeuralNetFastAI_BAG_L1.
Fitting model: WeightedEnsemble_L2 ... Training model for up to 360.00s of the 131.12s of remaining time.
Fitting 1 model on all data | Fitting with cpus=2, gpus=0, mem=0.0/8.2 GB
Ensemble Weights: {'RandomForestMSE_BAG_L1': 1.0}
-116.6254 = Validation score (-root_mean_squared_error)
0.03s = Training runtime
0.0s = Validation runtime
Fitting 106 L2 models, fit_strategy="parallel" ...
Note: fit_strategy='parallel', but `num_cpus=2`. Running parallel mode with fewer than 12 CPUs is not recommended and has been disabled. You can override this by specifying `os.environ["AG_FORCE_PARALLEL"] = "True"`. Falling back to fit_strategy='sequential' ...
Fitting model: LightGBMXT_BAG_L2 ... Training model for up to 131.07s of the 131.04s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.86%)
-115.2423 = Validation score (-root_mean_squared_error)
34.1s = Training runtime
0.5s = Validation runtime
Fitting model: LightGBM_BAG_L2 ... Training model for up to 92.69s of the 92.65s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.85%)
-115.4569 = Validation score (-root_mean_squared_error)
31.29s = Training runtime
0.2s = Validation runtime
Fitting model: RandomForestMSE_BAG_L2 ... Training model for up to 57.33s of the 57.29s of remaining time.
Fitting 1 model on all data (use_child_oof=True) | Fitting with cpus=2, gpus=0, mem=0.1/8.3 GB
-117.3336 = Validation score (-root_mean_squared_error)
39.42s = Training runtime
0.65s = Validation runtime
Fitting model: CatBoost_BAG_L2 ... Training model for up to 16.20s of the 16.17s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=4.59%)
-115.7023 = Validation score (-root_mean_squared_error)
27.33s = Training runtime
0.25s = Validation runtime
Fitting model: WeightedEnsemble_L3 ... Training model for up to 360.00s of the -14.81s of remaining time.
Fitting 1 model on all data | Fitting with cpus=2, gpus=0, mem=0.0/8.3 GB
Ensemble Weights: {'LightGBMXT_BAG_L2': 0.471, 'LightGBM_BAG_L2': 0.235, 'RandomForestMSE_BAG_L2': 0.176, 'RandomForestMSE_BAG_L1': 0.118}
-114.6282 = Validation score (-root_mean_squared_error)
0.04s = Training runtime
0.0s = Validation runtime
AutoGluon training complete, total runtime = 431.4s ... Best model: WeightedEnsemble_L3 | Estimated inference throughput: 128.6 rows/s (1361 batch size)
TabularPredictor saved. To load, use: predictor = TabularPredictor.load("/content/AutogluonModels/ag-20261004_182830")

```

```

In [28]: import torch

print("CUDA disponível?", torch.cuda.is_available())
print("Dispositivo:", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "Nenhum")

```

```

CUDA disponível? True
Dispositivo: Tesla T4

```

Review AutoGluon's training run with ranking of models that did the best.

```

In [31]: predictor.leaderboard(silent=False)

```

	model	score_val	eval_metric	pred_time_val	fit_time	pred_time_val_marginal	fit_time_ma
0	WeightedEnsemble_L3	-114.628178	root_mean_squared_error	12.129280	360.516042	0.000876	0.
042912	3	True	11				
1	LightGBMXT_BAG_L2	-115.242330	root_mean_squared_error	11.273526	289.767082	0.503703	34.
100850	2	True	7				
2	LightGBM_BAG_L2	-115.456884	root_mean_squared_error	10.972936	286.953676	0.203114	31.
287443	2	True	8				
3	CatBoost_BAG_L2	-115.702331	root_mean_squared_error	11.022127	282.994496	0.252305	27.
328264	2	True	10				
4	RandomForestMSE_BAG_L1	-116.625366	root_mean_squared_error	0.545028	15.402647	0.545028	15.
402647	1	True	3				
5	WeightedEnsemble_L2	-116.625366	root_mean_squared_error	0.545789	15.432832	0.000761	0.
030185	2	True	6				
6	RandomForestMSE_BAG_L2	-117.333641	root_mean_squared_error	11.421588	295.084837	0.651765	39.
418604	2	True	9				
7	ExtraTreesMSE_BAG_L1	-124.564085	root_mean_squared_error	0.573553	6.454430	0.573553	6.
454430	1	True	5				
8	LightGBM_BAG_L1	-130.675014	root_mean_squared_error	2.342306	38.468436	2.342306	38.
468436	1	True	2				
9	LightGBMXT_BAG_L1	-131.159249	root_mean_squared_error	7.087052	68.048435	7.087052	68.
048435	1	True	1				
10	CatBoost_BAG_L1	-132.384648	root_mean_squared_error	0.221884	127.292284	0.221884	127.
292284	1	True	4				

Out[31]:

	model	score_val	eval_metric	pred_time_val	fit_time	pred_time_val_marginal	fit_time_marginal	s
0	WeightedEnsemble_L3	-114.628178	root_mean_squared_error	12.129280	360.516042	0.000876	0.042912	
1	LightGBMXT_BAG_L2	-115.242330	root_mean_squared_error	11.273526	289.767082	0.503703	34.100850	
2	LightGBM_BAG_L2	-115.456884	root_mean_squared_error	10.972936	286.953676	0.203114	31.287443	
3	CatBoost_BAG_L2	-115.702331	root_mean_squared_error	11.022127	282.994496	0.252305	27.328264	
4	RandomForestMSE_BAG_L1	-116.625366	root_mean_squared_error	0.545028	15.402647	0.545028	15.402647	
5	WeightedEnsemble_L2	-116.625366	root_mean_squared_error	0.545789	15.432832	0.000761	0.030185	
6	RandomForestMSE_BAG_L2	-117.333641	root_mean_squared_error	11.421588	295.084837	0.651765	39.418604	
7	ExtraTreesMSE_BAG_L1	-124.564085	root_mean_squared_error	0.573553	6.454430	0.573553	6.454430	
8	LightGBM_BAG_L1	-130.675014	root_mean_squared_error	2.342306	38.468436	2.342306	38.468436	
9	LightGBMXT_BAG_L1	-131.159249	root_mean_squared_error	7.087052	68.048435	7.087052	68.048435	
10	CatBoost_BAG_L1	-132.384648	root_mean_squared_error	0.221884	127.292284	0.221884	127.292284	

Create predictions from test dataset

```
In [33]: predictions = predictor.predict(test)
predictions.head()
```

Out[33]:

	count
0	91.504036
1	99.419029
2	99.419029
3	100.806122
4	100.806122

dtype: float32

NOTE: Kaggle will reject the submission if we don't set everything to be > 0.

```
In [34]: # Describe the `predictions` series to see if there are any negative values
predictions.describe()
```

```
Out[34]:
```

	count
count	6493.000000
mean	221.969513
std	126.163551
min	17.277086
25%	124.126068
50%	198.192932
75%	305.954346
max	624.935669

dtype: float64

```
In [35]: # How many negative values do we have?
print(f"Negative values count:{sum(predictions<0)}")
```

Negative values count:0

```
In [36]: # Set them to zero
predictions[predictions<0] = 0
```

Set predictions to submission dataframe, save, and submit

```
In [37]: submission["count"] = predictions
submission.to_csv("submission.csv", index=False)
```

```
In [38]: !pip install kaggle
```

```
Requirement already satisfied: kaggle in /usr/local/lib/python3.11/dist-packages (1.7.4.5)
Requirement already satisfied: bleach in /usr/local/lib/python3.11/dist-packages (from kaggle) (6.2.0)
Requirement already satisfied: certifi>=14.05.14 in /usr/local/lib/python3.11/dist-packages (from kaggle) (2025.7.14)
Requirement already satisfied: charset-normalizer in /usr/local/lib/python3.11/dist-packages (from kaggle) (3.4.2)
Requirement already satisfied: idna in /usr/local/lib/python3.11/dist-packages (from kaggle) (3.10)
Requirement already satisfied: protobuf in /usr/local/lib/python3.11/dist-packages (from kaggle) (5.29.5)
Requirement already satisfied: python-dateutil>=2.5.3 in /usr/local/lib/python3.11/dist-packages (from kaggle) (2.9.0.post0)
Requirement already satisfied: python-slugify in /usr/local/lib/python3.11/dist-packages (from kaggle) (8.0.4)
Requirement already satisfied: requests in /usr/local/lib/python3.11/dist-packages (from kaggle) (2.32.3)
Requirement already satisfied: setuptools>=21.0.0 in /usr/local/lib/python3.11/dist-packages (from kaggle) (81.0.0)
Requirement already satisfied: six>=1.10 in /usr/local/lib/python3.11/dist-packages (from kaggle) (1.17.0)
Requirement already satisfied: text-unidecode in /usr/local/lib/python3.11/dist-packages (from kaggle) (1.3)
Requirement already satisfied: tqdm in /usr/local/lib/python3.11/dist-packages (from kaggle) (4.67.1)
Requirement already satisfied: urllib3>=1.15.1 in /usr/local/lib/python3.11/dist-packages (from kaggle) (2.4.0)
Requirement already satisfied: webencodings in /usr/local/lib/python3.11/dist-packages (from kaggle) (0.5.1)
```

```
In [45]: !mkdir -p ~/.kaggle
#!cp kaggle.json ~/.kaggle/
!touch ~/.kaggle/kaggle.json
!echo '{"username":"michaellesouza","key":"55781e9c097b8a6821635b6a7d4634a9"}' > ~/.kaggle/kaggle.json
!chmod 600 ~/.kaggle/kaggle.json
```

```
In [46]: !kaggle competitions submit -c bike-sharing-demand -f submission.csv -m "first raw submission"
```

100% 188k/188k [00:00<00:00, 286kB/s]
Successfully submitted to Bike Sharing Demand

View submission via the command line or in the web browser under the competition's page - [My Submissions](#)

```
In [47]: !kaggle competitions submissions -c bike-sharing-demand | tail -n +1 | head -n 6
```

fileName	date	description	status	publicScore
submission.csv	2026-10-04 19:07:43.593000	first raw submission	SubmissionStatus.COMPLETE	1.43583
submission_new_hpo.csv	2026-09-20 19:21:12.517000	new features with hyperparameters	SubmissionStatus.COMPLETE	0.57774
submission_new_features.csv	2026-09-20 18:56:09.673000	new features	SubmissionStatus.COMPLETE	0.46971
submission.csv	2026-09-20 17:22:17.870000	first raw submission	SubmissionStatus.COMPLETE	1.43525

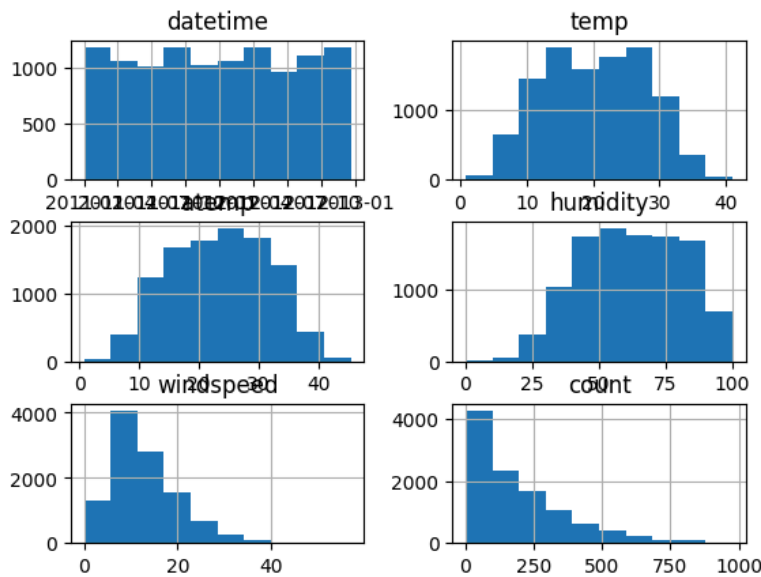
Initial score of 1.43583

Step 4: Exploratory Data Analysis and Creating an additional feature

- Any additional feature will do, but a great suggestion would be to separate out the datetime into hour, day, or month parts.

```
In [48]: # Create a histogram of all features to show the distribution of each one relative to the data. This is part of the explorator
train.hist()
```

```
Out[48]: array([[<Axes: title={'center': 'datetime'}>,
  <Axes: title={'center': 'temp'}>],
  [<Axes: title={'center': 'atemp'}>,
  <Axes: title={'center': 'humidity'}>],
  [<Axes: title={'center': 'windspeed'}>,
  <Axes: title={'center': 'count'}>]], dtype=object)
```



```
In [49]: train.columns
```

```
Out[49]: Index(['datetime', 'season', 'holiday', 'workingday', 'weather', 'temp',
  'atemp', 'humidity', 'windspeed', 'count'],
  dtype='object')
```

```
In [50]: train.head()
```

```
Out[50]:
```

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	count
0	2011-01-01 00:00:00	1	0	0	1	9.84	14.395	81	0.0	16
1	2011-01-01 01:00:00	1	0	0	1	9.02	13.635	80	0.0	40
2	2011-01-01 02:00:00	1	0	0	1	9.02	13.635	80	0.0	32
3	2011-01-01 03:00:00	1	0	0	1	9.84	14.395	75	0.0	13
4	2011-01-01 04:00:00	1	0	0	1	9.84	14.395	75	0.0	1

```
In [72]: # create new features
def create_features(df):
    """
    Creates new features from the datetime column in the dataframe.
    """
    # Categorical features from the datetime column
    df['hour'] = df['datetime'].dt.hour
    df['dayofweek'] = df['datetime'].dt.dayofweek
    df['month'] = df['datetime'].dt.month
    df['year'] = df['datetime'].dt.year

    # Categorical features from hour
    df['day_period'] = df['hour'].apply(lambda x: 'morning' if x <= 10 else 'lunch' if 11 <= x < 13 else 'afternoon' if 13 <= x < 18 else 'evening')

    # Categorical feature from temp
    df['feel_temp_category'] = df['temp'].apply(lambda x: 'cold' if x < 10 else 'cool' if 10 <= x < 20 else 'warm' if 20 <= x < 30 else 'hot')

    # Categorical feature from humidity
    df['humidity_category'] = df['humidity'].apply(lambda x: 'dry' if x < 30 else 'comfortable' if 30 <= x < 60 else 'humid')

    return df
```

```
In [73]: # create a new feature
```

```
train = create_features(train)
test = create_features(test)
```

```
In [66]: train.head()
```

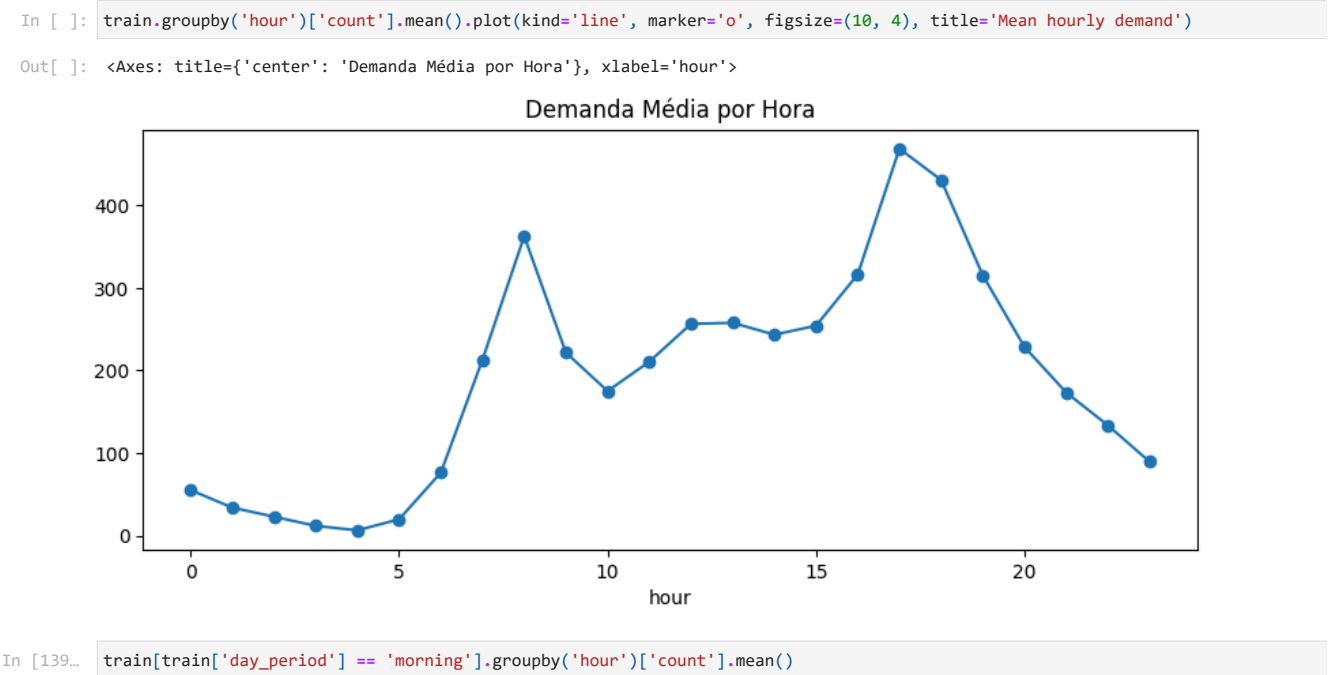
Out[66]:

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	count	hour	dayofweek	month	year	day_pe
0	2011-01-01 00:00:00	1	0	0	1	9.84	14.395	81	0.0	16	0	5	1	2011	mor
1	2011-01-01 01:00:00	1	0	0	1	9.02	13.635	80	0.0	40	1	5	1	2011	mor
2	2011-01-01 02:00:00	1	0	0	1	9.02	13.635	80	0.0	32	2	5	1	2011	mor
3	2011-01-01 03:00:00	1	0	0	1	9.84	14.395	75	0.0	13	3	5	1	2011	mor
4	2011-01-01 04:00:00	1	0	0	1	9.84	14.395	75	0.0	1	4	5	1	2011	mor

In [67]: train.describe()

Out[67]:

	datetime	temp	atemp	humidity	windspeed	count	hour	dayofweek	month
count	10886	10886.00000	10886.000000	10886.000000	10886.000000	10886.000000	10886.000000	10886.000000	10886.000000
mean	2011-12-27 05:56:22.399411968	20.23086	23.655084	61.886460	12.799395	191.574132	11.541613	3.013963	6.521495
min	2011-01-01 00:00:00	0.82000	0.760000	0.000000	0.000000	1.000000	0.000000	0.000000	1.000000
25%	2011-07-02 07:15:00	13.94000	16.665000	47.000000	7.001500	42.000000	6.000000	1.000000	4.000000
50%	2012-01-01 20:30:00	20.50000	24.240000	62.000000	12.998000	145.000000	12.000000	3.000000	7.000000
75%	2012-07-01 12:45:00	26.24000	31.060000	77.000000	16.997900	284.000000	18.000000	5.000000	10.000000
max	2012-12-19 23:00:00	41.00000	45.455000	100.000000	56.996900	977.000000	23.000000	6.000000	12.000000
std	NaN	7.79159	8.474601	19.245033	8.164537	181.144454	6.915838	2.004585	3.444373



Out[139...

	count
hour	
0	55.138462
1	33.859031
2	22.899554
3	11.757506
4	6.407240
5	19.767699
6	76.259341
7	213.116484
8	362.769231
9	221.780220
10	175.092308

dtype: float64

```
In [137... train[train['day_period'] == 'evening'].groupby('hour')['count'].mean()
```

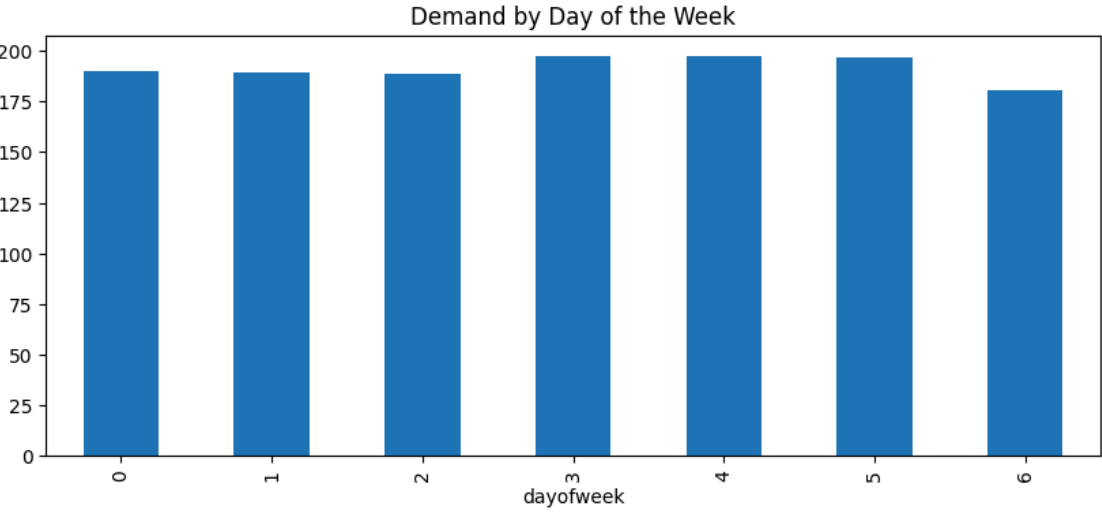
Out[137...

	count
hour	
17	468.765351
18	430.859649

dtype: float64

```
In [134... train.groupby('dayofweek')['count'].mean().plot(kind='bar', figsize=(10, 4), title='Demand by Day of the Week')
```

Out[134... <Axes: title={ 'center': 'Demand by Day of the Week'}, xlabel='dayofweek'>



```
In [146... train.groupby('dayofweek').aggregate({'count': 'mean'})
```

Out[146...

	count
dayofweek	
0	190.390716
1	189.723847
2	188.411348
3	197.296201
4	197.844343
5	196.665404
6	180.839772

Make category types for these so models know they are not just numbers

- AutoGluon originally sees these as ints, but in reality they are int representations of a category.
- Setting the dtype to category will classify these as categories in AutoGluon.

```
In [89]: season = {1:'spring',
                 2:'summer',
                 3:'fall',
                 4:'winter'}

train["season"] = train['season'].apply(lambda x:season[x])
```

```
In [90]: test["season"] = test['season'].apply(lambda x:season[x])
```

```
In [91]: weather ={
1: 'Clear, Few clouds, Partly cloudy, Partly cloudy',
2: 'Mist + Cloudy, Mist + Broken clouds, Mist + Few clouds, Mist',
3: 'Light Snow, Light Rain + Thunderstorm + Scattered clouds, Light Rain + Scattered clouds',
4: 'Heavy Rain + Ice Pallets + Thunderstorm + Mist, Snow + Fog ',
}

train["weather"] = train['weather'].apply(lambda x: weather[x])
test["weather"] = test['weather'].apply(lambda x: weather[x])
```

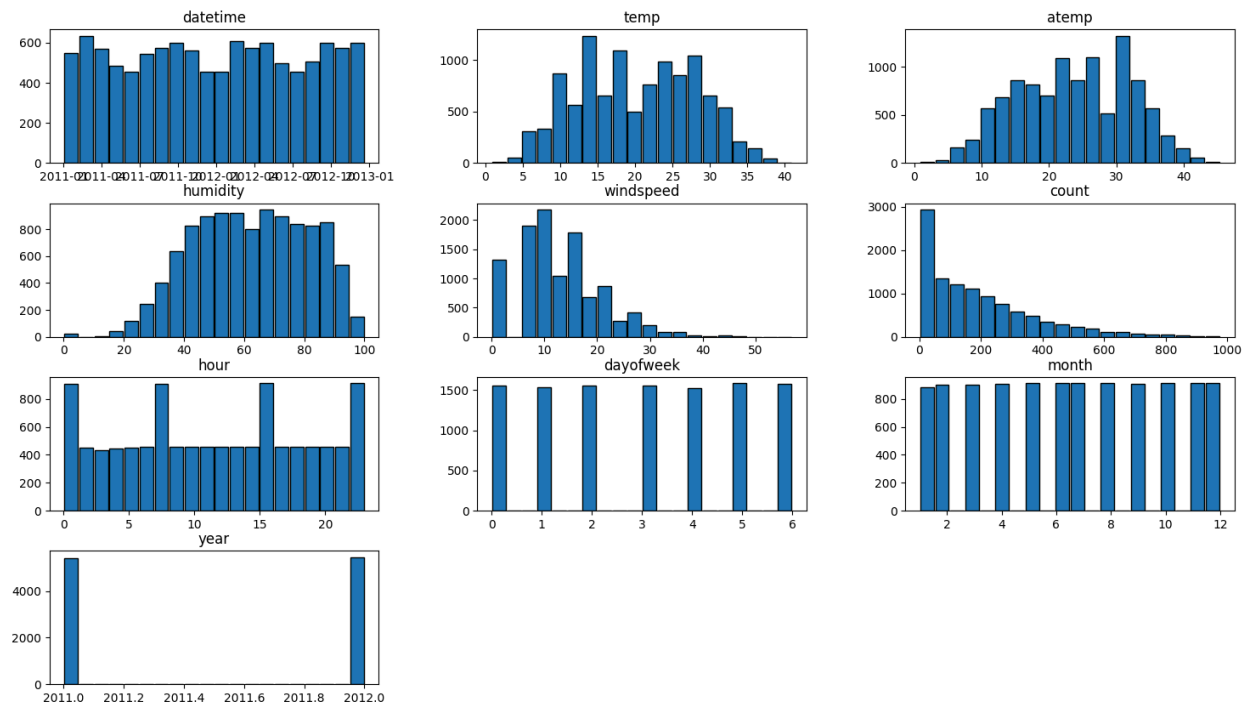
```
In [93]: # View are new feature
train.head()
```

```
Out[93]:
```

	datetime	season	holiday	workingday	weather	temp	atemp	humidity	windspeed	count	hour	dayofweek	month	year	day_pe
0	2011-01-01 00:00:00	spring	0	0	Clear, Few clouds, Partly cloudy, Partly cloudy	9.84	14.395	81	0.0	16	0	5	1	2011	mor
1	2011-01-01 01:00:00	spring	0	0	Clear, Few clouds, Partly cloudy, Partly cloudy	9.02	13.635	80	0.0	40	1	5	1	2011	mor
2	2011-01-01 02:00:00	spring	0	0	Clear, Few clouds, Partly cloudy, Partly cloudy	9.02	13.635	80	0.0	32	2	5	1	2011	mor
3	2011-01-01 03:00:00	spring	0	0	Clear, Few clouds, Partly cloudy, Partly cloudy	9.84	14.395	75	0.0	13	3	5	1	2011	mor
4	2011-01-01 04:00:00	spring	0	0	Clear, Few clouds, Partly cloudy, Partly cloudy	9.84	14.395	75	0.0	1	4	5	1	2011	mor

```
In [102... # View histogram of all features again now with the hour feature
train.hist(figsize=(18, 10), layout=(4, 3), bins=20, edgecolor='black',grid=False, rwidth=0.9)
```

```
Out[102... array([[<Axes: title={'center': 'datetime'}>,
        <Axes: title={'center': 'temp'}>,
        <Axes: title={'center': 'atemp'}>],
       [<Axes: title={'center': 'humidity'}>,
        <Axes: title={'center': 'windspeed'}>,
        <Axes: title={'center': 'count'}>],
       [<Axes: title={'center': 'hour'}>,
        <Axes: title={'center': 'dayofweek'}>,
        <Axes: title={'center': 'month'}>],
       [<Axes: title={'center': 'year'}>, <Axes: >, <Axes: >]],
      dtype=object)
```



Step 5: Rerun the model with the same settings as before, just with more features

```
In [103... predictor_new_features = TabularPredictor(
    label="count", eval_metric="root_mean_squared_error"
).fit(
    train_data=train,
    time_limit=600,
    presets="best_quality",
    fit_strategy="parallel",
    num_gpus=1
)
```

```

No path specified. Models will be saved in: "AutogluonModels/ag-20261004_194157"
Verbosity: 2 (Standard Logging)
===== System Info =====
AutoGluon Version: 1.6.3
Python Version: 3.11.13
Operating System: Linux
Platform Machine: x86_64
Platform Version: #1 SMP Thu Apr 30 18:17:14 UTC 2026
CPU Count: 2
Pytorch Version: 2.13.0+cu130
CUDA Version: 13.0
GPU Memory: GPU 0: 14.56/14.56 GB
Total GPU Memory: Free: 14.56 GB, Allocated: 0.00 GB, Total: 14.56 GB
GPU Count: 1
Memory Avail: 7.65 GB / 12.67 GB (60.4%)
Disk Space Avail: 187.25 GB / 241.91 GB (77.4%)
=====
Presets specified: ['best_quality']
Using hyperparameters preset: hyperparameters='zeroshot'
Setting dynamic_stacking from 'auto' to True. Reason: Enable dynamic_stacking when use_bag_holdout is disabled. (use_bag_holdout=False)
Stack configuration (auto_stack=True): num_stack_levels=1, num_bag_folds=8, num_bag_sets=1
DyStack is enabled (dynamic_stacking=True). AutoGluon will try to determine whether the input data is affected by stacked overfitting and enable or disable stacking as a consequence.
    This is used to identify the optimal `num_stack_levels` value. Copies of AutoGluon will be fit on subsets of the data. Then holdout validation data is used to detect stacked overfitting.
    Running DyStack for up to 150s of the 600s of remaining time (25%).
Running DyStack sub-fit ...
Beginning AutoGluon training ... Time limit = 150s
AutoGluon will save models to "/content/AutogluonModels/ag-20261004_194157/ds_sub_fit/sub_fit_ho"
Train Data Rows: 9676
Train Data Columns: 16
Label Column: count
Problem Type: regression
Preprocessing data...
Using Feature Generators to preprocess the data ...
Fitting AutoMLPipelineFeatureGenerator...
    Available Memory: 7833.03 MB
    Train Data (Original) Memory Usage: 2.30 MB (0.0% of available memory)
    Inferring data type of each feature based on column values. Set feature_metadata_in to manually specify special dtypes of the features.
    Stage 1 Generators:
        Fitting AsTypeFeatureGenerator...
        Note: Converting 3 features to boolean dtype as they only contain 2 unique values.
    Stage 2 Generators:
        Fitting FillNaFeatureGenerator...
    Stage 3 Generators:
        Fitting IdentityFeatureGenerator...
        Fitting CategoryFeatureGenerator...
        Fitting CategoryMemoryMinimizeFeatureGenerator...
        Fitting DatetimeFeatureGenerator...
    Stage 4 Generators:
        Fitting DropUniqueFeatureGenerator...
    Stage 5 Generators:
        Fitting DropDuplicatesFeatureGenerator...
    Types of features in original data (raw dtype, special dtypes):
        ('category', []) : 4 | ['season', 'holiday', 'workingday', 'weather']
        ('datetime', []) : 1 | ['datetime']
        ('float', []) : 3 | ['temp', 'atemp', 'windspeed']
        ('int', []) : 5 | ['humidity', 'hour', 'dayofweek', 'month', 'year']
        ('object', []) : 3 | ['day_period', 'feel_temp_category', 'humidity_category']
    Types of features in processed data (raw dtype, special dtypes):
        ('category', []) : 5 | ['season', 'weather', 'day_period', 'feel_temp_category', 'humidity_category']
        ('float', []) : 3 | ['temp', 'atemp', 'windspeed']
        ('int', []) : 4 | ['humidity', 'hour', 'dayofweek', 'month']
        ('int', ['bool']) : 3 | ['holiday', 'workingday', 'year']
        ('int', ['datetime_as_int']) : 3 | ['datetime', 'datetime.year', 'datetime.day']
    0.1s = Fit runtime
    16 features in original data used to generate 18 features in processed data.
    Train Data (Processed) Memory Usage: 0.70 MB (0.0% of available memory)
Data preprocessing and feature engineering runtime = 0.16s ...
AutoGluon will gauge predictive performance using evaluation metric: 'root_mean_squared_error'
    This metric's sign has been flipped to adhere to being higher_is_better. The metric score can be multiplied by -1 to get the metric value.
    To change this, specify the eval_metric parameter of Predictor()
Large model count detected (110 configs) ... Only displaying the first 3 models of each family. To see all, set `verbosity=3`.
User-specified model hyperparameters to be fit:
{
    'NN_TORCH': [{}, {'activation': 'elu', 'dropout_prob': 0.10077639529843717, 'hidden_size': 108, 'learning_rate': 0.002735937344002146, 'num_layers': 4, 'use_batchnorm': True, 'weight_decay': 1.356433327634438e-12, 'ag_args': {'name_suffix': '_r79', 'priority': -2}}, {'activation': 'elu', 'dropout_prob': 0.11897478034205347, 'hidden_size': 213, 'learning_rate': 0.0010474382260641949, 'num_layers': 4, 'use_batchnorm': False, 'weight_decay': 5.594471067786272e-10, 'ag_args': {'name_suffix': '_r22', 'priority': -7}}],
    'GBM': [{'extra_trees': True, 'ag_args': {'name_suffix': 'XT'}}, {}, {'learning_rate': 0.03, 'num_leaves': 128, 'feature_fraction': 0.9, 'min_data_in_leaf': 3, 'ag_args': {'name_suffix': 'Large', 'priority': 0, 'hyperparameter_tune_kwargs': None}}],
    'CAT': [{}, {'depth': 6, 'grow_policy': 'SymmetricTree', 'l2_leaf_reg': 2.1542798306067823, 'learning_rate': 0.06864209

```

```

415792857, 'max_ctr_complexity': 4, 'one_hot_max_size': 10, 'ag_args': {'name_suffix': '_r177', 'priority': -1}}, {'depth': 8,
'grow_policy': 'Depthwise', 'l2_leaf_reg': 2.799799596449104, 'learning_rate': 0.031375015734637225, 'max_ctr_complexity': 2,
'one_hot_max_size': 3, 'ag_args': {'name_suffix': '_r9', 'priority': -5}}],
'XGB': [{}, {'colsample_bytree': 0.6917311125174739, 'enable_categorical': False, 'learning_rate': 0.01806387608752396
7, 'max_depth': 10, 'min_child_weight': 0.6028633586934382, 'ag_args': {'name_suffix': '_r33', 'priority': -8}}, {'colsample_by
tree': 0.6628423832084077, 'enable_categorical': False, 'learning_rate': 0.08775715546881824, 'max_depth': 5, 'min_child_weigh
t': 0.629412337422513, 'ag_args': {'name_suffix': '_r89', 'priority': -16}}],
'FASTAI': [{}, {'bs': 256, 'emb_drop': 0.5411770367537934, 'epochs': 43, 'layers': [800, 400], 'lr': 0.0151984885831815
9, 'ps': 0.23782946566604385, 'ag_args': {'name_suffix': '_r191', 'priority': -4}}, {'bs': 2048, 'emb_drop': 0.0507041132260581
1, 'epochs': 29, 'layers': [200, 100], 'lr': 0.08974235041576624, 'ps': 0.10393466140748028, 'ag_args': {'name_suffix': '_r10
2', 'priority': -11}}],
'RF': [{'criterion': 'gini', 'ag_args': {'name_suffix': 'Gini', 'problem_types': ['binary', 'multiclass']}}, {'criterio
n': 'entropy', 'ag_args': {'name_suffix': 'Entr', 'problem_types': ['binary', 'multiclass']}}, {'criterion': 'squared_error',
'ag_args': {'name_suffix': 'MSE', 'problem_types': ['regression', 'quantile']}},
'XT': [{'criterion': 'gini', 'ag_args': {'name_suffix': 'Gini', 'problem_types': ['binary', 'multiclass']}}, {'criterio
n': 'entropy', 'ag_args': {'name_suffix': 'Entr', 'problem_types': ['binary', 'multiclass']}}, {'criterion': 'squared_error',
'ag_args': {'name_suffix': 'MSE', 'problem_types': ['regression', 'quantile']}},
}
AutoGluon will fit 2 stack levels (L1 to L2) ...
Fitting 106 L1 models, fit_strategy="parallel" ...
Note: fit_strategy='parallel', but `num_cpus=2`. Running parallel mode with fewer than 12 CPUs is not recommended and has been
disabled. You can override this by specifying `os.environ["AG_FORCE_PARALLEL"] = "True"`. Falling back to fit_strategy='sequen
tial' ...
Fitting model: LightGBMXT_BAG_L1 ... Training model for up to 99.86s of the 149.82s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.90%)
-35.6504 = Validation score (-root_mean_squared_error)
82.37s = Training runtime
7.89s = Validation runtime
Fitting model: LightGBM_BAG_L1 ... Training model for up to 12.47s of the 62.43s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.93%)
-34.7954 = Validation score (-root_mean_squared_error)
39.29s = Training runtime
1.96s = Validation runtime
Fitting model: WeightedEnsemble_L2 ... Training model for up to 149.84s of the 18.64s of remaining time.
Fitting 1 model on all data | Fitting with cpus=2, gpus=1, mem=0.0/7.3 GB
Ensemble Weights: {'LightGBM_BAG_L1': 0.571, 'LightGBMXT_BAG_L1': 0.429}
-33.7336 = Validation score (-root_mean_squared_error)
0.02s = Training runtime
0.0s = Validation runtime
Fitting 106 L2 models, fit_strategy="parallel" ...
Note: fit_strategy='parallel', but `num_cpus=2`. Running parallel mode with fewer than 12 CPUs is not recommended and has been
disabled. You can override this by specifying `os.environ["AG_FORCE_PARALLEL"] = "True"`. Falling back to fit_strategy='sequen
tial' ...
Fitting model: LightGBMXT_BAG_L2 ... Training model for up to 18.61s of the 18.59s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.96%)
-35.341 = Validation score (-root_mean_squared_error)
33.42s = Training runtime
0.44s = Validation runtime
Fitting model: WeightedEnsemble_L3 ... Training model for up to 149.84s of the -19.13s of remaining time.
Fitting 1 model on all data | Fitting with cpus=2, gpus=1, mem=0.0/7.3 GB
Ensemble Weights: {'LightGBM_BAG_L1': 0.571, 'LightGBMXT_BAG_L1': 0.429}
-33.7336 = Validation score (-root_mean_squared_error)
0.02s = Training runtime
0.0s = Validation runtime
AutoGluon training complete, total runtime = 169.17s ... Best model: WeightedEnsemble_L2 | Estimated inference throughput: 122.
8 rows/s (1210 batch size)
TabularPredictor saved. To load, use: predictor = TabularPredictor.load("/content/AutogluonModels/ag-20261004_194157/ds_sub_fi
t/sub_fit_ho")
Deleting DyStack predictor artifacts (clean_up_fits=True) ...
Leaderboard on holdout data (DyStack):

```

	model	score_holdout	score_val	eval_metric	pred_time_test	pred_time_val	fit_time	pred_time
	_test_marginal	pred_time_val_marginal	fit_time_marginal	stack_level	can_infer	fit_order		
0	WeightedEnsemble_L3	-33.011919	-33.733640	root_mean_squared_error	8.164552	9.853265	121.674694	
0.001808		0.000662	0.021069	3	True	5		
1	WeightedEnsemble_L2	-33.011919	-33.733640	root_mean_squared_error	8.164979	9.853313	121.670965	
0.002236		0.000710	0.017340	2	True	3		
2	LightGBMXT_BAG_L2	-33.319079	-35.340990	root_mean_squared_error	8.428525	10.293352	155.077349	
0.265781		0.440749	33.423724	2	True	4		
3	LightGBM_BAG_L1	-33.329339	-34.795372	root_mean_squared_error	1.533853	1.959491	39.285025	
1.533853		1.959491	39.285025	1	True	2		
4	LightGBMXT_BAG_L1	-35.184596	-35.650397	root_mean_squared_error	6.628891	7.893111	82.368600	
6.628891		7.893111	82.368600	1	True	1		
1								= Optimal num_stack_levels (Stacked Overfitting Occurred: False)
180s								= DyStack runtime 420s = Remaining runtime

```

Starting main fit with num_stack_levels=1.
For future fit calls on this dataset, you can skip DyStack to save time: `predictor.fit(..., dynamic_stacking=False, nu
m_stack_levels=1)`
Beginning AutoGluon training ... Time limit = 420s
AutoGluon will save models to "/content/AutogluonModels/ag-20261004_194157"
Train Data Rows: 10886
Train Data Columns: 16
Label Column: count
Problem Type: regression
Preprocessing data...
Using Feature Generators to preprocess the data ...

```

```

Fitting AutoMLPipelineFeatureGenerator...
  Available Memory: 7871.41 MB
  Train Data (Original) Memory Usage: 2.59 MB (0.0% of available memory)
  Inferring data type of each feature based on column values. Set feature_metadata_in to manually specify special dtypes
of the features.
  Stage 1 Generators:
    Fitting AsTypeFeatureGenerator...
    Note: Converting 3 features to boolean dtype as they only contain 2 unique values.
  Stage 2 Generators:
    Fitting FillNaFeatureGenerator...
  Stage 3 Generators:
    Fitting IdentityFeatureGenerator...
    Fitting CategoryFeatureGenerator...
    Fitting CategoryMemoryMinimizeFeatureGenerator...
    Fitting DatetimeFeatureGenerator...
  Stage 4 Generators:
    Fitting DropUniqueFeatureGenerator...
  Stage 5 Generators:
    Fitting DropDuplicatesFeatureGenerator...
  Types of features in original data (raw dtype, special dtypes):
    ('category', []) : 4 | ['season', 'holiday', 'workingday', 'weather']
    ('datetime', []) : 1 | ['datetime']
    ('float', []) : 3 | ['temp', 'atemp', 'windspeed']
    ('int', []) : 5 | ['humidity', 'hour', 'dayofweek', 'month', 'year']
    ('object', []) : 3 | ['day_period', 'feel_temp_category', 'humidity_category']
  Types of features in processed data (raw dtype, special dtypes):
    ('category', []) : 5 | ['season', 'weather', 'day_period', 'feel_temp_category', 'humidity_categor
y']
    ('float', []) : 3 | ['temp', 'atemp', 'windspeed']
    ('int', []) : 4 | ['humidity', 'hour', 'dayofweek', 'month']
    ('int', ['bool']) : 3 | ['holiday', 'workingday', 'year']
    ('int', ['datetime_as_int']) : 3 | ['datetime', 'datetime.year', 'datetime.day']
  0.1s = Fit runtime
  16 features in original data used to generate 18 features in processed data.
  Train Data (Processed) Memory Usage: 0.79 MB (0.0% of available memory)
Data preprocessing and feature engineering runtime = 0.16s ...
AutoGluon will gauge predictive performance using evaluation metric: 'root_mean_squared_error'
  This metric's sign has been flipped to adhere to being higher_is_better. The metric score can be multiplied by -1 to ge
t the metric value.
  To change this, specify the eval_metric parameter of Predictor()
Large model count detected (110 configs) ... Only displaying the first 3 models of each family. To see all, set `verbosity=3`.
User-specified model hyperparameters to be fit:
{
  'NN_TORCH': [{}, {'activation': 'elu', 'dropout_prob': 0.10077639529843717, 'hidden_size': 108, 'learning_rate': 0.0027
35937344002146, 'num_layers': 4, 'use_batchnorm': True, 'weight_decay': 1.356433327634438e-12, 'ag_args': {'name_suffix': '_r7
9', 'priority': -2}}, {'activation': 'elu', 'dropout_prob': 0.11897478034205347, 'hidden_size': 213, 'learning_rate': 0.0010474
382260641949, 'num_layers': 4, 'use_batchnorm': False, 'weight_decay': 5.594471067786272e-10, 'ag_args': {'name_suffix': '_r2
2', 'priority': -7}}],
  'GBM': [{'extra_trees': True, 'ag_args': {'name_suffix': 'XT'}}, {}, {'learning_rate': 0.03, 'num_leaves': 128, 'featur
e_fraction': 0.9, 'min_data_in_leaf': 3, 'ag_args': {'name_suffix': 'Large', 'priority': 0, 'hyperparameter_tune_kwargs': Non
e}}],
  'CAT': [{}, {'depth': 6, 'grow_policy': 'SymmetricTree', 'l2_leaf_reg': 2.1542798306067823, 'learning_rate': 0.06864209
415792857, 'max_ctr_complexity': 4, 'one_hot_max_size': 10, 'ag_args': {'name_suffix': '_r177', 'priority': -1}}, {'depth': 8,
'grow_policy': 'Depthwise', 'l2_leaf_reg': 2.7997999596449104, 'learning_rate': 0.031375015734637225, 'max_ctr_complexity': 2,
'one_hot_max_size': 3, 'ag_args': {'name_suffix': '_r9', 'priority': -5}}],
  'XGB': [{}, {'colsample_bytree': 0.6917311125174739, 'enable_categorical': False, 'learning_rate': 0.01806387608752396
7, 'max_depth': 10, 'min_child_weight': 0.6028633586934382, 'ag_args': {'name_suffix': '_r33', 'priority': -8}}, {'colsample_by
tree': 0.6628423832084077, 'enable_categorical': False, 'learning_rate': 0.08775715546881824, 'max_depth': 5, 'min_child_weigh
t': 0.6294123374222513, 'ag_args': {'name_suffix': '_r89', 'priority': -16}}],
  'FASTAI': [{}, {'bs': 256, 'emb_drop': 0.5411770367537934, 'epochs': 43, 'layers': [800, 400], 'lr': 0.0151984885831815
9, 'ps': 0.2378294656604385, 'ag_args': {'name_suffix': '_r191', 'priority': -4}}, {'bs': 2048, 'emb_drop': 0.0507041132260581
1, 'epochs': 29, 'layers': [200, 100], 'lr': 0.08974235041576624, 'ps': 0.10393466140748028, 'ag_args': {'name_suffix': '_r10
2', 'priority': -11}}],
  'RF': [{'criterion': 'gini', 'ag_args': {'name_suffix': 'Gini', 'problem_types': ['binary', 'multiclass']}}, {'criterio
n': 'entropy', 'ag_args': {'name_suffix': 'Entr', 'problem_types': ['binary', 'multiclass']}}, {'criterion': 'squared_error',
'ag_args': {'name_suffix': 'MSE', 'problem_types': ['regression', 'quantile']}},
  'XT': [{'criterion': 'gini', 'ag_args': {'name_suffix': 'Gini', 'problem_types': ['binary', 'multiclass']}}, {'criterio
n': 'entropy', 'ag_args': {'name_suffix': 'Entr', 'problem_types': ['binary', 'multiclass']}}, {'criterion': 'squared_error',
'ag_args': {'name_suffix': 'MSE', 'problem_types': ['regression', 'quantile']}},
  }
AutoGluon will fit 2 stack levels (L1 to L2) ...
Fitting 106 L1 models, fit_strategy="parallel" ...
Note: fit_strategy='parallel', but `num_cpus=2`. Running parallel mode with fewer than 12 CPUs is not recommended and has been
disabled. You can override this by specifying `os.environ["AG_FORCE_PARALLEL"] = "True"`. Falling back to fit_strategy='sequen
tial' ...
Fitting model: LightGBMXT_BAG_L1 ... Training model for up to 279.86s of the 419.89s of remaining time.
  Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.90%)
  -34.771 = Validation score (-root_mean_squared_error)
  97.84s = Training runtime
  12.32s = Validation runtime
Fitting model: LightGBM_BAG_L1 ... Training model for up to 173.09s of the 313.11s of remaining time.
  Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.93%)
  -33.6342 = Validation score (-root_mean_squared_error)
  66.02s = Training runtime
  5.92s = Validation runtime
Fitting model: RandomForestMSE_BAG_L1 ... Training model for up to 102.10s of the 242.12s of remaining time.

```

```

Fitting 1 model on all data (use_child_oof=True) | Fitting with cpus=2, gpus=0, mem=0.1/7.4 GB
-38.458 = Validation score (-root_mean_squared_error)
20.56s = Training runtime
0.61s = Validation runtime
Fitting model: CatBoost_BAG_L1 ... Training model for up to 79.86s of the 219.89s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, mem=5.17%)
-38.4997 = Validation score (-root_mean_squared_error)
76.09s = Training runtime
0.29s = Validation runtime
Fitting model: ExtraTreesMSE_BAG_L1 ... Training model for up to 0.36s of the 140.39s of remaining time.
Fitting 1 model on all data (use_child_oof=True) | Fitting with cpus=2, gpus=0, mem=0.1/7.3 GB
-38.3484 = Validation score (-root_mean_squared_error)
10.37s = Training runtime
0.56s = Validation runtime
Fitting model: WeightedEnsemble_L2 ... Training model for up to 360.00s of the 128.31s of remaining time.
Fitting 1 model on all data | Fitting with cpus=2, gpus=1, mem=0.0/7.6 GB
Ensemble Weights: {'LightGBM_BAG_L1': 0.476, 'LightGBMXT_BAG_L1': 0.333, 'RandomForestMSE_BAG_L1': 0.143, 'CatBoost_BAG_L1': 0.048}
-32.3626 = Validation score (-root_mean_squared_error)
0.02s = Training runtime
0.0s = Validation runtime
Fitting 106 L2 models, fit_strategy="parallel" ...
Note: fit_strategy='parallel', but `num_cpus=2`. Running parallel mode with fewer than 12 CPUs is not recommended and has been disabled. You can override this by specifying `os.environ["AG_FORCE_PARALLEL"] = "True"`. Falling back to fit_strategy='sequential' ...
Fitting model: LightGBMXT_BAG_L2 ... Training model for up to 128.28s of the 128.25s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, mem=0.93%)
-33.6135 = Validation score (-root_mean_squared_error)
34.7s = Training runtime
0.44s = Validation runtime
Fitting model: LightGBM_BAG_L2 ... Training model for up to 90.01s of the 89.99s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, mem=0.99%)
-32.8932 = Validation score (-root_mean_squared_error)
31.41s = Training runtime
0.25s = Validation runtime
Fitting model: RandomForestMSE_BAG_L2 ... Training model for up to 54.66s of the 54.64s of remaining time.
Fitting 1 model on all data (use_child_oof=True) | Fitting with cpus=2, gpus=0, mem=0.1/7.2 GB
-33.0731 = Validation score (-root_mean_squared_error)
40.8s = Training runtime
0.64s = Validation runtime
Fitting model: CatBoost_BAG_L2 ... Training model for up to 12.37s of the 12.35s of remaining time.
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, mem=5.32%)
-34.2272 = Validation score (-root_mean_squared_error)
24.46s = Training runtime
0.14s = Validation runtime
Fitting model: WeightedEnsemble_L3 ... Training model for up to 360.00s of the -15.67s of remaining time.
Fitting 1 model on all data | Fitting with cpus=2, gpus=1, mem=0.0/7.2 GB
Ensemble Weights: {'LightGBM_BAG_L1': 0.304, 'RandomForestMSE_BAG_L2': 0.261, 'LightGBMXT_BAG_L1': 0.217, 'RandomForestMSE_BAG_L1': 0.087, 'LightGBM_BAG_L2': 0.087, 'CatBoost_BAG_L2': 0.043}
-32.197 = Validation score (-root_mean_squared_error)
0.04s = Training runtime
0.0s = Validation runtime
AutoGluon training complete, total runtime = 435.81s ... Best model: WeightedEnsemble_L3 | Estimated inference throughput: 71.0 rows/s (1361 batch size)
TabularPredictor saved. To load, use: predictor = TabularPredictor.load("/content/AutogluonModels/ag-20261004_194157")

```

```
In [106... predictor_new_features.leaderboard()
```

```
Out[106...
```

	model	score_val	eval_metric	pred_time_val	fit_time	pred_time_val_marginal	fit_time_marginal	st
0	WeightedEnsemble_L3	-32.196982	root_mean_squared_error	20.739971	367.585089	0.000847	0.042005	
1	WeightedEnsemble_L2	-32.362592	root_mean_squared_error	19.141901	260.526054	0.000582	0.022637	
2	LightGBM_BAG_L2	-32.893209	root_mean_squared_error	19.954914	302.281716	0.254721	31.410632	
3	RandomForestMSE_BAG_L2	-33.073118	root_mean_squared_error	20.343991	311.668437	0.643798	40.797353	
4	LightGBMXT_BAG_L2	-33.613493	root_mean_squared_error	20.136414	305.567797	0.436221	34.696713	
5	LightGBM_BAG_L1	-33.634169	root_mean_squared_error	5.921896	66.015301	5.921896	66.015301	
6	CatBoost_BAG_L2	-34.227213	root_mean_squared_error	19.840605	295.335100	0.140412	24.464016	
7	LightGBMXT_BAG_L1	-34.771022	root_mean_squared_error	12.321824	97.837913	12.321824	97.837913	
8	ExtraTreesMSE_BAG_L1	-38.348406	root_mean_squared_error	0.558875	10.367667	0.558875	10.367667	
9	RandomForestMSE_BAG_L1	-38.457978	root_mean_squared_error	0.606615	20.564800	0.606615	20.564800	
10	CatBoost_BAG_L1	-38.499695	root_mean_squared_error	0.290983	76.085402	0.290983	76.085402	

```
In [107... predictions_new_features = predictor_new_features.predict(test)
predictions_new_features.head()
```

```
Out[107...
      count
0  13.694748
1   2.700119
2   1.706871
3   2.472302
4   2.061820
```

dtype: float32

```
In [108...
# Remember to set all negative values to zero
predictions_new_features[predictions_new_features<0]=0
```

```
In [109...
# Same thing as train and test dataset
submission_new_features = pd.read_csv(path+'/sampleSubmission.csv')
submission_new_features.head()
```

```
Out[109...
      datetime  count
0  2011-01-20 00:00:00    0
1  2011-01-20 01:00:00    0
2  2011-01-20 02:00:00    0
3  2011-01-20 03:00:00    0
4  2011-01-20 04:00:00    0
```

```
In [110...
# Same submitting predictions
submission_new_features["count"] = predictions_new_features
submission_new_features.to_csv("submission_new_features.csv", index=False)
```

```
In [111...
!kaggle competitions submit -c bike-sharing-demand -f submission_new_features.csv -m "new features"

100% 188k/188k [00:00<00:00, 272kB/s]
Successfully submitted to Bike Sharing Demand
```

```
In [112...
!kaggle competitions submissions -c bike-sharing-demand | tail -n +1 | head -n 6
```

fileName	date	description	status	publicScore
submission_new_features.csv	2026-10-04 20:05:27.113000	new features	SubmissionStatus.COMPLETE	0.48565
submission.csv	2026-10-04 19:07:43.593000	first raw submission	SubmissionStatus.COMPLETE	1.43583
submission_new_hpo.csv	2026-09-20 19:21:12.517000	new features with hyperparameters	SubmissionStatus.COMPLETE	0.57774
submission_new_features.csv	2026-09-20 18:56:09.673000	new features	SubmissionStatus.COMPLETE	0.46971

New Score of 0.48565

Step 6: Hyper parameter optimization

- There are many options for hyper parameter optimization.
- Options are to change the AutoGluon higher level parameters or the individual model hyperparameters.
- The hyperparameters of the models themselves that are in AutoGluon. Those need the `hyperparameter` and `hyperparameter_tune_kwargs` arguments.

```
In [113...
from autogluon.common import space

hyperparameters = {
    "CAT": {
        "learning_rate": space.Real(0.01, 0.2, default=0.05, log=True),
        "depth": space.Int(4, 10, default=6),
        "l2_leaf_reg": space.Real(1, 20, default=3, log=True),
        "iterations": 10000, # early stopping cuida do resto
    },
    "GBM": {
        "learning_rate": space.Real(0.01, 0.2, default=0.05, log=True),
        "num_leaves": space.Int(16, 128, default=31),
        "min_data_in_leaf": space.Int(5, 100, default=20),
        "feature_fraction": space.Real(0.5, 1.0, default=0.9),
        "lambda_l2": space.Real(0.0, 10.0, default=0.0),
    },
    "XGB": {
        "learning_rate": space.Real(0.01, 0.2, default=0.05, log=True),
```

```

        "max_depth": space.Int(3, 10, default=6),
        "subsample": space.Real(0.5, 1.0, default=0.8),
        "colsample_bytree": space.Real(0.5, 1.0, default=0.8),
    },
}

```

```

In [114... time_limit = 10*60 # train various models for ~10 min
num_trials = 5 # try at most 5 different hyperparameter configurations for each type of model
search_strategy = 'auto' # to tune hyperparameters using random search routine with a Local scheduler

hyperparameter_tune_kwargs = { # HPO is not performed unless hyperparameter_tune_kwargs is specified
    'num_trials': num_trials,
    'scheduler' : 'local',
    'searcher': search_strategy,
} # Refer to TabularPredictor.fit docstring for all valid values

```

```

In [115... predictor_new_hpo = TabularPredictor(
    label="count", eval_metric="root_mean_squared_error"
).fit(
    train_data=train,
    presets="best_quality",
    fit_strategy="parallel",
    num_gpus=1,
    hyperparameter_tune_kwargs=hyperparameter_tune_kwargs,
    hyperparameters=hyperparameters
)

```

```

No path specified. Models will be saved in: "AutogluonModels/ag-20261004_203601"
Verbosity: 2 (Standard Logging)
===== System Info =====
AutoGluon Version: 1.6.3
Python Version: 3.11.13
Operating System: Linux
Platform Machine: x86_64
Platform Version: #1 SMP Thu Apr 30 18:17:14 UTC 2026
CPU Count: 2
Pytorch Version: 2.13.0+cu130
CUDA Version: 13.0
GPU Memory: GPU 0: 14.56/14.56 GB
Total GPU Memory: Free: 14.56 GB, Allocated: 0.00 GB, Total: 14.56 GB
GPU Count: 1
Memory Avail: 7.54 GB / 12.67 GB (59.5%)
Disk Space Avail: 186.27 GB / 241.91 GB (77.0%)
=====
Presets specified: ['best_quality']
Warning: hyperparameter tuning is currently experimental and may cause the process to hang.
Setting dynamic_stacking from 'auto' to True. Reason: Enable dynamic_stacking when use_bag_holdout is disabled. (use_bag_holdout=False)
Stack configuration (auto_stack=True): num_stack_levels=1, num_bag_folds=8, num_bag_sets=1
DyStack is enabled (dynamic_stacking=True). AutoGluon will try to determine whether the input data is affected by stacked overfitting and enable or disable stacking as a consequence.
    This is used to identify the optimal `num_stack_levels` value. Copies of AutoGluon will be fit on subsets of the data. Then holdout validation data is used to detect stacked overfitting.
    Running DyStack for up to 900s of the 3600s of remaining time (25%).
Running DyStack sub-fit ...
Beginning AutoGluon training ... Time limit = 900s
AutoGluon will save models to "/content/AutogluonModels/ag-20261004_203601/ds_sub_fit/sub_fit_ho"
Train Data Rows: 9676
Train Data Columns: 16
Label Column: count
Problem Type: regression
Preprocessing data...
Using Feature Generators to preprocess the data ...
Fitting AutoMLPipelineFeatureGenerator...
    Available Memory: 7726.22 MB
    Train Data (Original) Memory Usage: 2.30 MB (0.0% of available memory)
    Inferring data type of each feature based on column values. Set feature_metadata_in to manually specify special dtypes of the features.
    Stage 1 Generators:
        Fitting AsTypeFeatureGenerator...
        Note: Converting 3 features to boolean dtype as they only contain 2 unique values.
    Stage 2 Generators:
        Fitting FillNaFeatureGenerator...
    Stage 3 Generators:
        Fitting IdentityFeatureGenerator...
        Fitting CategoryFeatureGenerator...
        Fitting CategoryMemoryMinimizeFeatureGenerator...
        Fitting DatetimeFeatureGenerator...
    Stage 4 Generators:
        Fitting DropUniqueFeatureGenerator...
    Stage 5 Generators:
        Fitting DropDuplicatesFeatureGenerator...
    Types of features in original data (raw dtype, special dtypes):
        ('category', []) : 4 | ['season', 'holiday', 'workingday', 'weather']
        ('datetime', []) : 1 | ['datetime']
        ('float', []) : 3 | ['temp', 'atemp', 'windspeed']
        ('int', []) : 5 | ['humidity', 'hour', 'dayofweek', 'month', 'year']
        ('object', []) : 3 | ['day_period', 'feel_temp_category', 'humidity_category']
    Types of features in processed data (raw dtype, special dtypes):
        ('category', []) : 5 | ['season', 'weather', 'day_period', 'feel_temp_category', 'humidity_category']
        ('float', []) : 3 | ['temp', 'atemp', 'windspeed']
        ('int', []) : 4 | ['humidity', 'hour', 'dayofweek', 'month']
        ('int', ['bool']) : 3 | ['holiday', 'workingday', 'year']
        ('int', ['datetime_as_int']) : 3 | ['datetime', 'datetime.year', 'datetime.day']
    0.2s = Fit runtime
    16 features in original data used to generate 18 features in processed data.
    Train Data (Processed) Memory Usage: 0.70 MB (0.0% of available memory)
Data preprocessing and feature engineering runtime = 0.18s ...
AutoGluon will gauge predictive performance using evaluation metric: 'root_mean_squared_error'
    This metric's sign has been flipped to adhere to being higher_is_better. The metric score can be multiplied by -1 to get the metric value.
    To change this, specify the eval_metric parameter of Predictor()
User-specified model hyperparameters to be fit:
{
    'CAT': [{'learning_rate': Real: lower=0.01, upper=0.2, 'depth': Int: lower=4, upper=10, 'l2_leaf_reg': Real: lower=1, upper=20, 'iterations': 10000}],
    'GBM': [{'learning_rate': Real: lower=0.01, upper=0.2, 'num_leaves': Int: lower=16, upper=128, 'min_data_in_leaf': Int: lower=5, upper=100, 'feature_fraction': Real: lower=0.5, upper=1.0, 'lambda_l2': Real: lower=0.0, upper=10.0}],
    'XGB': [{'learning_rate': Real: lower=0.01, upper=0.2, 'max_depth': Int: lower=3, upper=10, 'subsample': Real: lower=0.5, upper=1.0, 'colsample_bytree': Real: lower=0.5, upper=1.0}],
}
AutoGluon will fit 2 stack levels (L1 to L2) ...
Fitting 3 L1 models, fit_strategy="parallel" ...
WARNING: fit_strategy='parallel', but `hyperparameter_tune_kwargs` is specified for model 'LightGBM_BAG_L1' with value {'num_tr

```

```
ials': 5, 'scheduler': 'local', 'searcher': 'auto'}. Hyperparameter tuning does not yet support `parallel` fit_strategy. Falling
g back to fit_strategy='sequential' ...
Hyperparameter tuning model: LightGBM_BAG_L1 ... Tuning model for up to 179.92s of the 899.81s of remaining time.
0%|          | 0/5 [00:00<?, ?it/s]
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.92%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=1.62%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=1.50%)
    Stopping HPO to satisfy time limit...
Fitted model: LightGBM_BAG_L1/T1 ...
    -34.1818      = Validation score    (-root_mean_squared_error)
    57.52s      = Training  runtime
    3.1s        = Validation runtime
Fitted model: LightGBM_BAG_L1/T2 ...
    -34.4201      = Validation score    (-root_mean_squared_error)
    64.17s      = Training  runtime
    4.28s      = Validation runtime
Fitted model: LightGBM_BAG_L1/T3 ...
    -34.3128      = Validation score    (-root_mean_squared_error)
    82.69s      = Training  runtime
    6.76s      = Validation runtime
Hyperparameter tuning model: CatBoost_BAG_L1 ... Tuning model for up to 179.92s of the 695.32s of remaining time.
0%|          | 0/5 [00:00<?, ?it/s]
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=5.34%)
    Stopping HPO to satisfy time limit...
Fitted model: CatBoost_BAG_L1/T1 ...
    -35.5758      = Validation score    (-root_mean_squared_error)
    157.47s     = Training  runtime
    0.23s       = Validation runtime
Hyperparameter tuning model: XGBoost_BAG_L1 ... Tuning model for up to 179.92s of the 537.79s of remaining time.
0%|          | 0/5 [00:00<?, ?it/s]
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.22%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.21%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.31%)
    Stopping HPO to satisfy time limit...
Fitted model: XGBoost_BAG_L1/T1 ...
    -34.3347      = Validation score    (-root_mean_squared_error)
    60.26s      = Training  runtime
    1.08s      = Validation runtime
Fitted model: XGBoost_BAG_L1/T2 ...
    -34.4531      = Validation score    (-root_mean_squared_error)
    40.3s       = Training  runtime
    0.68s      = Validation runtime
Fitted model: XGBoost_BAG_L1/T3 ...
    -34.6088      = Validation score    (-root_mean_squared_error)
    43.59s      = Training  runtime
    0.86s      = Validation runtime
Fitting model: WeightedEnsemble_L2 ... Training model for up to 360.00s of the 393.50s of remaining time.
    Fitting 1 model on all data | Fitting with cpus=2, gpus=1, mem=0.0/7.4 GB
    Ensemble Weights: {'LightGBM_BAG_L1/T1': 0.24, 'LightGBM_BAG_L1/T2': 0.16, 'CatBoost_BAG_L1/T1': 0.16, 'XGBoost_BAG_L1/
T2': 0.16, 'LightGBM_BAG_L1/T3': 0.12, 'XGBoost_BAG_L1/T3': 0.12, 'XGBoost_BAG_L1/T1': 0.04}
    -33.0479      = Validation score    (-root_mean_squared_error)
    0.03s        = Training  runtime
    0.0s         = Validation runtime
Fitting 3 L2 models, fit_strategy="parallel" ...
WARNING: fit_strategy='parallel', but `hyperparameter_tune_kwargs` is specified for model 'LightGBM_BAG_L2' with value {'num_tr
ials': 5, 'scheduler': 'local', 'searcher': 'auto'}. Hyperparameter tuning does not yet support `parallel` fit_strategy. Fallin
g back to fit_strategy='sequential' ...
Hyperparameter tuning model: LightGBM_BAG_L2 ... Tuning model for up to 118.04s of the 393.45s of remaining time.
0%|          | 0/5 [00:00<?, ?it/s]
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=0.97%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=1.74%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, m
emory=1.61%)
    Stopping HPO to satisfy time limit...
Fitted model: LightGBM_BAG_L2/T1 ...
    -33.8098      = Validation score    (-root_mean_squared_error)
    34.48s      = Training  runtime
    0.17s      = Validation runtime
Fitted model: LightGBM_BAG_L2/T2 ...
    -33.821      = Validation score    (-root_mean_squared_error)
    40.27s      = Training  runtime
    0.17s      = Validation runtime
Fitted model: LightGBM_BAG_L2/T3 ...
    -33.8343      = Validation score    (-root_mean_squared_error)
    40.06s      = Training  runtime
    0.49s      = Validation runtime
Hyperparameter tuning model: CatBoost_BAG_L2 ... Tuning model for up to 118.04s of the 278.51s of remaining time.
0%|          | 0/5 [00:00<?, ?it/s]
```

```
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=5.60%)
Stopping HPO to satisfy time limit...
Fitted model: CatBoost_BAG_L2/T1 ...
-33.4057 = Validation score (-root_mean_squared_error)
62.78s = Training runtime
0.17s = Validation runtime
Hyperparameter tuning model: XGBoost_BAG_L2 ... Tuning model for up to 118.04s of the 215.67s of remaining time.
0%|          | 0/5 [00:00<?, ?it/s]
```

```

    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.27%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.28%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.41%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.18%)
    Stopping HPO to satisfy time limit...
Fitted model: XGBoost_BAG_L2/T1 ...
    -33.8017 = Validation score (-root_mean_squared_error)
    27.54s = Training runtime
    0.3s = Validation runtime
Fitted model: XGBoost_BAG_L2/T2 ...
    -33.7545 = Validation score (-root_mean_squared_error)
    24.64s = Training runtime
    0.22s = Validation runtime
Fitted model: XGBoost_BAG_L2/T3 ...
    -34.05 = Validation score (-root_mean_squared_error)
    27.97s = Training runtime
    0.21s = Validation runtime
Fitted model: XGBoost_BAG_L2/T4 ...
    -34.6398 = Validation score (-root_mean_squared_error)
    42.5s = Training runtime
    0.17s = Validation runtime
Fitting model: WeightedEnsemble_L3 ... Training model for up to 360.00s of the 92.83s of remaining time.
    Fitting 1 model on all data | Fitting with cpus=2, gpus=1, mem=0.0/7.2 GB
    Ensemble Weights: {'CatBoost_BAG_L2/T1': 0.522, 'XGBoost_BAG_L2/T2': 0.217, 'CatBoost_BAG_L1/T1': 0.13, 'LightGBM_BAG_L2/T2': 0.087, 'XGBoost_BAG_L2/T3': 0.043}
    -33.2391 = Validation score (-root_mean_squared_error)
    0.04s = Training runtime
    0.0s = Validation runtime
AutoGluon training complete, total runtime = 807.24s ... Best model: WeightedEnsemble_L2 | Estimated inference throughput: 71.3 rows/s (1210 batch size)
TabularPredictor saved. To load, use: predictor = TabularPredictor.load("/content/AutogluonModels/ag-20261004_203601/ds_sub_fit/sub_fit_ho")
Deleting DyStack predictor artifacts (clean_up_fits=True) ...
Leaderboard on holdout data (DyStack):

```

	model	score_holdout	score_val	eval_metric	pred_time_test	pred_time_val	fit_time	pred_time
	e_test_marginal	pred_time_val_marginal	fit_time_marginal	stack_level	can_infer	fit_order		
0	WeightedEnsemble_L2	-33.058678	-33.047887	root_mean_squared_error	14.995734	16.982720	506.031822	
0.005800		0.000777	0.032474	2	True	8		
1	LightGBM_BAG_L1/T1	-33.074187	-34.181798	root_mean_squared_error	3.593002	3.096177	57.520046	
3.593002		3.096177	57.520046	1	True	1		
2	WeightedEnsemble_L3	-33.133407	-33.239078	root_mean_squared_error	15.746187	17.756997	661.698132	
0.003551		0.000798	0.038866	3	True	17		
3	LightGBM_BAG_L2/T1	-33.148234	-33.809771	root_mean_squared_error	15.131399	17.152885	540.481136	
0.141465		0.170942	34.481787	2	True	9		
4	XGBoost_BAG_L2/T1	-33.200171	-33.801742	root_mean_squared_error	15.232242	17.280964	533.536566	
0.242308		0.299021	27.537217	2	True	13		
5	XGBoost_BAG_L2/T3	-33.226191	-34.050040	root_mean_squared_error	15.224621	17.191123	533.968424	
0.234688		0.209180	27.969075	2	True	15		
6	XGBoost_BAG_L2/T2	-33.346111	-33.754468	root_mean_squared_error	15.194039	17.203133	530.642663	
0.204105		0.221190	24.643314	2	True	14		
7	LightGBM_BAG_L2/T2	-33.347051	-33.820990	root_mean_squared_error	15.150732	17.155784	546.270089	
0.160798		0.173840	40.270740	2	True	10		
8	CatBoost_BAG_L2/T1	-33.392368	-33.405736	root_mean_squared_error	15.143046	17.151990	568.776136	
0.153112		0.170047	62.776787	2	True	12		
9	LightGBM_BAG_L2/T3	-33.431620	-33.834324	root_mean_squared_error	15.262915	17.469916	546.056289	
0.272982		0.487973	40.056941	2	True	11		
10	XGBoost_BAG_L2/T4	-33.432251	-34.639796	root_mean_squared_error	15.266198	17.152496	548.498301	
0.276264		0.170553	42.498952	2	True	16		
11	XGBoost_BAG_L1/T1	-33.799386	-34.334723	root_mean_squared_error	1.023542	1.080935	60.263149	
1.023542		1.080935	60.263149	1	True	5		
12	XGBoost_BAG_L1/T3	-33.957003	-34.608822	root_mean_squared_error	0.661616	0.859962	43.594690	
0.661616		0.859962	43.594690	1	True	7		
13	XGBoost_BAG_L1/T2	-34.046252	-34.453113	root_mean_squared_error	0.523539	0.676354	40.300704	
0.523539		0.676354	40.300704	1	True	6		
14	LightGBM_BAG_L1/T3	-34.399871	-34.312844	root_mean_squared_error	5.466192	6.757896	82.687372	
5.466192		6.757896	82.687372	1	True	3		
15	LightGBM_BAG_L1/T2	-34.533855	-34.420103	root_mean_squared_error	3.578889	4.276225	64.167034	
3.578889		4.276225	64.167034	1	True	2		
16	CatBoost_BAG_L1/T1	-34.667111	-35.575793	root_mean_squared_error	0.143155	0.234395	157.466352	
0.143155		0.234395	157.466352	1	True	4		

```

    1 = Optimal num_stack_levels (Stacked Overfitting Occurred: False)
    826s = DyStack runtime | 2774s = Remaining runtime
Starting main fit with num_stack_levels=1.
    For future fit calls on this dataset, you can skip DyStack to save time: `predictor.fit(..., dynamic_stacking=False, num_stack_levels=1)`
Beginning AutoGluon training ... Time limit = 2774s
AutoGluon will save models to "/content/AutogluonModels/ag-20261004_203601"
Train Data Rows: 10886
Train Data Columns: 16
Label Column: count
Problem Type: regression
Preprocessing data...
Using Feature Generators to preprocess the data ...
Fitting AutoMLPipelineFeatureGenerator...

```

```

Available Memory: 7799.02 MB
Train Data (Original) Memory Usage: 2.59 MB (0.0% of available memory)
Inferring data type of each feature based on column values. Set feature_metadata_in to manually specify special dtypes
of the features.
Stage 1 Generators:
    Fitting AsTypeFeatureGenerator...
        Note: Converting 3 features to boolean dtype as they only contain 2 unique values.
Stage 2 Generators:
    Fitting FillNaFeatureGenerator...
Stage 3 Generators:
    Fitting IdentityFeatureGenerator...
    Fitting CategoryFeatureGenerator...
    Fitting CategoryMemoryMinimizeFeatureGenerator...
    Fitting DatetimeFeatureGenerator...
Stage 4 Generators:
    Fitting DropUniqueFeatureGenerator...
Stage 5 Generators:
    Fitting DropDuplicatesFeatureGenerator...
Types of features in original data (raw dtype, special dtypes):
('category', []) : 4 | ['season', 'holiday', 'workingday', 'weather']
('datetime', []) : 1 | ['datetime']
('float', []) : 3 | ['temp', 'atemp', 'windspeed']
('int', []) : 5 | ['humidity', 'hour', 'dayofweek', 'month', 'year']
('object', []) : 3 | ['day_period', 'feel_temp_category', 'humidity_category']
Types of features in processed data (raw dtype, special dtypes):
('category', []) : 5 | ['season', 'weather', 'day_period', 'feel_temp_category', 'humidity_category']
('float', []) : 3 | ['temp', 'atemp', 'windspeed']
('int', []) : 4 | ['humidity', 'hour', 'dayofweek', 'month']
('int', ['bool']) : 3 | ['holiday', 'workingday', 'year']
('int', ['datetime_as_int']) : 3 | ['datetime', 'datetime.year', 'datetime.day']
0.2s = Fit runtime
16 features in original data used to generate 18 features in processed data.
Train Data (Processed) Memory Usage: 0.79 MB (0.0% of available memory)
Data preprocessing and feature engineering runtime = 0.24s ...
AutoGluon will gauge predictive performance using evaluation metric: 'root_mean_squared_error'
This metric's sign has been flipped to adhere to being higher_is_better. The metric score can be multiplied by -1 to get the metric value.
To change this, specify the eval_metric parameter of Predictor()
User-specified model hyperparameters to be fit:
{
    'CAT': [{'learning_rate': Real: lower=0.01, upper=0.2, 'depth': Int: lower=4, upper=10, 'l2_leaf_reg': Real: lower=1, upper=20, 'iterations': 10000}],
    'GBM': [{'learning_rate': Real: lower=0.01, upper=0.2, 'num_leaves': Int: lower=16, upper=128, 'min_data_in_leaf': Int: lower=5, upper=100, 'feature_fraction': Real: lower=0.5, upper=1.0, 'lambda_l2': Real: lower=0.0, upper=10.0}],
    'XGB': [{'learning_rate': Real: lower=0.01, upper=0.2, 'max_depth': Int: lower=3, upper=10, 'subsample': Real: lower=0.5, upper=1.0, 'colsample_bytree': Real: lower=0.5, upper=1.0}],
}
AutoGluon will fit 2 stack levels (L1 to L2) ...
Fitting 3 L1 models, fit_strategy="parallel" ...
WARNING: fit_strategy="parallel", but `hyperparameter_tune_kwargs` is specified for model 'LightGBM_BAG_L1' with value {'num_trials': 5, 'scheduler': 'local', 'searcher': 'auto'}. Hyperparameter tuning does not yet support `parallel` fit_strategy. Falling back to fit_strategy='sequential' ...
Hyperparameter tuning model: LightGBM_BAG_L1 ... Tuning model for up to 554.62s of the 2773.81s of remaining time.
0% | 0/5 [00:00<?, ?it/s]

Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.91%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.61%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.50%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.18%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.00%)
Fitted model: LightGBM_BAG_L1/T1 ...
-33.535 = Validation score (-root_mean_squared_error)
66.03s = Training runtime
5.35s = Validation runtime
Fitted model: LightGBM_BAG_L1/T2 ...
-33.7567 = Validation score (-root_mean_squared_error)
68.21s = Training runtime
5.85s = Validation runtime
Fitted model: LightGBM_BAG_L1/T3 ...
-33.688 = Validation score (-root_mean_squared_error)
96.85s = Training runtime
8.3s = Validation runtime
Fitted model: LightGBM_BAG_L1/T4 ...
-34.5608 = Validation score (-root_mean_squared_error)
56.82s = Training runtime
4.04s = Validation runtime
Fitted model: LightGBM_BAG_L1/T5 ...
-33.3262 = Validation score (-root_mean_squared_error)
142.04s = Training runtime
20.83s = Validation runtime
Hyperparameter tuning model: CatBoost_BAG_L1 ... Tuning model for up to 554.62s of the 2343.7s of remaining time.
0% | 0/5 [00:00<?, ?it/s]

```

```
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=5.36%)
Stopping HPO to satisfy time limit...
Fitted model: CatBoost_BAG_L1/T1 ...
-33.4497 = Validation score (-root_mean_squared_error)
458.52s = Training runtime
0.78s = Validation runtime
Hyperparameter tuning model: XGBoost_BAG_L1 ... Tuning model for up to 554.62s of the 1885.1s of remaining time.
0% | 0/5 [00:00<?, ?it/s]

Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.22%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.22%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.31%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.86%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.32%)
Fitted model: XGBoost_BAG_L1/T1 ...
-33.8136 = Validation score (-root_mean_squared_error)
76.52s = Training runtime
2.58s = Validation runtime
Fitted model: XGBoost_BAG_L1/T2 ...
-34.4497 = Validation score (-root_mean_squared_error)
47.38s = Training runtime
1.2s = Validation runtime
Fitted model: XGBoost_BAG_L1/T3 ...
-33.9353 = Validation score (-root_mean_squared_error)
56.65s = Training runtime
1.2s = Validation runtime
Fitted model: XGBoost_BAG_L1/T4 ...
-36.0632 = Validation score (-root_mean_squared_error)
72.73s = Training runtime
1.22s = Validation runtime
Fitted model: XGBoost_BAG_L1/T5 ...
-33.22 = Validation score (-root_mean_squared_error)
81.9s = Training runtime
1.83s = Validation runtime
Fitting model: WeightedEnsemble_L2 ... Training model for up to 360.00s of the 1549.75s of remaining time.
Fitting 1 model on all data | Fitting with cpus=2, gpus=1, mem=0.0/7.4 GB
Ensemble Weights: {'CatBoost_BAG_L1/T1': 0.318, 'LightGBM_BAG_L1/T1': 0.182, 'XGBoost_BAG_L1/T5': 0.182, 'LightGBM_BAG_L1/T2': 0.091, 'LightGBM_BAG_L1/T3': 0.091, 'LightGBM_BAG_L1/T5': 0.091, 'XGBoost_BAG_L1/T3': 0.045}
-32.2172 = Validation score (-root_mean_squared_error)
0.05s = Training runtime
0.0s = Validation runtime
Fitting 3 L2 models, fit_strategy="parallel" ...
WARNING: fit_strategy='parallel', but `hyperparameter_tune_kwargs` is specified for model 'LightGBM_BAG_L2' with value {'num_trials': 5, 'scheduler': 'local', 'searcher': 'auto'}. Hyperparameter tuning does not yet support `parallel` fit_strategy. Falling back to fit_strategy='sequential' ...
Hyperparameter tuning model: LightGBM_BAG_L2 ... Tuning model for up to 464.9s of the 1549.66s of remaining time.
0% | 0/5 [00:00<?, ?it/s]

Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.99%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.80%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.66%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.30%)
Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.07%)
Fitted model: LightGBM_BAG_L2/T1 ...
-33.0298 = Validation score (-root_mean_squared_error)
38.07s = Training runtime
0.25s = Validation runtime
Fitted model: LightGBM_BAG_L2/T2 ...
-33.0149 = Validation score (-root_mean_squared_error)
43.89s = Training runtime
0.28s = Validation runtime
Fitted model: LightGBM_BAG_L2/T3 ...
-33.1251 = Validation score (-root_mean_squared_error)
42.94s = Training runtime
0.52s = Validation runtime
Fitted model: LightGBM_BAG_L2/T4 ...
-33.1763 = Validation score (-root_mean_squared_error)
39.24s = Training runtime
0.13s = Validation runtime
Fitted model: LightGBM_BAG_L2/T5 ...
-32.8535 = Validation score (-root_mean_squared_error)
43.57s = Training runtime
0.61s = Validation runtime
Hyperparameter tuning model: CatBoost_BAG_L2 ... Tuning model for up to 464.9s of the 1341.78s of remaining time.
0% | 0/5 [00:00<?, ?it/s]
```

```

    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=5.51%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=5.87%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=5.52%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=6.61%)
    Stopping HPO to satisfy time limit...
Fitted model: CatBoost_BAG_L2/T1 ...
    -32.5674 = Validation score (-root_mean_squared_error)
    73.85s = Training runtime
    0.17s = Validation runtime
Fitted model: CatBoost_BAG_L2/T2 ...
    -32.959 = Validation score (-root_mean_squared_error)
    98.02s = Training runtime
    0.15s = Validation runtime
Fitted model: CatBoost_BAG_L2/T3 ...
    -32.5313 = Validation score (-root_mean_squared_error)
    103.29s = Training runtime
    0.16s = Validation runtime
Fitted model: CatBoost_BAG_L2/T4 ...
    -32.5619 = Validation score (-root_mean_squared_error)
    163.81s = Training runtime
    0.37s = Validation runtime
Hyperparameter tuning model: XGBoost_BAG_L2 ... Tuning model for up to 464.9s of the 902.66s of remaining time.
0%|          | 0/5 [00:00<?, ?it/s]

    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.33%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.31%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.47%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=1.36%)
    Fitting 8 child models (S1F1 - S1F8) | Fitting with ParallelLocalFoldFittingStrategy (2 workers, per: cpus=1, gpus=0, memory=0.48%)
Fitted model: XGBoost_BAG_L2/T1 ...
    -33.1862 = Validation score (-root_mean_squared_error)
    28.71s = Training runtime
    0.21s = Validation runtime
Fitted model: XGBoost_BAG_L2/T2 ...
    -33.0497 = Validation score (-root_mean_squared_error)
    26.79s = Training runtime
    0.15s = Validation runtime
Fitted model: XGBoost_BAG_L2/T3 ...
    -33.4496 = Validation score (-root_mean_squared_error)
    29.38s = Training runtime
    0.34s = Validation runtime
Fitted model: XGBoost_BAG_L2/T4 ...
    -34.3575 = Validation score (-root_mean_squared_error)
    47.65s = Training runtime
    0.22s = Validation runtime
Fitted model: XGBoost_BAG_L2/T5 ...
    -33.1997 = Validation score (-root_mean_squared_error)
    33.08s = Training runtime
    0.3s = Validation runtime
Fitting model: WeightedEnsemble_L3 ... Training model for up to 360.00s of the 736.85s of remaining time.
    Fitting 1 model on all data | Fitting with cpus=2, gpus=1, mem=0.0/7.3 GB
    Ensemble Weights: {'CatBoost_BAG_L2/T1': 0.273, 'CatBoost_BAG_L2/T3': 0.273, 'XGBoost_BAG_L1/T5': 0.136, 'LightGBM_BAG_L2/T5': 0.136, 'CatBoost_BAG_L2/T4': 0.091, 'XGBoost_BAG_L2/T2': 0.091}
    -32.3899 = Validation score (-root_mean_squared_error)
    0.05s = Training runtime
    0.0s = Validation runtime
AutoGluon training complete, total runtime = 2037.28s ... Best model: WeightedEnsemble_L2 | Estimated inference throughput: 30.8 rows/s (1361 batch size)
TabularPredictor saved. To load, use: predictor = TabularPredictor.load("/content/AutogluonModels/ag-20261004_203601")

```

In [119... predictor_new_hpo.leaderboard()]

Out[119...

	model	score_val	eval_metric	pred_time_val	fit_time	pred_time_val_marginal	fit_time_marginal	stack
0	WeightedEnsemble_L2	-32.217247	root_mean_squared_error	44.139306	970.254645	0.000748	0.054783	
1	WeightedEnsemble_L3	-32.389932	root_mean_squared_error	49.378852	1505.452359	0.000692	0.051726	
2	CatBoost_BAG_L2/T3	-32.531252	root_mean_squared_error	48.078591	1197.390141	0.159307	103.291187	
3	CatBoost_BAG_L2/T4	-32.561924	root_mean_squared_error	48.291876	1257.905307	0.372591	163.806352	
4	CatBoost_BAG_L2/T1	-32.567445	root_mean_squared_error	48.088339	1167.945280	0.169055	73.846325	
5	LightGBM_BAG_L2/T5	-32.853545	root_mean_squared_error	48.530510	1137.664110	0.611226	43.565155	
6	CatBoost_BAG_L2/T2	-32.959024	root_mean_squared_error	48.064410	1192.122505	0.145126	98.023550	
7	LightGBM_BAG_L2/T2	-33.014879	root_mean_squared_error	48.194550	1137.993551	0.275266	43.894596	
8	LightGBM_BAG_L2/T1	-33.029782	root_mean_squared_error	48.169108	1132.170493	0.249824	38.071538	
9	XGBoost_BAG_L2/T2	-33.049665	root_mean_squared_error	48.065982	1120.891615	0.146698	26.792660	
10	LightGBM_BAG_L2/T3	-33.125070	root_mean_squared_error	48.442214	1137.036803	0.522929	42.937848	
11	LightGBM_BAG_L2/T4	-33.176264	root_mean_squared_error	48.051472	1133.334836	0.132188	39.235881	
12	XGBoost_BAG_L2/T1	-33.186225	root_mean_squared_error	48.124877	1122.811527	0.205593	28.712572	
13	XGBoost_BAG_L2/T5	-33.199713	root_mean_squared_error	48.223953	1127.174942	0.304669	33.075988	
14	XGBoost_BAG_L1/T5	-33.220013	root_mean_squared_error	1.834283	81.902466	1.834283	81.902466	
15	LightGBM_BAG_L1/T5	-33.326196	root_mean_squared_error	20.826915	142.038612	20.826915	142.038612	
16	XGBoost_BAG_L2/T3	-33.449611	root_mean_squared_error	48.254313	1123.482401	0.335029	29.383446	
17	CatBoost_BAG_L1/T1	-33.449744	root_mean_squared_error	0.778120	458.515389	0.778120	458.515389	
18	LightGBM_BAG_L1/T1	-33.534970	root_mean_squared_error	5.350332	66.034115	5.350332	66.034115	
19	LightGBM_BAG_L1/T3	-33.688010	root_mean_squared_error	8.299894	96.853023	8.299894	96.853023	
20	LightGBM_BAG_L1/T2	-33.756716	root_mean_squared_error	5.852969	68.210052	5.852969	68.210052	
21	XGBoost_BAG_L1/T1	-33.813569	root_mean_squared_error	2.582469	76.524085	2.582469	76.524085	
22	XGBoost_BAG_L1/T3	-33.935278	root_mean_squared_error	1.196046	56.646205	1.196046	56.646205	
23	XGBoost_BAG_L2/T4	-34.357468	root_mean_squared_error	48.142688	1141.744500	0.223404	47.645545	
24	XGBoost_BAG_L1/T2	-34.449669	root_mean_squared_error	1.198257	47.375008	1.198257	47.375008	
25	LightGBM_BAG_L1/T4	-34.560787	root_mean_squared_error	4.038150	56.817691	4.038150	56.817691	
26	XGBoost_BAG_L1/T4	-36.063174	root_mean_squared_error	1.220205	72.730691	1.220205	72.730691	



```
In [120...] predictions_new_hpo = predictor_new_hpo.predict(test)
```

```
In [121...] # Remember to set all negative values to zero
predictions_new_hpo[predictions_new_hpo<0]=0
```

```
In [122...] # Same thing as train and test dataset
submission_new_hpo = pd.read_csv(path+'/sampleSubmission.csv')
submission_new_hpo.head()
```

Out[122...

	datetime	count
0	2011-01-20 00:00:00	0
1	2011-01-20 01:00:00	0
2	2011-01-20 02:00:00	0
3	2011-01-20 03:00:00	0
4	2011-01-20 04:00:00	0

```
In [123...] # Same submitting predictions
submission_new_hpo["count"] = predictions_new_hpo
submission_new_hpo.to_csv("submission_new_hpo.csv", index=False)
```

```
In [124...] !kaggle competitions submit -c bike-sharing-demand -f submission_new_hpo.csv -m "new features with hyperparameters"
```

```
100% 187k/187k [00:00<00:00, 288kB/s]
Successfully submitted to Bike Sharing Demand
```

```
In [129...] !kaggle competitions submissions -c bike-sharing-demand | tail -n +1 | head -n 6
```

fileName	date	description	status	publicScore
submission_new_hpo.csv	2026-10-04 21:31:23.407000	new features with hyperparameters	SubmissionStatus.COMPLETE	0.55715
submission_new_features.csv	2026-10-04 20:05:27.113000	new features	SubmissionStatus.COMPLETE	0.48565
submission.csv	2026-10-04 19:07:43.593000	first raw submission	SubmissionStatus.COMPLETE	1.43583
submission_new_hpo.csv	2026-09-20 19:21:12.517000	new features with hyperparameters	SubmissionStatus.COMPLETE	0.57774

New Score of 0.55715

Step 7: Write a Report

Refer to the markdown file for the full report

Creating plots and table for report

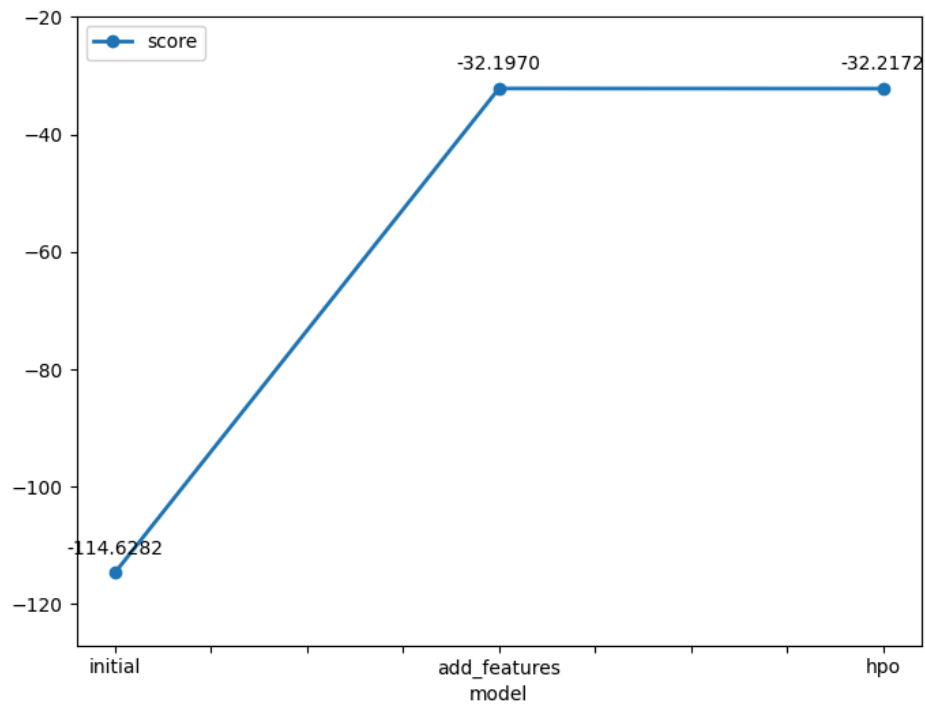
```
In [187... # Taking the top model score from each training run and creating a line plot to show improvement
# You can create these in the notebook and save them to PNG or use some other tool (e.g. google sheets, excel)
# score_val (-RMSE) of the best model in each predictor's Leaderboard(); AutoGluon flips the sign, so higher is better
fig = pd.DataFrame(
    {
        "model": ["initial", "add_features", "hpo"],
        "score": [-114.628178, -32.196982, -32.217247]
    }
)

ax = fig.plot(
    x="model",
    y="score",
    figsize=(8, 6),
    marker="o",
    linestyle="-",
    linewidth=2
)

for i, score in enumerate(fig["score"]):
    ax.annotate(
        f"{score:.4f}",
        (i, score),
        textcoords="offset points",
        xytext=(0, 10),
        ha="center",
    )

ax.margins(y=0.15)

fig = ax.get_figure()
fig.savefig('model_test_RMSE.png', bbox_inches="tight")
```



In [168..

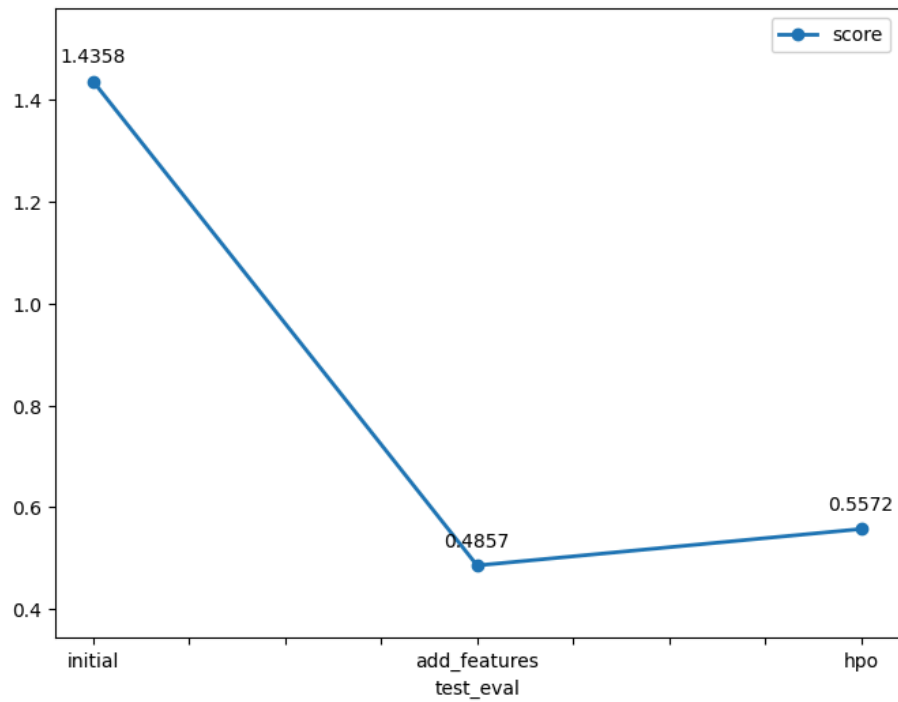
```
df = pd.DataFrame(
    {
        "test_eval": ["initial", "add_features", "hpo"],
        "score": [1.43583, 0.48565, 0.55715]
    }
)

ax = df.plot(
    x="test_eval",
    y="score",
    figsize=(8, 6),
    marker="o",
    linestyle="-",
    linewidth=2
)

for i, score in enumerate(df["score"]):
    ax.annotate(
        f"{score:.4f}",
        (i, score),
        textcoords="offset points",
        xytext=(0, 10),
        ha="center",
    )

ax.margins(y=0.15)

fig = ax.get_figure()
fig.savefig("model_test_score.png", bbox_inches="tight")
```



Hyperparameter table

```
In [186... # The 3 hyperparameters we tuned with the kaggle score as the result
pd.DataFrame({
    "model": ["initial", "add_features", "hpo"],
    "hpo1": ["time_limit=600", "time_limit=600", "time_limit=600"],
    "hpo2": ["presets=best_quality", "presets=best_quality", "search_strategy=auto"],
    "hpo3": ["fit_strategy=parallel", "fit_strategy=parallel", "num_trials=5 "],
    "score": [1.43525, 0.46971, 0.57774]
})
```

```
Out[186...
   model  hpo1  hpo2  hpo3  score
0  initial  time_limit=600  presets=best_quality  fit_strategy=parallel  1.43525
1  add_features  time_limit=600  presets=best_quality  fit_strategy=parallel  0.46971
2    hpo  time_limit=600  search_strategy=auto  num_trials=5  0.57774
```

```
In [183... predictor_new_hpo.model_info(predictor_new_hpo.model_best)
```

```

Out[183... {'name': 'WeightedEnsemble_L2',
             'model_type': 'WeightedEnsembleModel',
             'problem_type': 'regression',
             'eval_metric': 'root_mean_squared_error',
             'stopping_metric': 'root_mean_squared_error',
             'fit_time': 0.05478334426879883,
             'num_classes': None,
             'quantile_levels': None,
             'predict_time': 0.0007481575012207031,
             'val_score': np.float64(-32.21724730922139),
             'hyperparameters': {'use_orig_features': False,
                                  'valid_stack': True,
                                  'max_base_models': 0,
                                  'max_base_models_per_type': 'auto',
                                  'save_bag_folds': True,
                                  'stratify': 'auto',
                                  'bin': 'auto',
                                  'n_bins': None,
                                  'vary_seed_across_folds': False,
                                  'custom_splits': None,
                                  'model_random_seed': 0},
             'hyperparameters_user': {'save_bag_folds': True},
             'hyperparameters_fit': {},
             'hyperparameters_nondefault': ['save_bag_folds'],
             'ag_args_fit': {'max_memory_usage_ratio': 1.0,
                              'max_gpu_memory_usage_ratio': 1.0,
                              'max_time_limit_ratio': 1.0,
                              'max_time_limit': None,
                              'min_time_limit': 0,
                              'valid_raw_types': None,
                              'valid_special_types': None,
                              'ignored_type_group_special': None,
                              'ignored_type_group_raw': None,
                              'get_features_kwargs': None,
                              'get_features_kwargs_extra': None,
                              'predict_1_batch_size': None,
                              'temperature_scalar': None,
                              'drop_unique': False},
             'num_features': 7,
             'features': ['LightGBM_BAG_L1/T5',
                           'XGBoost_BAG_L1/T5',
                           'CatBoost_BAG_L1/T1',
                           'LightGBM_BAG_L1/T1',
                           'XGBoost_BAG_L1/T3',
                           'LightGBM_BAG_L1/T2',
                           'LightGBM_BAG_L1/T3'],
             'feature_metadata': <autogluon.common.features.feature_metadata.FeatureMetadata at 0x78a37b0d91d0>,
             'memory_size': 4418,
             'compile_time': None,
             'is_initialized': True,
             'is_fit': True,
             'is_valid': True,
             'can_infer': True,
             'has_learning_curves': False,
             'num_samples': 10886,
             'val_in_fit': False,
             'unlabeled_in_fit': False,
             'num_cpus': 2,
             'num_gpus': 1,
             'bagged_info': {'child_model_type': 'GreedyWeightedEnsembleModel',
                              'num_child_models': 1,
                              'child_model_names': ['S1F1'],
                              '_n_repeats': 1,
                              '_k_per_n_repeat': [1],
                              '_random_state': 2,
                              'low_memory': False,
                              'bagged_mode': False,
                              'max_memory_size': 4418,
                              'min_memory_size': 4418,
                              'child_hyperparameters': {'ensemble_size': 25, 'subsample_size': 100000},
                              'child_hyperparameters_user': {},
                              'child_hyperparameters_fit': {'ensemble_size': 22},
                              'child_ag_args_fit': {'max_memory_usage_ratio': 1.0,
                                                       'max_gpu_memory_usage_ratio': 1.0,
                                                       'max_time_limit_ratio': 1.0,
                                                       'max_time_limit': None,
                                                       'min_time_limit': 0,
                                                       'valid_raw_types': None,
                                                       'valid_special_types': None,
                                                       'ignored_type_group_special': None,
                                                       'ignored_type_group_raw': None,
                                                       'get_features_kwargs': None,
                                                       'get_features_kwargs_extra': None,
                                                       'predict_1_batch_size': None,
                                                       'temperature_scalar': None,
                                                       'drop_unique': False}},
             'stacker_info': {'num_base_models': 7,
                               'base_model_names': ['LightGBM_BAG_L1/T1',

```

```

'LightGBM_BAG_L1/T2',
'LightGBM_BAG_L1/T3',
'LightGBM_BAG_L1/T5',
'CatBoost_BAG_L1/T1',
'XGBoost_BAG_L1/T3',
'XGBoost_BAG_L1/T5']]],
'children_info': {'S1F1': {'name': 'S1F1',
'model_type': 'GreedyWeightedEnsembleModel',
'problem_type': 'regression',
'eval_metric': 'root_mean_squared_error',
'stopping_metric': 'root_mean_squared_error',
'fit_time': 0.05478334426879883,
'num_classes': None,
'quantile_levels': None,
'predict_time': None,
'val_score': None,
'hyperparameters': {'ensemble_size': 25, 'subsample_size': 1000000},
'hyperparameters_user': {}},
'hyperparameters_fit': {'ensemble_size': 22},
'hyperparameters_nondefault': [],
'ag_args_fit': {'max_memory_usage_ratio': 1.0,
'max_gpu_memory_usage_ratio': 1.0,
'max_time_limit_ratio': 1.0,
'max_time_limit': None,
'min_time_limit': 0,
'valid_raw_types': None,
'valid_special_types': None,
'ignored_type_group_special': None,
'ignored_type_group_raw': None,
'get_features_kwargs': None,
'get_features_kwargs_extra': None,
'predict_1_batch_size': None,
'temperature_scalar': None,
'drop_unique': False},
'num_features': 7,
'features': ['LightGBM_BAG_L1/T1',
'LightGBM_BAG_L1/T2',
'LightGBM_BAG_L1/T3',
'LightGBM_BAG_L1/T5',
'CatBoost_BAG_L1/T1',
'XGBoost_BAG_L1/T3',
'XGBoost_BAG_L1/T5'],
'feature_metadata': <autogluon.common.features.feature_metadata.FeatureMetadata at 0x78a38d91a990>,
'memory_size': 9343,
'compile_time': None,
'is_initialized': True,
'is_fit': True,
'is_valid': True,
'can_infer': True,
'has_learning_curves': False,
'num_samples': 10886,
'val_in_fit': False,
'unlabeled_in_fit': False,
'num_cpus': 2,
'num_gpus': 1,
'model_weights': {'LightGBM_BAG_L1/T1': np.float64(0.18181818181818182),
'LightGBM_BAG_L1/T2': np.float64(0.09090909090909091),
'LightGBM_BAG_L1/T3': np.float64(0.09090909090909091),
'LightGBM_BAG_L1/T5': np.float64(0.09090909090909091),
'CatBoost_BAG_L1/T1': np.float64(0.3181818181818182),
'XGBoost_BAG_L1/T3': np.float64(0.045454545454545456),
'XGBoost_BAG_L1/T5': np.float64(0.18181818181818182)}}}]

```

In [173.. predictor_new_hpo.leaderboard().head(10)

	model	score_val	eval_metric	pred_time_val	fit_time	pred_time_val_marginal	fit_time_marginal	stack_l
0	WeightedEnsemble_L2	-32.217247	root_mean_squared_error	44.139306	970.254645	0.000748	0.054783	
1	WeightedEnsemble_L3	-32.389932	root_mean_squared_error	49.378852	1505.452359	0.000692	0.051726	
2	CatBoost_BAG_L2/T3	-32.531252	root_mean_squared_error	48.078591	1197.390141	0.159307	103.291187	
3	CatBoost_BAG_L2/T4	-32.561924	root_mean_squared_error	48.291876	1257.905307	0.372591	163.806352	
4	CatBoost_BAG_L2/T1	-32.567445	root_mean_squared_error	48.088339	1167.945280	0.169055	73.846325	
5	LightGBM_BAG_L2/T5	-32.853545	root_mean_squared_error	48.530510	1137.664110	0.611226	43.565155	
6	CatBoost_BAG_L2/T2	-32.959024	root_mean_squared_error	48.064410	1192.122505	0.145126	98.023550	
7	LightGBM_BAG_L2/T2	-33.014879	root_mean_squared_error	48.194550	1137.993551	0.275266	43.894596	
8	LightGBM_BAG_L2/T1	-33.029782	root_mean_squared_error	48.169108	1132.170493	0.249824	38.071538	
9	XGBoost_BAG_L2/T2	-33.049665	root_mean_squared_error	48.065982	1120.891615	0.146698	26.792660	

In [170.. predictor.leaderboard()

Out[170...

	model	score_val	eval_metric	pred_time_val	fit_time	pred_time_val_marginal	fit_time_marginal	s
0	WeightedEnsemble_L3	-114.628178	root_mean_squared_error	12.129280	360.516042	0.000876	0.042912	
1	LightGBMXT_BAG_L2	-115.242330	root_mean_squared_error	11.273526	289.767082	0.503703	34.100850	
2	LightGBM_BAG_L2	-115.456884	root_mean_squared_error	10.972936	286.953676	0.203114	31.287443	
3	CatBoost_BAG_L2	-115.702331	root_mean_squared_error	11.022127	282.994496	0.252305	27.328264	
4	RandomForestMSE_BAG_L1	-116.625366	root_mean_squared_error	0.545028	15.402647	0.545028	15.402647	
5	WeightedEnsemble_L2	-116.625366	root_mean_squared_error	0.545789	15.432832	0.000761	0.030185	
6	RandomForestMSE_BAG_L2	-117.333641	root_mean_squared_error	11.421588	295.084837	0.651765	39.418604	
7	ExtraTreesMSE_BAG_L1	-124.564085	root_mean_squared_error	0.573553	6.454430	0.573553	6.454430	
8	LightGBM_BAG_L1	-130.675014	root_mean_squared_error	2.342306	38.468436	2.342306	38.468436	
9	LightGBMXT_BAG_L1	-131.159249	root_mean_squared_error	7.087052	68.048435	7.087052	68.048435	
10	CatBoost_BAG_L1	-132.384648	root_mean_squared_error	0.221884	127.292284	0.221884	127.292284	

In [184...

```
predictor.model_info(predictor.model_best)
```

```
Out[184...: {'name': 'WeightedEnsemble_L3',
  'model_type': 'WeightedEnsembleModel',
  'problem_type': 'regression',
  'eval_metric': 'root_mean_squared_error',
  'stopping_metric': 'root_mean_squared_error',
  'fit_time': 0.04291248321533203,
  'num_classes': None,
  'quantile_levels': None,
  'predict_time': 0.0008757114410400391,
  'val_score': np.float64(-114.62817830628732),
  'hyperparameters': {'use_orig_features': False,
    'valid_stack': True,
    'max_base_models': 0,
    'max_base_models_per_type': 'auto',
    'save_bag_folds': True,
    'stratify': 'auto',
    'bin': 'auto',
    'n_bins': None,
    'vary_seed_across_folds': False,
    'custom_splits': None,
    'model_random_seed': 0},
  'hyperparameters_user': {'save_bag_folds': True},
  'hyperparameters_fit': {},
  'hyperparameters_nondefault': ['save_bag_folds'],
  'ag_args_fit': {'max_memory_usage_ratio': 1.0,
    'max_gpu_memory_usage_ratio': 1.0,
    'max_time_limit_ratio': 1.0,
    'max_time_limit': None,
    'min_time_limit': 0,
    'valid_raw_types': None,
    'valid_special_types': None,
    'ignored_type_group_special': None,
    'ignored_type_group_raw': None,
    'get_features_kwargs': None,
    'get_features_kwargs_extra': None,
    'predict_1_batch_size': None,
    'temperature_scalar': None,
    'drop_unique': False},
  'num_features': 4,
  'features': ['RandomForestMSE_BAG_L1',
    'LightGBM_BAG_L2',
    'RandomForestMSE_BAG_L2',
    'LightGBMXT_BAG_L2'],
  'feature_metadata': <autogluon.common.features.feature_metadata.FeatureMetadata at 0x78a37f05e010>,
  'memory_size': 4237,
  'compile_time': None,
  'is_initialized': True,
  'is_fit': True,
  'is_valid': True,
  'can_infer': True,
  'has_learning_curves': False,
  'num_samples': 10886,
  'val_in_fit': False,
  'unlabeled_in_fit': False,
  'num_cpus': 2,
  'num_gpus': 0,
  'bagged_info': {'child_model_type': 'GreedyWeightedEnsembleModel',
    'num_child_models': 1,
    'child_model_names': ['S1F1'],
    'n_repeats': 1,
    'k_per_n_repeat': [1],
    'random_state': 3,
    'low_memory': False,
    'bagged_mode': False,
    'max_memory_size': 4237,
    'min_memory_size': 4237,
    'child_hyperparameters': {'ensemble_size': 25, 'subsample_size': 1000000},
    'child_hyperparameters_user': {},
    'child_hyperparameters_fit': {'ensemble_size': 17},
    'child_ag_args_fit': {'max_memory_usage_ratio': 1.0,
      'max_gpu_memory_usage_ratio': 1.0,
      'max_time_limit_ratio': 1.0,
      'max_time_limit': None,
      'min_time_limit': 0,
      'valid_raw_types': None,
      'valid_special_types': None,
      'ignored_type_group_special': None,
      'ignored_type_group_raw': None,
      'get_features_kwargs': None,
      'get_features_kwargs_extra': None,
      'predict_1_batch_size': None,
      'temperature_scalar': None,
      'drop_unique': False}},
  'stacker_info': {'num_base_models': 4,
    'base_model_names': ['RandomForestMSE_BAG_L1',
      'LightGBMXT_BAG_L2',
      'LightGBM_BAG_L2',
      'RandomForestMSE_BAG_L2']},
  'stacker_type': 'WeightedEnsembleModel',
  'stacker_problem_type': 'regression',
  'stacker_eval_metric': 'root_mean_squared_error',
  'stacker_stopping_metric': 'root_mean_squared_error',
  'stacker_fit_time': 0.04291248321533203,
  'stacker_val_score': np.float64(-114.62817830628732),
  'stacker_hyperparameters': {'use_orig_features': False,
    'valid_stack': True,
    'max_base_models': 0,
    'max_base_models_per_type': 'auto',
    'save_bag_folds': True,
    'stratify': 'auto',
    'bin': 'auto',
    'n_bins': None,
    'vary_seed_across_folds': False,
    'custom_splits': None,
    'model_random_seed': 0},
  'stacker_hyperparameters_user': {'save_bag_folds': True},
  'stacker_hyperparameters_fit': {},
  'stacker_hyperparameters_nondefault': ['save_bag_folds'],
  'stacker_ag_args_fit': {'max_memory_usage_ratio': 1.0,
    'max_gpu_memory_usage_ratio': 1.0,
    'max_time_limit_ratio': 1.0,
    'max_time_limit': None,
    'min_time_limit': 0,
    'valid_raw_types': None,
    'valid_special_types': None,
    'ignored_type_group_special': None,
    'ignored_type_group_raw': None,
    'get_features_kwargs': None,
    'get_features_kwargs_extra': None,
    'predict_1_batch_size': None,
    'temperature_scalar': None,
    'drop_unique': False},
  'stacker_num_features': 4,
  'stacker_features': ['RandomForestMSE_BAG_L1',
    'LightGBM_BAG_L2',
    'RandomForestMSE_BAG_L2',
    'LightGBMXT_BAG_L2'],
  'stacker_feature_metadata': <autogluon.common.features.feature_metadata.FeatureMetadata at 0x78a37f05e010>,
  'stacker_memory_size': 4237,
  'stacker_compile_time': None,
  'stacker_is_initialized': True,
  'stacker_is_fit': True,
  'stacker_is_valid': True,
  'stacker_can_infer': True,
  'stacker_has_learning_curves': False,
  'stacker_num_samples': 10886,
  'stacker_val_in_fit': False,
  'stacker_unlabeled_in_fit': False,
  'stacker_num_cpus': 2,
  'stacker_num_gpus': 0,
  'stacker_bagged_info': {'child_model_type': 'GreedyWeightedEnsembleModel',
    'num_child_models': 1,
    'child_model_names': ['S1F1'],
    'n_repeats': 1,
    'k_per_n_repeat': [1],
    'random_state': 3,
    'low_memory': False,
    'bagged_mode': False,
    'max_memory_size': 4237,
    'min_memory_size': 4237,
    'child_hyperparameters': {'ensemble_size': 25, 'subsample_size': 1000000},
    'child_hyperparameters_user': {},
    'child_hyperparameters_fit': {'ensemble_size': 17},
    'child_ag_args_fit': {'max_memory_usage_ratio': 1.0,
      'max_gpu_memory_usage_ratio': 1.0,
      'max_time_limit_ratio': 1.0,
      'max_time_limit': None,
      'min_time_limit': 0,
      'valid_raw_types': None,
      'valid_special_types': None,
      'ignored_type_group_special': None,
      'ignored_type_group_raw': None,
      'get_features_kwargs': None,
      'get_features_kwargs_extra': None,
      'predict_1_batch_size': None,
      'temperature_scalar': None,
      'drop_unique': False}},
  'stacker_stacker_info': {'num_base_models': 4,
    'base_model_names': ['RandomForestMSE_BAG_L1',
      'LightGBMXT_BAG_L2',
      'LightGBM_BAG_L2',
      'RandomForestMSE_BAG_L2']},
  'stacker_stacker_type': 'WeightedEnsembleModel',
  'stacker_stacker_problem_type': 'regression',
  'stacker_stacker_eval_metric': 'root_mean_squared_error',
  'stacker_stacker_stopping_metric': 'root_mean_squared_error',
  'stacker_stacker_fit_time': 0.04291248321533203,
  'stacker_stacker_val_score': np.float64(-114.62817830628732),
  'stacker_stacker_hyperparameters': {'use_orig_features': False,
    'valid_stack': True,
    'max_base_models': 0,
    'max_base_models_per_type': 'auto',
    'save_bag_folds': True,
    'stratify': 'auto',
    'bin': 'auto',
    'n_bins': None,
    'vary_seed_across_folds': False,
    'custom_splits': None,
    'model_random_seed': 0},
  'stacker_stacker_hyperparameters_user': {'save_bag_folds': True},
  'stacker_stacker_hyperparameters_fit': {},
  'stacker_stacker_hyperparameters_nondefault': ['save_bag_folds'],
  'stacker_stacker_ag_args_fit': {'max_memory_usage_ratio': 1.0,
    'max_gpu_memory_usage_ratio': 1.0,
    'max_time_limit_ratio': 1.0,
    'max_time_limit': None,
    'min_time_limit': 0,
    'valid_raw_types': None,
    'valid_special_types': None,
    'ignored_type_group_special': None,
    'ignored_type_group_raw': None,
    'get_features_kwargs': None,
    'get_features_kwargs_extra': None,
    'predict_1_batch_size': None,
    'temperature_scalar': None,
    'drop_unique': False},
  'stacker_stacker_num_features': 4,
  'stacker_stacker_features': ['RandomForestMSE_BAG_L1',
    'LightGBM_BAG_L2',
    'RandomForestMSE_BAG_L2',
    'LightGBMXT_BAG_L2'],
  'stacker_stacker_feature_metadata': <autogluon.common.features.feature_metadata.FeatureMetadata at 0x78a37f05e010>,
  'stacker_stacker_memory_size': 4237,
  'stacker_stacker_compile_time': None,
  'stacker_stacker_is_initialized': True,
  'stacker_stacker_is_fit': True,
  'stacker_stacker_is_valid': True,
  'stacker_stacker_can_infer': True,
  'stacker_stacker_has_learning_curves': False,
  'stacker_stacker_num_samples': 10886,
  'stacker_stacker_val_in_fit': False,
  'stacker_stacker_unlabeled_in_fit': False,
  'stacker_stacker_num_cpus': 2,
  'stacker_stacker_num_gpus': 0,
  'stacker_stacker_bagged_info': {'child_model_type': 'GreedyWeightedEnsembleModel',
    'num_child_models': 1,
    'child_model_names': ['S1F1'],
    'n_repeats': 1,
    'k_per_n_repeat': [1],
    'random_state': 3,
    'low_memory': False,
    'bagged_mode': False,
    'max_memory_size': 4237,
    'min_memory_size': 4237,
    'child_hyperparameters': {'ensemble_size': 25, 'subsample_size': 1000000},
    'child_hyperparameters_user': {},
    'child_hyperparameters_fit': {'ensemble_size': 17},
    'child_ag_args_fit': {'max_memory_usage_ratio': 1.0,
      'max_gpu_memory_usage_ratio': 1.0,
      'max_time_limit_ratio': 1.0,
      'max_time_limit': None,
      'min_time_limit': 0,
      'valid_raw_types': None,
      'valid_special_types': None,
      'ignored_type_group_special': None,
      'ignored_type_group_raw': None,
      'get_features_kwargs': None,
      'get_features_kwargs_extra': None,
      'predict_1_batch_size': None,
      'temperature_scalar': None,
      'drop_unique': False}},
  'stacker_stacker_stacker_info': {'num_base_models': 4,
    'base_model_names': ['RandomForestMSE_BAG_L1',
      'LightGBMXT_BAG_L2',
      'LightGBM_BAG_L2',
      'RandomForestMSE_BAG_L2']},
  'stacker_stacker_stacker_type': 'WeightedEnsembleModel',
  'stacker_stacker_stacker_problem_type': 'regression',
  'stacker_stacker_stacker_eval_metric': 'root_mean_squared_error',
  'stacker_stacker_stacker_stopping_metric': 'root_mean_squared_error',
  'stacker_stacker_stacker_fit_time': 0.04291248321533203,
  'stacker_stacker_stacker_val_score': np.float64(-114.62817830628732),
  'stacker_stacker_stacker_hyperparameters': {'use_orig_features': False,
    'valid_stack': True,
    'max_base_models': 0,
    'max_base_models_per_type': 'auto',
    'save_bag_folds': True,
    'stratify': 'auto',
    'bin': 'auto',
    'n_bins': None,
    'vary_seed_across_folds': False,
    'custom_splits': None,
    'model_random_seed': 0},
  'stacker_stacker_stacker_hyperparameters_user': {'save_bag_folds': True},
  'stacker_stacker_stacker_hyperparameters_fit': {},
  'stacker_stacker_stacker_hyperparameters_nondefault': ['save_bag_folds'],
  'stacker_stacker_stacker_ag_args_fit': {'max_memory_usage_ratio': 1.0,
    'max_gpu_memory_usage_ratio': 1.0,
    'max_time_limit_ratio': 1.0,
    'max_time_limit': None,
    'min_time_limit': 0,
    'valid_raw_types': None,
    'valid_special_types': None,
    'ignored_type_group_special': None,
    'ignored_type_group_raw': None,
    'get_features_kwargs': None,
    'get_features_kwargs_extra': None,
    'predict_1_batch_size': None,
    'temperature_scalar': None,
    'drop_unique': False},
  'stacker_stacker_stacker_num_features': 4,
  'stacker_stacker_stacker_features': ['RandomForestMSE_BAG_L1',
    'LightGBM_BAG_L2',
    'RandomForestMSE_BAG_L2',
    'LightGBMXT_BAG_L2'],
  'stacker_stacker_stacker_feature_metadata': <autogluon.common.features.feature_metadata.FeatureMetadata at 0x78a37f05e010>,
  'stacker_stacker_stacker_memory_size': 4237,
  'stacker_stacker_stacker_compile_time': None,
  'stacker_stacker_stacker_is_initialized': True,
  'stacker_stacker_stacker_is_fit': True,
  'stacker_stacker_stacker_is_valid': True,
  'stacker_stacker_stacker_can_infer': True,
  'stacker_stacker_stacker_has_learning_curves': False,
  'stacker_stacker_stacker_num_samples': 10886,
  'stacker_stacker_stacker_val_in_fit': False,
  'stacker_stacker_stacker_unlabeled_in_fit': False,
  'stacker_stacker_stacker_num_cpus': 2,
  'stacker_stacker_stacker_num_gpus': 0,
  'stacker_stacker_stacker_bagged_info': {'child_model_type': 'GreedyWeightedEnsembleModel',
    'num_child_models': 1,
    'child_model_names': ['S1F1'],
    'n_repeats': 1,
    'k_per_n_repeat': [1],
    'random_state': 3,
    'low_memory': False,
    'bagged_mode': False,
    'max_memory_size': 
```

```
'children_info': {'S1F1': {'name': 'S1F1',
'problem_type': 'GreedyWeightedEnsembleModel',
'eval_metric': 'root_mean_squared_error',
'stopping_metric': 'root_mean_squared_error',
'fit_time': 0.04291248321533203,
'num_classes': None,
'quantile_levels': None,
'predict_time': None,
'val_score': None,
'hyperparameters': {'ensemble_size': 25, 'subsample_size': 1000000},
'hyperparameters_user': {},
'hyperparameters_fit': {'ensemble_size': 17},
'hyperparameters_nondefault': []},
'ag_args_fit': {'max_memory_usage_ratio': 1.0,
'max_gpu_memory_usage_ratio': 1.0,
'max_time_limit_ratio': 1.0,
'max_time_limit': None,
'min_time_limit': 0,
'valid_raw_types': None,
'valid_special_types': None,
'ignored_type_group_special': None,
'ignored_type_group_raw': None,
'get_features_kwargs': None,
'get_features_kwargs_extra': None,
'predict_1_batch_size': None,
'temperature_scalar': None,
'drop_unique': False},
'num_features': 4,
'features': ['RandomForestMSE_BAG_L1',
'LightGBMXT_BAG_L2',
'LightGBM_BAG_L2',
'RandomForestMSE_BAG_L2'],
'feature_metadata': <autogluon.common.features.feature_metadata.FeatureMetadata at 0x78a3a289ef10>,
'memory_size': 8961,
'compile_time': None,
'is_initialized': True,
'is_fit': True,
'is_valid': True,
'can_infer': True,
'has_learning_curves': False,
'num_samples': 10886,
'val_in_fit': False,
'unlabeled_in_fit': False,
'num_cpus': 2,
'num_gpus': 0,
'model_weights': {'RandomForestMSE_BAG_L1': np.float64(0.11764705882352941),
'LightGBMXT_BAG_L2': np.float64(0.47058823529411764),
'LightGBM_BAG_L2': np.float64(0.23529411764705882),
'RandomForestMSE_BAG_L2': np.float64(0.17647058823529413)}}}
```

```
In [171... predictor_new_features.leaderboard()
```

		model	score_val	eval_metric	pred_time_val	fit_time	pred_time_val_marginal	fit_time_marginal	st
0		WeightedEnsemble_L3	-32.196982	root_mean_squared_error	20.739971	367.585089	0.000847	0.042005	
1		WeightedEnsemble_L2	-32.362592	root_mean_squared_error	19.141901	260.526054	0.000582	0.022637	
2		LightGBM_BAG_L2	-32.893209	root_mean_squared_error	19.954914	302.281716	0.254721	31.410632	
3		RandomForestMSE_BAG_L2	-33.073118	root_mean_squared_error	20.343991	311.668437	0.643798	40.797353	
4		LightGBMXT_BAG_L2	-33.613493	root_mean_squared_error	20.136414	305.567797	0.436221	34.696713	
5		LightGBM_BAG_L1	-33.634169	root_mean_squared_error	5.921896	66.015301	5.921896	66.015301	
6		CatBoost_BAG_L2	-34.227213	root_mean_squared_error	19.840605	295.335100	0.140412	24.464016	
7		LightGBMXT_BAG_L1	-34.771022	root_mean_squared_error	12.321824	97.837913	12.321824	97.837913	
8		ExtraTreesMSE_BAG_L1	-38.348406	root_mean_squared_error	0.558875	10.367667	0.558875	10.367667	
9		RandomForestMSE_BAG_L1	-38.457978	root_mean_squared_error	0.606615	20.564800	0.606615	20.564800	
10		CatBoost_BAG_L1	-38.499695	root_mean_squared_error	0.290983	76.085402	0.290983	76.085402	



```
In [185... predictor_new_features.model_info(predictor_new_features.model_best)
```

```
Out[185... {'name': 'WeightedEnsemble_L3',
'model_type': 'WeightedEnsembleModel',
'problem_type': 'regression',
'eval_metric': 'root_mean_squared_error',
'stopping_metric': 'root_mean_squared_error',
'fit_time': 0.04200458526611328,
'num_classes': None,
'quantile_levels': None,
'predict_time': 0.0008473396301269531,
'val_score': np.float64(-32.19698212440195),
'hyperparameters': {'use_orig_features': False,
'valid_stack': True,
'max_base_models': 0,
'max_base_models_per_type': 'auto',
'save_bag_folds': True,
'strainify': 'auto',
'bin': 'auto',
'n_bins': None,
'vary_seed_across_folds': False,
'custom_splits': None,
'model_random_seed': 0},
'hyperparameters_user': {'save_bag_folds': True},
'hyperparameters_fit': {},
'hyperparameters_nondefault': ['save_bag_folds'],
'ag_args_fit': {'max_memory_usage_ratio': 1.0,
'max_gpu_memory_usage_ratio': 1.0,
'max_time_limit_ratio': 1.0,
'max_time_limit': None,
'min_time_limit': 0,
'valid_raw_types': None,
'valid_special_types': None,
'ignored_type_group_special': None,
'ignored_type_group_raw': None,
'get_features_kwargs': None,
'get_features_kwargs_extra': None,
'predict_1_batch_size': None,
'temperature_scalar': None,
'drop_unique': False},
'num_features': 6,
'features': ['LightGBM_BAG_L2',
'CatBoost_BAG_L2',
'LightGBM_BAG_L1',
'LightGBMXt_BAG_L1',
'RandomForestMSE_BAG_L1',
'RandomForestMSE_BAG_L2'],
'feature_metadata': <autogluon.common.features.feature_metadata.FeatureMetadata at 0x78a38113d250>,
'memory_size': 4283,
'compile_time': None,
'is_initialized': True,
'is_fit': True,
'is_valid': True,
'can_infer': True,
'has_learning_curves': False,
'num_samples': 10886,
'val_in_fit': False,
'unlabeled_in_fit': False,
'num_cpus': 2,
'num_gpus': 1,
'bagged_info': {'child_model_type': 'GreedyWeightedEnsembleModel',
'num_child_models': 1,
'child_model_names': ['S1F1'],
'_n_repeats': 1,
'_k_per_n_repeat': [1],
'_random_state': 3,
'low_memory': False,
'bagged_mode': False,
'max_memory_size': 4283,
'min_memory_size': 4283,
'child_hyperparameters': {'ensemble_size': 25, 'subsampling_size': 1000000},
'child_hyperparameters_user': {},
'child_hyperparameters_fit': {'ensemble_size': 23},
'child_ag_args_fit': {'max_memory_usage_ratio': 1.0,
'max_gpu_memory_usage_ratio': 1.0,
'max_time_limit_ratio': 1.0,
'max_time_limit': None,
'min_time_limit': 0,
'valid_raw_types': None,
'valid_special_types': None,
'ignored_type_group_special': None,
'ignored_type_group_raw': None,
'get_features_kwargs': None,
'get_features_kwargs_extra': None,
'predict_1_batch_size': None,
'temperature_scalar': None,
'drop_unique': False}},
'stacker_info': {'num_base_models': 6,
'base_model_names': ['LightGBMXt_BAG_L1',
'LightGBM BAG L1',
```

```

'RandomForestMSE_BAG_L1',
'LightGBM_BAG_L2',
'RandomForestMSE_BAG_L2',
'CatBoost_BAG_L2']]),
'children_info': {'S1F1': {'name': 'S1F1',
'model_type': 'GreedyWeightedEnsembleModel',
'problem_type': 'regression',
'eval_metric': 'root_mean_squared_error',
'stopping_metric': 'root_mean_squared_error',
'fit_time': 0.04200458526611328,
'num_classes': None,
'quantile_levels': None,
'predict_time': None,
'val_score': None,
'hyperparameters': {'ensemble_size': 25, 'subsample_size': 1000000},
'hyperparameters_user': {},
'hyperparameters_fit': {'ensemble_size': 23},
'hyperparameters_nondefault': [],
'ag_args_fit': {'max_memory_usage_ratio': 1.0,
'max_gpu_memory_usage_ratio': 1.0,
'max_time_limit_ratio': 1.0,
'max_time_limit': None,
'min_time_limit': 0,
'valid_raw_types': None,
'valid_special_types': None,
'ignored_type_group_special': None,
'ignored_type_group_raw': None,
'get_features_kwargs': None,
'get_features_kwargs_extra': None,
'predict_1_batch_size': None,
'temperature_scalar': None,
'drop_unique': False},
'num_features': 6,
'features': ['LightGBMXT_BAG_L1',
'LightGBM_BAG_L1',
'RandomForestMSE_BAG_L1',
'LightGBM_BAG_L2',
'RandomForestMSE_BAG_L2',
'CatBoost_BAG_L2']},
'feature_metadata': <autogluon.common.features.feature_metadata.FeatureMetadata at 0x78a383e3ea90>,
'memory_size': 9357,
'compile_time': None,
'is_initialized': True,
'is_fit': True,
'is_valid': True,
'can_infer': True,
'has_learning_curves': False,
'num_samples': 10886,
'val_in_fit': False,
'unlabeled_in_fit': False,
'num_cpus': 2,
'num_gpus': 1,
'model_weights': {'LightGBMXT_BAG_L1': np.float64(0.21739130434782608),
'LightGBM_BAG_L1': np.float64(0.30434782608695654),
'RandomForestMSE_BAG_L1': np.float64(0.08695652173913043),
'LightGBM_BAG_L2': np.float64(0.08695652173913043),
'RandomForestMSE_BAG_L2': np.float64(0.2608695652173913),
'CatBoost_BAG_L2': np.float64(0.043478260869565216)}}})

```